

# Formal Architecture and Machinery of the Normative Formal Core and Appendices G, I, J, K, and L

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|                                                 |                          |                         |                      |                               |
|-------------------------------------------------|--------------------------|-------------------------|----------------------|-------------------------------|
| <b>CORE/G</b><br>Shared semantics +<br>contract | <b>I</b><br>Local replay | <b>J</b><br>Safety glue | <b>K</b><br>Liveness | <b>L</b><br>Witness selection |
|-------------------------------------------------|--------------------------|-------------------------|----------------------|-------------------------------|

Based on: Catapult HLS User View Scheduling Rules, 3 September 2026 revision

*The latest version of the scheduling rules document is available here:*

[https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib\\_examples/doc/catapult\\_user\\_view\\_scheduling\\_rules.pdf](https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/catapult_user_view_scheduling_rules.pdf)

The scheduling rules document is the normative source: it defines the scheduling contract, formal models, assumptions, theorem statements, proofs, and evidence obligations. This formal architecture guide adds value by reorganizing that material into a coherent map of how the Core and Appendices G, I, J, K, and L depend on one another, what information crosses each proof boundary, and how design choices connect to the safety, witness-selection, and deadlock arguments. It therefore serves as a navigation and conceptual-integration aid rather than an alternate specification; when its simplified explanation and the scheduling rules differ, the scheduling rules remain authoritative.

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# 1. Executive orientation

## PURPOSE

**Provide a single mental model for the Normative Formal Core and five appendices that otherwise appear as separate layers of formal detail.**

**Engineer takeaway:** Treat the architecture as a proof stack: the Core fixes the shared operational semantics; G states the observable contract; I generates local evidence; J constructs the global safety witness; K proves a separate conditional internal-deadlock result; and L selects source witnesses when latency-sensitive choices matter.

The formal architecture is best understood as one pairwise correspondence engine in Appendix J, two different ways to obtain implementation-wide safety coverage from it, and a separate K-facing liveness/deadlock pipeline. The pairwise J.1 construction is exact-history/occurrence-relative on the source side and is deliberately B1-free. The standard QSync safety route obtains universal RTL coverage by cycle/carrier induction under QualifiedSafetyCarrierModel\_RTL\_B and, for arbitrary finite prefixes, Core.RTLPartialCycleClosure/J.CycleIdealCover. The alternate B1-run route instead assumes bucketed B1 at the selected outer alphabet  $\Sigma_{\text{DUT}}$ , one finite reachable DeclaredTerminal outer-occurrence maximum Core.B1RunMax, and narrowly triggered reset/elaboration congruence; it then reduces universal RTL safety to one selected J.TraceCertCov package. K-facing observation remains a sibling branch rather than the object extended by either safety route: Core.L3CloseView/J.L3CloseView-H close a source construction checkpoint only for KObs normalization and K.NormStable. On RTL, the three-level QSync hierarchy remains unchanged for the standard carrier/K-facing routes: QualifiedSafetyCarrierModel\_RTL\_B = QS1 + QS2 + QS2a-D, QualifiedSynchronousCarrierModel\_RTL\_B adds QS3-QS5, and QualifiedSynchronousModel\_RTL\_B adds QS2a-R.

- Safety pipeline (Core  $\rightarrow$  G  $\rightarrow$  I  $\rightarrow$  J). Plain Theorem J.1 constructs or validates one reachable source witness for one selected reachable RTL prefix; it does not require Definition B1. Core.17 supplies canonical occurrence/value matching and Appendix G supplies R1-R5/E1-E5. J.LocalGWR/J.GWR-Comp uses source-side QualifiedSynchronousModel\_Sys\_ref/Core.QSync-Ref on the normal compositional route, or a directly validated J.GWR package may carry equivalent per-Horizon settling/bridge evidence. From that B1-free pairwise engine, the rules expose two universal safety routes for J.IdealPrefixClosed properties: (a) standard QSync cycle/carrier induction under QualifiedSafetyCarrierModel\_RTL\_B plus, when needed, Core.RTLPartialCycleClosure/J.CycleIdealCover; or (b) the alternate B1-run route, where bucketed outer B1 + Core.B1RunMax + J.B1OuterHistCongruence derive J.B1PostProjCover and Corollary J.1-Safe-B1Run needs only singleton J.TraceCertCov( $\{p\_post^0\}$ ). The B1-run route does not require RTL QSync carrier/origin induction, RTLPartialCycleClosure, J.LocalGWRExtCert/GWR-CompExt, or K-only Offer/KObs/drain evidence.
- Liveness/deadlock pipeline (Core  $\rightarrow$  G  $\rightarrow$  I  $\rightarrow$  J  $\rightarrow$  K). Starts from the exact J-selected source/RTL package and its common typed Core.18 KObs record, then adds qualifications that KObs deliberately does not contain. Core.CycleClosure/Core.CycleChoiceExt/FairCycleDovetail and Core.InFlightCycleExt/J.InFlightPathSound are occurrence-relative: every successor/path edge consumes the exact prefix/carrier occurrence produced by the previous step. K.InFlightQual now separates record-level K.InFlightRelationPrecision from each negative-search K.InFlightSearchStatus, and K.InFlightProfileCover is supplied by the consumer for every exact occurrence it uses rather than being stored in a route-neutral certified cutpoint. The primary Policy-S reduction additionally binds K.PolicySProgress, exact-prefix K.CVPred provenance (including ObsKey/version identity, ResetSupport\_CV and CVP-M), SR3-G generalized-drain coverage, and K.SerialCandidateApplicability over the complete counterexample domain. A nested Continue candidate fixes an explicit valid K.DomResetScope and one prospective K.PolicySCompareDomains= $\langle$ SFControl,Handoff,Response $\rangle$  package before SF/Q. K.SFOutputEnvelope and the conservative K.RespSerialBlockerClassSet describe all possible SF outputs without predicting the actual blocker; CD3-C proves Handoff completeness for every relevant dynamic monitor instance, and CD4 is derived only after SF by instantiating that universal fact. Response extraction then classifies post-handoff affecting RESET first, DeclaredTerminal second, and timeout third. In G2, K.Low.TimeStutterQual is required when lowering consumes real cycles; K.Low.ModalQual MQ2 is instance-wide over every relevant normalization window and MQ5 includes both model-indexed InFlightQual records plus paired-occurrence ProfileCover. Universal RTL preservation still separately consumes Core.KCutClosure\_post and K.CertCov.
- **Witness-selection adapter (L).** Supplies the WC witness needed by I and J when failed non-blocking polls or other latency-sensitive epsilon choices affect later observable behavior. It selects an admissible source execution aligned with the free-running RTL; it does not redefine the source model or stall the RTL.

**KEY ARCHITECTURAL FACT: Safety and liveness are not one theorem**

Appendix J establishes a global observable-safety correspondence from local evidence. Appendix K does not strengthen that correspondence into liveness automatically. It adds a different argument with different assumptions: wait-for-graph reconstruction, drain discipline, fairness/progress, cutpoint coverage, and the scheduler-robust source-liveness premise A5-L.

## 1.1 What the formal framework is trying to achieve

The user-facing goal is a verification flow in which most functional verification and debug can remain at the pre-HLS SystemC level, while the RTL is checked against a precise implementation contract. The formal appendices make that promise conditional and auditable: they identify exactly which source rules, implementation properties, witness choices, boundary abstractions, and liveness assumptions must hold.

How to interpret the qualification detail. The formal architecture names many conditions that can look like new work when they are listed explicitly. Many design- and implementation-conformance conditions were already implicit or explicit in conventional HLS-generated and manual RTL practice: clock/reset discipline, boundary/capacity correctness, request persistence, registered or synchronized interface behavior, and faithful implementation-boundary mapping still have to be valid if source-level conclusions are to be trusted. The safety pipeline also needs proof-construction, provenance, and coverage artifacts that conventional RTL flows do not ordinarily produce. In the standard QSync route those include exact carrier/origin provenance, incremental cross-prefix witness extension, and exhaustive cycle-successor coverage. In the alternate B1-run route the major RTL-side coverage burden moves instead to observable-projection determinism B1, Core.B1RunMax/BRM2 outer-occurrence maximality, and the triggered Core.ResetScheduleCongruence / Q.OuterProjectionCongruence interfaces, while only one selected J correspondence package is needed. The intended architecture is to amortize these through reusable tool/library qualification, translation validation, generated proof artifacts, and independent checking rather than separate manual design-user tasks. Appendix F still contains genuine theorem-scope design restrictions, and Appendix K adds richer liveness-only provenance, release-family, drain, lowering, and response evidence.

The central idea is to compare only architecturally meaningful observations: committed message transfers, synchronization operations, and anchored signal reads/writes at selected boundaries. Internal behavior outside  $\Sigma$  is not all represented by the same formal object. Source evaluation and other proof-hidden micro-steps may be  $\epsilon$ -steps; same-cycle settling is classified separately as  $\delta$ -settling; a real clock cycle with hidden registered progress but no selected  $\Sigma$  event is a silent  $\tau$ -cycle; and a B6 idle tick performs no  $\epsilon$  work. Section 4 distinguishes these cases.

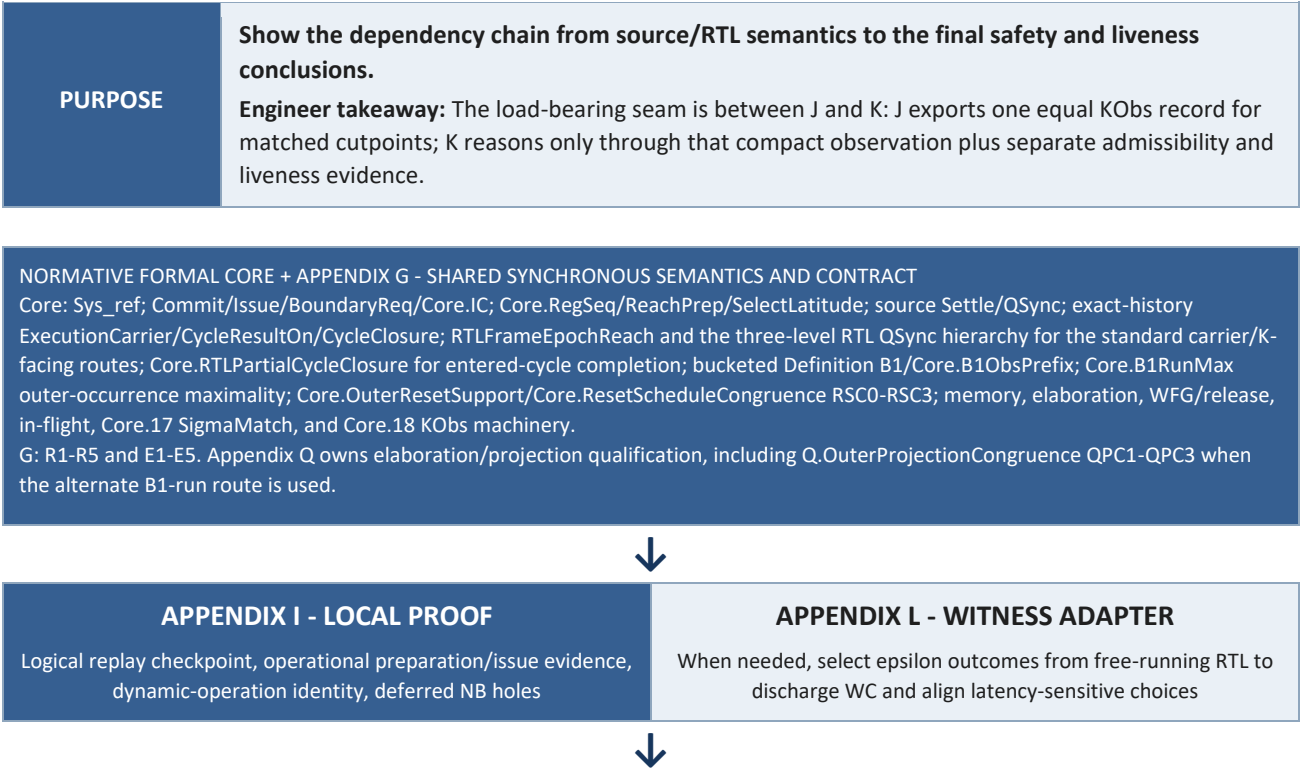
## 1.2 What the framework does not claim

- It does not claim identical cycle timing between pre-HLS and post-HLS models. Core.SelectLatitude deliberately permits finite non-maximal source selection at L1 issue and L3 boundary-action opportunities; J.HCanon discards incidental source-versus-RTL clock-bucket placement unless the selected environment or other cycle-sensitive evidence makes that spacing semantically relevant.
- It does not claim equality of all internal state, requests, or pipeline registers.
- It does not automatically prove that a particular HLS run or RTL instance conforms to E3, E4, E5/Core.SyncFence, Core.IC, B-faithfulness, Core.RegSeq, Core.ReachPrep, Core.ElabProj, Core.MemFaithful, Core.MemSyncOrder, Core.MemProfile, or the applicable settling/coverage qualification. For the standard QSync safety route, RTL qualification is QualifiedSafetyCarrierModel\_RTL\_B = QS1 + QS2 + QS2a-D, with Core.RTLPartialCycleClosure added only for the all-finite-prefix ideal-cover step; canonical post settling adds QS3-QS5 and queried-KCut recovery adds QS2a-R. The alternate B1-run route does not consume that RTL carrier hierarchy merely for safety coverage, but instead requires B1 at outer  $\Sigma_{DUT}$ , Core.B1RunMax, ordinary R/E/Core.17 conformance, and any triggered RSC0-RSC3 / QPC1-QPC3 qualification. Source-side Core.QSync remains required on the normal J.LocalGWR  $\rightarrow$  J.GWR-Comp route in either safety profile; a directly validated J.GWR route may carry equivalent checked per-Horizon settling evidence.
- It does not claim that the alternate B1-run route is a general replacement for the standard QSync safety route. Corollary J.1-Safe-B1Run is safety-only and fixed-stimulus; it requires one finite reachable DeclaredTerminal run satisfying

Core.B1RunMax and applies only to safety predicates that are extensional on CanonHist\_ $\Sigma$ \_DUT and satisfy J.IdealPrefixClosed. It does not establish Appendix-K/K-facing package coverage, does not cover nonterminating or service instances lacking such a finite outer-occurrence maximum, and does not transfer latency/performance or other non-ideal-prefix-closed properties. Those cases remain on the standard QSync route or another separately qualified coverage route.

- It does not infer global deadlock freedom from the source-order no-reverse rule alone.
- The standard compositional QSync route, and the standard K-facing/universal G-K results, do not cover unqualified non-QSync settling semantics. Pure combinational feedback that is not broken by explicit registered or finite-capacity state, non-deterministic or order-dependent same-cycle settling, and multi-clock instances without a separately specified semantic layer are outside that standard scope. Plain finite-prefix J.1 may instead use a directly validated J.GWR package carrying the required checked per-Horizon L2-settling evidence. A non-QSync design is not thereby asserted incorrect; broader K-facing or universal use requires an explicitly defined and separately qualified theorem extension.
- It does not treat an arbitrary finite or unqualified Policy-S simulation as a complete liveness proof. The primary route uses one functional SDet-selected artifact, but SDet is only selection: K.PolicySProgress separately binds Core.IssueFair\_{Policy S} and the applicable B2/B3 obligations, and K.CompleteS packages the complete/maximal run. K.SerialRoutePremises adds SR3-G generalized-drain qualification/profile/query coverage and universal K.SerialCandidateApplicability; K.CVPred is exact-prefix/provenance based and recurses on WellFounded(prec\_CV), with a numeric rank optional rather than normative. Reset/terminal applicability binds an explicitly selected valid K.DomResetScope and a prospectively fixed PolicySCompareDomains package; the actual post-SF handoff may not be used to carve that package. K.RespCov/K.RespClean cover literal and static-derived canonical logical monitors. Generalized drain adds model-specific InFlightQual, per-consumer InFlightProfileCover, and typed accepted negative queries; UnknownIncomplete is diagnostic only. In G2, K.Low.TimeStutterQual and K.Low.ModalQual are separate model-transport qualifications, including instance-wide MQ2 normalization opportunity and paired-occurrence MQ5 profile coverage. A single RTL or Policy-S run does not establish universal KCut/package totality.
- It does not make latency-sensitive polling safe merely by hiding failed polls as epsilon steps; such sites require NB-CF, isolation, direct witness construction, or Appendix L snooping.

## 2. Architecture at a glance



## APPENDIX J - GLOBAL SAFETY CONSTRUCTION

LocalGWR -> typed construction/dependency machinery -> exact source bridges -> J.GWR-Comp -> Proposition J.0 -> plain Theorem J.1. J.1 is the B1-free pairwise Post -> Ref engine.

STANDARD QSYNC UNIVERSAL SAFETY: QualifiedSafetyCarrierModel\_RTL\_B + ExecutionCarrier/RTLFrameEpochReachOn -> J.QSyncTraceInd -> J.TraceCertCov-QSyncCycle -> J.CycleIdealCover/J.1-Safe-QSync, with Core.RTLPartialCycleClosure only for an already-entered partial cycle.

ALTERNATE B1-RUN UNIVERSAL SAFETY: outer B1 + Core.B1RunMax + triggered RSC/QPC + J.B1OuterHistCongruence -> J.B1PostProjCover-RunMax -> singleton J.TraceCertCov({p\_post^0}) -> Corollary J.1-Safe-B1Run. No RTL carrier/QSync induction, RTLPartialCycleClosure, or LocalGWRExtCert is required by this alternate coverage proof.



## SAFETY OUTPUT

Plain J.1 gives Post -> Ref compatibility for a selected RTL prefix and is B1-free. For implementation-wide ideal-prefix-closed safety, choose a coverage route:

- 1) Standard QSync: cycle-aligned package induction under QualifiedSafetyCarrierModel\_RTL\_B, then Core.RTLPartialCycleClosure/J.CycleIdealCover when arbitrary finite prefixes may end inside an entered cycle.
- 2) Alternate B1-run: one finite reachable cycle-closed DeclaredTerminal p\_post^0 satisfying Core.B1RunMax, B1 at outer  $\Sigma_{DUT}$ , triggered RSC0-RSC3/QPC1-QPC3, and one selected J.TraceCertCov package. J.B1PostProjCover-RunMax makes every reachable RTL outer history an ideal of the selected maximum; J.1-Safe-B1Run transfers  $\Phi$  from the matched selected source witness.

Neither route by itself supplies K-facing deadlock evidence.

## KOBS BRIDGE

Common typed Core.18 record <E,w,H,O>. J constructs KObsPost\_J and proves equality for the exact selected package, then reconstructs process/channel/frontier/WFG facts plus the current synchronization carrier and current in-flight action relation. KObs does not contain an activation certificate, historical Issue times, or future action opportunities; J.InFlightPathSound is a separate level-3 qualification.



## APPENDIX K - CONDITIONAL LIVENESS

Equal KObs at matched KCuts -> exact message roots/release families plus current in-flight relation ->

K.WaitRoots/InFlightEscape/DrainClosedNow -> release-aware K.Dead -> A5-L contradiction. J.InFlightPathSound is universal over admitted profile starts and threads the exact operational prefix/carrier occurrence edge by edge. K.InFlightQual carries action/effect soundness, finite drain, continuation, path soundness and record-level relation precision; each consumer separately binds K.InFlightProfileCover plus accepted occurrence-bound or uniform-fiber negative-search status.

The Policy-S route fixes one SDet artifact but separately requires K.PolicySProgress, exact-prefix K.CVPred/CVP-M/ResetSupport\_CV provenance, SR3-G, universal K.SerialCandidateApplicability and an explicit selected valid K.DomResetScope. Nested Continue fixes K.PolicySCompareDomains=<SFControl,Handoff,Response> before SF/Q; K.SFOutputEnvelope and K.RespSerialClassExclusionSound conservatively determine class coverage; CD3-C makes Handoff prospectively complete; CD4 is post-SF instantiation only. K.2b.Q then evaluates Response with reset-first cancelled-pending, terminal maximal-pending, then timeout. G2 additionally uses TimeStutterQual, instance-wide MQ2 normalization opportunity, paired MQ5 ProfileCover, ResetStutterMatch/ResetKey-Stutter where LT2 maps a reset, and fresh cross-model negative-evidence transport. K.CertHdr/M.7 use a consumed-interface manifest; incompatible stored artifacts require checked InterfaceSchemaMigration MIG1-MIG5 or revalidation.

## APPENDIX M.7b / M.7c PACKAGING OVERLAYS

M.7a remains the definitive trigger/discharge ledger. M.7b packages the standard QSync safety/K-facing flow through four logical bundle interfaces. M.7c packages only Corollary J.1-Safe-B1Run: one RTL maximum/congruence record, one selected-run correspondence record, one WC variant, and one source-safety/property record. M.7c does not weaken any M.7a trigger; it simply changes the universal RTL coverage mechanism. K.CertHdr/K.CertPkg may bind either safety profile without fabricating K-only fields.

## 2.1 The main interfaces between layers

| Interface     | What crosses it                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core -> G/I   | The Core supplies Sys_ref, observable/operational semantics, source exact-history provenance, and the standard RTL carrier/QSync hierarchy. It now also supplies the alternate B1-run coverage vocabulary: bucketed B1/Core.B1ObsPrefix, Core.B1RunMax/Core.B1RunMax-Cover, and Core.OuterResetSupport/Core.ResetScheduleCongruence. G constrains the common observable contract through R1-R5/E1-E5. Appendix Q supplies elaboration/projection qualification, including Q.OuterProjectionCongruence for the B1-run route and Q.2 action/effect schemas where selected. Plain J.1 does not require B1 merely to construct the selected pair.                                                                                                                                                                                                                                                                                                                                                                                          |
| I -> J        | I supplies process-local J.PCert evidence: a logical replay checkpoint plus separate operational preparation/issue facts, exact dynamic identity, typed footprints, and deferred non-blocking holes. These local facts intentionally do not claim global channel enablement or one common source execution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| L -> I/J      | L supplies a witness selector proving WC when epsilon-modeled choices can affect later observable control, values, request attribution, or frontier identity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| J -> K        | For one K-facing use, J exports the exact extendible construction checkpoint tau_ref^G and then forms a fresh closed sibling tau_ref^cl through Core.L3CloseView/J.L3CloseView-H before KObsNormRef/K.NormStable/OfferReach. K.NormStable NS0-NS6 establishes that this close-view/observation branch does not reopen construction or create request/root/deferred-choice/reset/L4 changes. J.KObs-Proc/Chan/Offer/WFG and the synchronization/in-flight reconstruction lemmas recover current state. Future progress is separate: J.InFlightPathSound realizes every finite reconstructed path on the exact occurrence chain. K consumers bind their own InFlightProfileCover and accepted negative-search queries. For universal standard-QSync coverage, QualifiedSynchronousCarrierModel_RTL_B supplies the QS3-QS5 post-settling layer used by Core.QSync-Post to derive KCut closure; full QS2a-R/KOriginCarrier-KCut plus J.QSyncCertInd/J.CertExist-QSync/K.CertCov-QSync provide queried-KCut placement and package totality. |
| Policy-S -> K | The design-user artifact remains one selected complete instrumented Policy-S execution, but the current architecture separates selection, progress, applicability and response. SDet fixes one functional artifact; K.PolicySProgress supplies exact-run IssueFair/B2/B3; K.SerialRoutePremises includes SR3-G generalized-drain coverage; K.SerialCandidateApplicability covers every non-waived counterexample candidate; K.CVPred supplies exact-prefix/provenance/ResetSupport_CV evidence. Each nested Continue candidate binds an explicit valid K.DomResetScope and a prospective K.PolicySCompareDomains=<SFControl,Handoff,Response> package. SFOutputEnvelope/class coverage is fixed before SF; CD4 actual handoff membership is derived after SF from CD3-C. In G2, K.Low.TimeStutterQual and K.Low.ModalQual additionally carry reset-sensitive normalization opportunity and paired Sys_K/Sys_ref profile coverage before K.Low.A5 transports A5L_Sys_K to source A5-L.                                                  |



## 2.2 Generic theorem machinery versus design-specific evidence

The framework deliberately separates proof algorithms that should be mechanized once from evidence that must be supplied or checked for each design, theorem instance, or RTL build. Appendix T of the scheduling-rules document complements this distinction by identifying the scheduling-specific semantic and proof obligations that remain beyond generic transition-system, simulation, induction, and temporal-logic libraries. It is a conceptual gap map, not an additional theorem or certificate layer: the normative definitions and theorem interfaces remain in the Core and Appendices I-K/K-P, with evidence and qualification interfaces in Appendix M.7a and the M.7b/M.7c packaging profiles. Appendix S is likewise a selected foundational mechanization-spine/import-anchor inventory, not an exhaustive side-condition list; Appendix M.7a remains the definitive consolidated per-instance discharge ledger.

| Category                          | Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reusable formal machinery</b>  | Core exact-history/cycle/QSync semantics; ExecutionCarrier/ResetEpochKey/RTLFrameEpochReachOn/RTLPartialCycleClosure; Core.17/Core.18; Appendix-I replay; J.SourceBridgeClosure/Ext, PhaseRole_A5, ProofUnitConstructionSupport/COrdActPred/SupportInsertionExists, GWR/GWR-CompExt, Q.1(c-order) outer-group ideal machinery, QSync safety/package induction; Core.FlightStateRec/InFlightStepRec and Q.2 finite action/effect schemas; J.InFlightPathSound; K.WaitRoots/InFlightEscape plus relation-precision/search-status/profile-cover framework; and the K.Low TimeStutter/ModalQual/path-transport stack. The serial K.CV/WellFounded(prec_CV)/SegWork and prospective response-domain machinery are likewise reusable formal mechanisms once encoded.                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Per-design source evidence</b> | Source well-formedness, unique signal anchors including the final-Write-at-shared-anchor canonicalization, Core.RegSeq classification, ReachPrep coverage where eager preparation is admitted, source semantics using Core.9 operational Issue, Core.10 operation-specific BoundaryReq, and the constitutive Core.IC/Core.SyncFence legality rules, and - when a concrete executable reference model is the theorem carrier - conformance to Core.SelectLatitude rather than a hidden eager/maximal-progress rule. Also include source-side QSync structural conformance or direct per-Horizon settling evidence, WC/NB-CF/snooping coverage, Core.MemFaithful for hidden transformed mapped-memory activity, Core.MemSyncOrder where array accesses cross synchronization, J.Mem complete dependency-level shared-memory lowering when cross-process value flow exists, Core.MemProfile when a qualified transformation-invariant logical memory boundary is used, barrier well-formedness, and value/control provenance. Qualified Sys_B^elab templates may discharge reusable settling facts, but whole-source composition/adequacy remains instance-owned. |



| Category                            | Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Per-RTL/tool evidence               | <p>E3/E5/Core.IC/SyncFence and E4 boundary mapping; exact capacities/elaboration; B-faithfulness; Core.ElabProj including Q.1(c-order) projected group identity/order; mapped-memory qualifications; reset behavior; ClockScope; exact RTL ExecutionCarrier/reset provenance and QualifiedSafetyCarrierModel_RTL_B/QS2a-D designation. When no dynamic RESET can occur after the prescribed base, reset evidence reduces to the checked exclusion fact, prescribed initial epoch binding, and Core.ResetEpochAdvance_I(e,∅,e) identity threading; otherwise the applicable dynamic RW1-RW5/reset-key/epoch-advance evidence is supplied. The all-finite-prefix route additionally consumes Core.RTLPartialCycleClosure, but that property may be discharged once for a qualified RTL operational-semantics/translation class and then bound per instance by class/version/side-condition evidence rather than reproved exhaustively for each design; Core.RTLPartialCycleExt is derived. Uses of Core.QSync-Post/canonical post settling additionally supply QualifiedSynchronousCarrierModel_RTL_B/QS3-QS5. A KCut-starting recovery route further supplies full QS2a-R/RTLKCSettlingWindow and the pre-origin disposition. Stored/generated artifacts are bound to the authoritative current consumed-interface manifest. Prior synchronous-carrier/full qualification may project to the weaker current safety interface through the named projection theorem/checker identity without MIG1-MIG5 migration solely for that premise weakening; genuinely incompatible consumed-interface changes still require revalidation or InterfaceSchemaMigration.</p> |
| Per-liveness-instance evidence      | <p>The fixed capacities/elaboration/stimulus/environment/witness, exact execution-carrier occurrences, selected Policy-S artifact and K.PolicySProgress, K.SerialRoutePremises/SR3-G and universal K.SerialCandidateApplicability, exact selected DomResetScope, K.CVPred ObsKey/ResetSupport_CV/Avail_pi/CVP-M provenance, generalized-drain InFlightQual plus consumer-domain InFlightProfileCover and accepted search status, and prospective SFOutputEnvelope/response-class/PolicySCompareDomains evidence. G2 additionally binds TimeStutterQual, instance-wide MQ2 normalization opportunity, paired-occurrence MQ5 ProfileCover, ResetStutterMatch/ResetKey-Stutter where used, and fresh cross-model negative-evidence transport. Separately supply K.RespCov/K.CompleteS/A5-S, K-facing NormStable/DrainReach packages, universal QSync package coverage where claimed, and the selected A5-L route.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Per-B1-run safety-instance evidence | <p>When M.7c is selected: B1 at the outer <math>\Sigma</math>_DUT boundary; one reachable BRM1 cycle-closed DeclaredTerminal run <math>p\_post^0</math>; BRM2 outer-occurrence maximality (equivalently, under B1+BRM1, no reachable complete selected occurrence later than the selected final completed cycle); RSC0/ORS1 support conservativeness plus RSC1-RSC3 if dynamic reset can affect the outer quotient; QPC1-QPC3 if elaboration/multi-effect/erased proof paths can affect outer grouping/order; singleton J.TraceCertCov(<math>\{p\_post^0\}</math>); WC; and <math>\Phi</math> evidence on the selected <math>p\_ref^0</math> or a separately proved equal CanonHist. Much of B1/RSC/QPC can be reusable qualification plus lightweight per-instance binding.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### 3. Design choices that determine proof applicability

#### PURPOSE

**Connect common HLS design patterns to the proof layer and evidence route they affect.**

**Engineer takeaway:** The architect does not construct the formal certificates, but the design must expose communication, timing, reset, and state-transfer behavior clearly enough for the tool and verification flow to construct them. Appendix F of the scheduling-rules document is the architect-facing checklist; this section explains why those choices matter to the proof architecture.

This guide remains explanatory. Appendix F and the normative sections it references remain authoritative. The purpose here is to connect recognizable design patterns to Core, G, I, J, K, and L without requiring an architect to begin from certificate or theorem names.

#### 3.1 The architect/flow responsibility boundary

##### ARCHITECT / FLOW BOUNDARY

The architect chooses interfaces, reset topology, memory communication, buffering, and protocol structure that admit the required correspondence. The tool and verification flow construct and check the detailed replay, channel, attribution, footprint, reset, coverage, and liveness evidence. A design may synthesize and function correctly while remaining outside a selected theorem instance when proof-relevant behavior is implicit or modeled at the wrong boundary.

The useful question is therefore not whether the architect can write a certificate. It is whether the chosen architecture exposes enough explicit structure for a qualified flow to determine what happened, assign each request or value transfer to the correct dynamic operation and reset epoch, and compare the same abstract channel and observation boundaries in source and RTL. The G-K theorem stack is a single-global-clock formalism: a selected theorem instance must stay within one synchronous clock domain unless a separate multi-clock semantic layer is supplied; CDC structures are therefore outside the ordinary instance or must be represented by separately verified boundaries. Core.IC and Core.SyncFence illustrate this boundary directly: they define legal behavior of the prescribed source transition system, while the RTL/tool flow must prove the mapped request lifecycle, process-facing completion, and synchronization-fence behavior through the corresponding conformance evidence.

#### 3.2 Design choices affecting the safety and correspondence pipeline

These choices affect the Core  $\rightarrow$  G  $\rightarrow$  I  $\rightarrow$  J pipeline and the Post  $\rightarrow$  Ref safety result. They are not merely coding-style preferences: each determines whether the source and RTL provide a common observable contract and whether the local replay records can be composed into one reachable source witness.

Choosing the universal safety coverage route. (standard QSync versus alternate B1-run)

- **Design implication.** The standard QSync route is the general coverage architecture and works without observable determinism or a finite terminal maximum, but it requires RTL carrier/origin qualification and cycle-successor package induction. The alternate M.7c B1-run route is attractive for fixed-stimulus terminating instances whose selected outer behavior is B1-deterministic and admits one finite reachable Core.B1RunMax maximum. It is safety-only and applies only to predicates extensional on  $\text{CanonHist}_{\Sigma\_DUT}$  and closed under  $\text{J.IdealPrefixClosed}$ .
- **Why the proof cares.** M.7c replaces universal per-cycle correspondence-package induction with universal RTL outer-history coverage derived entirely on the RTL side. B1 fixes the selected cycle buckets; BRM1 supplies the cycle-closed terminal run; BRM2 rules out any reachable selected occurrence beyond its final observable horizon;  $\text{J.B1OuterHistCongruence}$  supplies down-closure of outer histories. The remaining J.1 correspondence obligation is only the selected run. This does not reduce the content of that one package and does not prove anything about a different stimulus.
- **Practical qualification.** Full internal RTL determinism is not required. Hidden state, epsilon paths, carrier provenance, and irrelevant micro-order may differ as long as the selected  $\Sigma\_DUT$  bucketed projection, reset quotient, and elaboration/group-order quotient satisfy the named conditions. With dynamic reset, RSC0/ORS1 proves that Core.OuterResetSupport is conservative; taking all reset scopes makes conservativeness cheap but enlarges RSC1-RSC3,

while a narrow support makes RSC cheaper but requires an explicit omitted-reset irrelevance proof. Elaborated/multi-effect boundaries similarly need reusable QPC1-QPC3. A service/nonterminating design, a property that is not ideal-prefix-closed, or a branch-dependent outer quotient stays on another coverage route.

**Qualified synchronous settling discipline.** (Core.RegSeq, Core.Settle, Core.QSync, Appendix Q)

- **Design implication.** Treat the selected theorem instance as ordinary synchronous hardware, but make history-sensitive provenance explicit. A qualified flow must distinguish repeated occurrences of an equal bare RTL/source state by Core.ExecutionCarrier and matched reset-epoch provenance, and must bind the actual CycleResultOn occurrence when a theorem consumes an environment/reset-dependent successor. If the compared executions permit no dynamic RESET after the prescribed initialization/reset-established base, bind that fact once and thread the prescribed initial epoch valuation by identity; otherwise bind the actual reset scopes/occurrences and Core.ResetEpochAdvance evidence. On RTL, establish RTLFrameEpochReachOn/RTLFrameEpochReach and QS2a-D before the safety-carrier theorems. If all-finite-prefix safety may stop after a complete unit inside an entered real cycle, establish Core.RTLPartialCycleClosure for finite completion of that same cycle, either directly for the instance or by binding the concrete RTL/tool semantics and version to an applicable independently checked operational-semantics/translation qualification; a per-instance checker then validates the class identity and side conditions rather than repeating exhaustive state exploration merely because the design instance changed. Core.RTLPartialCycleExt is then derived from Closure and adds no qualification premise. If canonical origin-indexed post settling is consumed, add the QS3-QS5 QualifiedSynchronousCarrierModel\_RTL\_B layer. A route that starts from an arbitrary reachable post KCut separately adds QS2a-R/RTLKCUTSettlingWindow through full QualifiedSynchronousModel\_RTL\_B; the pre-origin region must be explicitly resolved.
- **Why the proof cares.** The source L2 frame still freezes registered state and selected cycle inputs during same-cycle settling. The new exact-history interfaces prevent a proof from borrowing a legal EnvStep/reset successor from another occurrence of the same machine state. Core.QSync-Post remains a forward origin-indexed settling theorem under QualifiedSynchronousCarrierModel\_RTL\_B, not under the lighter safety-carrier predicate. Core.RTLPartialCycleClosure is not QSync or fairness and is used only for the later finite-prefix ideal-cover completion; Core.RTLPartialCycleExt merely selects the already-present result or the closure-supplied exact-history extension. On the source side, a safety construction checkpoint remains extendible; K-facing observation closes a fresh sibling through Core.L3CloseView and K.NormStable rather than extending the closed branch into the next cycle.
- **Supported alternatives.** Plain finite-prefix J.1 may still use a directly validated J.GWR package carrying checked per-Horizon settling and exact source-bridge evidence. Non-QSync designs require a separately defined qualified extension for broader K-facing/universal use. Tool reports are evidence only to the extent they are trusted or independently checked. A stored certificate produced against an incompatible older construction, bridge, projection, QSync, or K schema must be rechecked/rebound or accompanied by a checked K.InterfaceSchemaMigration; naming an old theorem is not sufficient.

**Source selection latitude and finite deferral.** (Core.SelectLatitude)

- **Design implication.** The selected Sys\_ref semantics must not impose hidden maximal progress at L1 issue or L3 boundary-action selection. Under both Policy L and Policy S, an otherwise legal action may remain unselected for a finite opportunity while all ordinary legality, reset, environment, arbitration, atomicity, and Core.16 restrictions continue to apply. A concrete executable reference model used as the theorem carrier must be shown to admit this latitude.
- **Why the proof cares.** Post -> Ref witness construction may need to postpone a reached-but-unissued request or an enabled boundary action in order to realize the selected RTL-derived observable sequence. This is finite-prefix legality, not fairness: Core.IssueFair and B2 are the separate complete-execution eventual-selection requirements. Nonselection does not imply that Core.16 disabled the action, and Core.SelectLatitude never licenses an action that Core.16 forbids.
- **Supported alternatives.** Use the normative abstract Sys\_ref or a translation-validated/symbolic source carrier that exposes the same finite non-maximal selection semantics. Do not use Core.SelectLatitude as a proof that a needed source cycle will exist or arrive at a particular state: J.SourceBridgeClosure/J.SourceBridgeExt carry that finite exact-history arrival obligation. J.RefProjCover needs latency coverage only when legal source delay changes canonical Sigma\_DUT history.

**Observation boundaries and signal timing.** (Appendix F: safety items 1-3)

- **Design implication.** For an HLS-pipelined loop in the standard theorem scope, the loop body contains no wait/ac\_wait statement, SyncChannel/ac\_sync call, or other explicit synchronization interface call; ordinary synchronization belongs outside the pipelined region or at explicit boundaries around it. Give every sequential signal read and write one

unambiguous synchronization anchor. Keep direct inputs stable for the applicable reset epoch or update them through the supported synchronization handshake. Environment and testbench decisions used by the proof should depend on prior observable causal history, not on an unmodeled absolute cycle count.

- **Why the proof cares.** Appendix G defines the selected signal observations, Appendix I replays the process state at those anchors, and Appendix J composes complete observable units. An ambiguous branch-dependent anchor or a hidden latency-triggered environment decision does not identify one stable observation for the construction to match.
- **Supported alternatives.** Move the signal operation to a unique boundary, use `hls_direct_input_sync`, expose the timing decision through an explicit protocol, isolate a cycle-accurate or latency-sensitive block, or verify the timing-dependent behavior directly at RTL rather than assuming it transfers from an arbitrary source run. If a pipelined loop requires an in-body synchronization call, rewrite the source to move that synchronization outside or around the pipelined region, preserve/treat the loop as non-pipelined, or use a separate proof/certificate outside the standard pipelined-loop theorem route.

#### Non-blocking calls and invisible polling behavior. (Appendix F: safety items 4-5)

- **Design implication.** A failed `PopNB` or `PushNB` may be hidden from the committed observable trace, but it is not harmless when its result or cadence determines arbitration, retries, timeout counters, payloads, endpoint choice, or the next visible action. An unbounded loop of failed polls is also not a stable wait in the ordinary replay fragment.
- **Why the proof cares.** Appendix I may carry a deferred non-blocking decision only past work proved independent through `NoDependentUse`. Appendix J later makes the shared snapshot decision. When the outcome selects later observable behavior, WC must identify a realizable source witness; Appendix L is the common implementation-alignment route.
- **Supported alternatives.** Prefer blocking communication when the architecture permits it. Otherwise prove NB-CF, use a valid Appendix L WC/L.Dir witness, directly construct the witness, or isolate and explicitly model the polling logic as a timing-sensitive protocol block.

#### Mapped memory, shared state, and the selected proof boundary. (Appendix F: safety item 6 and deadlock item 9)

- **Design implication.** Every mapped-memory instance in proof scope has a declared untransformed logical mapping semantics and selected logical boundary. Caching, merging, splitting, or reordering may occur below that boundary only when `Core.MemFaithful` shows that the transformed realization preserves the selected `Core.17/E1-E5` boundary behavior relative to the untransformed reference. An event already selected at a message or anchored-signal boundary cannot be hidden merely because it arose from memory mapping. `Core.MemSyncOrder` independently preserves synchronization fences; in particular, one merged storage commit may not represent accesses on opposite sides of the same synchronization boundary.
- **Why the proof cares.** `J.Mem` is triggered by every shared-memory dependency through which one process's write can affect another process's control flow, payload/value expressions, or the identity or order of its later operations. The audit is dependency-based rather than per physical RAM access, but it is complete: the dependency must be lowered through covered proof-visible operations or the separate memory-state extension, and no other qualifying unlowered memory-carried path may remain. If a transaction-changing transformation would alter the event stream used for that lowering, the standard reusable route is `Core.MemProfile`, whose logical boundary lies above the transformation and satisfies `Core.MemFaithful/J.Mem/Core.MemSyncOrder`.
- **Supported alternatives.** For J, use a qualified `Core.MemProfile` boundary, the separately supplied memory-state extension, or place the unsupported transformed shared-memory case outside Theorem J.1. For K, hidden memory realization must be WFG-inert or the qualified profile must expose a transformation-invariant K-faithful logical boundary that represents the relevant capacity, backpressure, and release dependencies. A persistently backpressuring realization with no such boundary cannot compose transaction-changing transformations with the Appendix-K proof; disable those transformations or exclude that memory from the K theorem instance. Multi-client profiles also inherit A6 explicit-arbiter/point-to-point ownership, RW reset disposition, and Appendix Q.1 when the exposed boundary is `Sys_B^elab`.

#### Channel realization and coupled structures. (Appendix F: safety items 7-8)

- **Design implication.** Model the channel behavior actually provided at the selected proof boundary. A positive-capacity boundary must satisfy the `Core.12` non-fall-through E4 semantics; internal bypass/fall-through may be hidden only when the chosen abstraction absorbs it without changing the selected boundary semantics. For coupled multi-input pipelines, bundles, joins, shared stages, couplers, or epoch gates, use the explicit `Sys_B^elab` structure rather than treating outer capacities as independent slack.

- Why the proof cares. E4 channel reasoning and J/K reconstruction are sound only when enablement, occupancy, transfer ordering, and release dependencies at the selected boundary match the implementation. Hidden bypass or coupled readiness can otherwise produce the wrong replay state, frontier, ReleaseOwnerSets\_E/WaitOwners\_E projection, or wait-for graph.
- Supported alternatives. Move the proof boundary to one that restores the declared E4 semantics; model a bypass as an explicit rendezvous path; or use a qualified Sys\_B^elab template with explicit finite B\_E(d)/occ\_d storage/rendezvous channels, the complete finite stateless ready/valid/data network, a checked coordinated Core.ElabProj package ( $\Pi_{\text{elab}}$ ,  $\pi_{\Sigma, E}$ ,  $\Pi_{\text{token}, E}$ , signed A\_E), and the required process release-support projection. Stateless coupler logic remains realization detail rather than a modeled process vertex. When Appendix K generalized drain can traverse proof-internal staged/join/synchronization/gate state, the same qualified template also declares the applicable Q.2 finite action/effect schema and its finite-state/rank evidence. The three Q.2 worked artifacts are architecture-validation examples, not automatic qualifications of a concrete template library.

#### Reset topology and cross-epoch transaction disposition. (*Appendix F: safety item 9*)

- **Design implication.** Reset both endpoints and the state of a live channel consistently, or define an explicit flush, abort, cancellation, or end-of-epoch protocol. Outstanding requests and in-flight data cannot be silently retained, discarded, or reattributed across a reset boundary.
- **Why the proof cares.** Appendix J represents reset as an atomic observable unit with a named scope, fresh dynamic identities, request clearing, reset-state correspondence, and framing outside the scope. Appendix K adds a second question: whether a reset can occur during the serial comparison without invalidating the pending operation used by the reduction.
- Supported alternatives. Use a channel-consistent reset scope, make cross-epoch disposition explicit, or supply a verified reset-state extension. For the Policy-S route, K.DomResetScope is now a predicate on an explicitly selected valid S\_dom, not a hidden single-valued function. The direct canonical K.DomResetScopeMin, leastness, valid-scope joins and only the named downward-transfer rules may be used. On a nested Continue candidate, reset and terminal exclusions apply to the prospectively fixed SFControl domain only; affecting RESET or DeclaredTerminal after the certified handoff are valid Response dispositions. A future proof may not reinterpret a post-SF reset/terminal result by retrospectively changing S\_dom or the comparison domains. For M.7c safety, dynamic reset that can affect a retained outer occurrence/group or contracted required-predecessor path additionally triggers Core.ResetScheduleCongruence over a checked conservative Core.OuterResetSupport. RSC0/ORS1 proves support conservativeness; RSC1-RSC3 then compare only the supported logical reset schedule/frontier/order. The no-dynamic-reset case is trivial for this interface.

### 3.3 Additional design choices affecting the deadlock pipeline

The following choices affect the additional Core → G → I → J → K liveness pipeline. They are not extra premises for every use of the Appendix J safety theorem. They matter when the claim is that the defined reachable closed internal message-passing wait-cycle cutpoint cannot occur.

#### Unambiguous wait ownership and persistent requests. (*Appendix F: deadlock items 1-3*)

- **Design implication.** Each source-level logical message channel should have one producer process owning its Push operations and one consumer process owning its Pop operations; shared buses and multi-client resources are represented by explicit arbiters and point-to-point process-facing channels. In Sys\_B^elab, an evaluated explicit boundary may instead meet a stateless coupler on one side. That coupler is not a process owner or WFG vertex: the qualified elaboration declares the modeled-process release supports for the settled blocking condition. Appendix K's deadlock observation boundary must include every WFG-relevant explicit abstract message-passing boundary; RTL micro-staging already absorbed into the selected B(c) remains internal. Multi-lane or burst behavior must be explicit. Once an ordinary blocking request is issued, Core.IC requires the same attributed dynamic instance (and stable Push payload) to remain pending until process-facing commit or an affecting RW2 reset; inserted latency is permitted under continued-request attribution. Independent nonterminal cancellation, withdrawal, retry, or reattribution is outside the present G-K theorem model.
- **Why the proof cares.** For each disabled frontier item, Core.WFG receives ReleaseOwnerSets\_E: a finite antichain of minimal nonempty modeled-process release supports. Processes inside one support are conjunctive requirements and different supports are alternatives. WaitOwners\_E is only their union for the conservative directed WFG; K.Dead keeps the unflattened family for exact release closure. Q.1(j) owns reconstructibility: the settled handshake and state-



instantiated release family must be reproducible from E and the reconstructed current state, with J.KObsConf separately validating the same equations/signals/schema against the RTL boundary mapping. Q.1(k) is the release-family correctness firewall relative to the declared template semantics: support soundness forbids spurious rescue alternatives, while support completeness forbids missed genuine rescue alternatives. Both directions matter because K.Dead consumes the exact family for release closure while Core.WFG consumes its WaitOwners union for connectivity. At an ordinary primitive point-to-point boundary the family reduces to  $\{\{Q\}\}$  for the unique complementary process owner. Hidden arbitration, an unqualified coupler dependency, or a speculative request that can disappear would invalidate the stable release/dependency and attribution facts used by the wait-cycle theorem.

- **Supported alternatives.** Insert an explicit arbiter process, expose lanes or beats in the abstract interface, or use a qualified Sys\_B^elab template whose Appendix-Q contract supplies the Q.1(j) reconstructible stateless handshake/release schema and the Q.1(k) sound/complete ReleaseOwnerSets\_E process supports. Do not create fictitious process vertices for local coupler logic or treat a shared/coupled resource as an ordinary scalar point-to-point dependency without that projection. For the primary Policy-S serial route, Q.1(j)/(k) qualification alone is not sufficient for every coupling-sensitive topology; apply the Q.1(l) requirement below when K.2b consumes such a boundary.
- **Serial-route coupling qualification.** When K.2b consumes a coupling-sensitive boundary, Q.1(l) additionally requires a finite handshake-support description and CouplerPersistence: once the earlier support facts and persistent request roots needed for the target transfer are established, every permitted pre-target action in the fixed frozen continuation must preserve those support facts or preserve ChannelEnabled at the next applicable settled opportunity. Q.1(d)-(k) well-formedness alone does not establish this for mutual-stall, shared-capacity, or arbitrated/non-monotone handshake topologies. If the selected template cannot prove CouplerPersistence, that candidate is outside K.PSF/CF-S and must use another A5-L discharge route.

#### Drain-only behavior after a blocking frontier is reached. (*Appendix F: deadlock item 4*)

- **Design implication.** Previously accepted work may finish only through the declared finite already-admitted action system. When generalized drain is used, the profile must enumerate the genuine Q.2 actions/effects, provide a decreasing InFlightDrainWF measure, and bind operational opportunity/path realization on the exact occurrence chain. A synchronization profile may include a ready SyncFire; lowered Sys\_K may include explicit guard/epoch administrative actions. None of these actions may launch source-later work behind the blocked frontier, and an action available only from another occurrence of an equal state is not an escape from the current occurrence.
- **Why the proof cares.** Core.16/Core.Drain/Core.SettleKBridge qualify the actually reached non-replenishing prefix. Generalized DrainClosedNow is a separate modal question about finite InFlightStepRec paths from the exact current K-facing occurrence. K.InFlightQual supplies relation-level soundness/precision and J.InFlightPathSound; each consumer additionally supplies K.InFlightProfileCover and, for every negative, an accepted K.InFlightSearchStatus on an OccurrenceBound or checked UniformFiber query. ExhaustiveExact requires proof-carrying K.InFlightRelationComplete; OmittedRootIrrelevant is query-specific; UnknownIncomplete cannot prove drain closure. Profile coverage belongs at K.1A/K.1, direct A5, relocation, modal correspondence, or the serial SR3-G consumer - not inside a route-neutral K.CertifiedCutpoint.
- **Supported alternatives.** Use automatic-flush pipelining or another proved no-new-later-work discipline together with a complete qualified Q.2 profile. Conservative over-approximation is allowed when it only adds genuine-effect-safe possibilities, but it can reduce the ability to prove a negative. Do not accept an incomplete relation as exact. If a custom template hides action identity, participant state, or future opportunity, expose the relevant proof state or prove a named preservation/irrelevance theorem; otherwise use a different liveness argument.

#### Scheduler robustness and the actual capacity configuration. (*Appendix F: deadlock items 5-6*)

- **Design implication.** Do not make deadlock freedom depend on favorable overlap, same-cycle issue, or a preferred arbitration choice when claiming scheduler-robust A5-L. Check the selected design against its actual back-annotated capacities and explicit elaborated boundaries, not an idealized unbounded network or a different rendezvous/buffered configuration.
- **Why the proof cares.** Policy-S serial dominance now has several explicit producer/consumer seams. K.CVPred uses exact-prefix Avail\_pi, typed historical ObsKey occurrence/version identity, finite ResetSupport\_CV and K.CVResetSupportSound; K.CompareFact-Mono persists history only through CVP-M rather than DomEpoch equality alone, and the predecessor stack recurses directly on WellFounded(prec\_CV). K.PolicySProgress is separate from SDet and K.CompleteS. SR3-G binds generalized-drain qualification/profile/query evidence over the exact serial-

counterexample occurrence domain. K.SerialCandidateApplicability closes the outer quantifier by requiring environment/release plus reset/terminal applicability for every non-waived counterexample candidate. Finally, all possible SF response classes are covered prospectively before the actual SF blocker is known.

- **Supported alternatives.** Rewrite or buffer the protocol, add explicit synchronization, change the selected capacity/elaboration and rebind the theorem instance, or use a scheduler-specific structural, invariant, exploration, or tool-certificate proof of A5-L.

#### Reset, environment, termination, and other excluded stalls. (*Appendix F: deadlock items 7-8*)

- **Design implication.** Treat reset/termination in two phases. Before K.2b.E/SF/Q, select one valid K.DomResetScope and resolve the early K.ResetEpoch then K.TerminalDisposition DirectFail-or-Continue gates. On nested Continue, bind K.PolicySCompareDomains=<SFControl,Handoff,Response> prospectively from the fixed SDet artifact, comparison/provenance schema, selected scope and conservative response-class set. R-S/T-S constrain SFControl through the canonical monitor trigger. After the actual SF blocker is obtained, do not change the package; K.2b.Q derives its Handoff membership from CD3-C and then treats affecting reset/terminal on Response as response dispositions rather than applicability failures.
- **Why the proof cares.** The anti-circularity boundary is load-bearing. K.SFOutputEnvelope names the pre-SF owner/frontier/provenance/key envelope but does not predict that an operation will become the SF blocker, trigger a monitor, or lie in Handoff. K.RespSerialClassExclusionSound is the only permitted basis for pruning the all-possible-output class set. CD3-C must already cover every relevant dynamic monitor instance on the fixed functional SDet artifact; CD4 is a theorem-side instantiation after SF. If CD4 fails, the prospective certificate/application is invalid - the flow may not enlarge Handoff, change the domains, or reroute that observed outcome into an early DirectFail. A concrete early DirectFail is itself K.RespFail and therefore cannot coexist with K.RespClean on the same successful clean run.
- **Supported alternatives.** Use a valid independently certified early Reset/Terminal DirectFail branch, a complete candidate-specific applicability proof, or the applicable uniform constructor. If the required prospectiveness, selected reset scope, class coverage, or Policy-S progress cannot be proved, use another A5-L route rather than weakening K.2b. Post-handoff Response classification is reset-first: affecting RW2 cancelled-pending, then DeclaredTerminal maximal-pending while the same monitor remains pending, then timeout at the finite bound.

### 3.4 How to use Appendix F with this guide

The parenthetical tags on the nine Appendix-F-tagged themes give the complete item-level mapping to the nine safety items and nine deadlock items in Appendix F, preserving Appendix F as the checklist. Use Appendix F first as the architect-facing design-pattern screen. For each applicable item, identify the proof layer and evidence family described above, select the intended theorem scope (safety only or safety plus internal-deadlock preservation), and ensure that the tool/verification flow records the corresponding discharge in Appendix M.7a. This guide explains the dependency; it does not replace the normative constraint or the per-instance evidence record.

## 4. Core vocabulary and proof boundaries

#### PURPOSE

**Define the small set of objects that repeatedly connect the Normative Formal Core and Appendices G, I, J, K, and L. Engineer takeaway:** keep four distinctions visible: observable events versus generic epsilon micro-steps; delta-settling versus silent tau-cycles versus B6 idle ticks; committed history versus current residual offers; and safety correspondence versus liveness assumptions.

| Term                               | Engineer-oriented meaning                                                                                                                                                                                                            |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sigma (<math>\Sigma</math>)</b> | The selected alphabet of observable events: committed Push/Pop transfers, synchronization events, and selected signal Read/Write events. The proof boundary determines which internal or DUT-boundary signals/channels are included. |



| Term                                        | Engineer-oriented meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Epsilon (<math>\epsilon</math>)</b>      | The generic proof-hidden micro-step relation. It includes source evaluation, failed non-blocking choices, and other internal behavior according to the model. It is not a synonym for a silent clock cycle. Delta-settling is a specific same-cycle settling classification; B5 may charge a silent tau-cycle to epsilon work without identifying the two.                                                                                                                                                                                                                              |
| <b><math>\delta</math>-settling (delta)</b> | Cycle-local same-cycle settling. On a phase-bearing source model, $\delta_{ref}$ is exactly the Core.Settle relation $\rightarrow_{\epsilon, L2}$ ; on RTL_B in the standard QSync scope, $\delta_{post}$ is the same-cycle settling relation of Core.QSync. Delta steps preserve cycle index and registered state. Other epsilon steps may also preserve the cycle but are not delta-settling.                                                                                                                                                                                         |
| <b>Silent <math>\tau</math>-cycle (tau)</b> | A real clock cycle in which no selected $\Sigma$ event commits while hidden registered/internal progress may occur. Tau is explanatory cycle-level terminology, not a transition label and not an epsilon step. B5 may prove that an internally advancing silent cycle consumes bounded epsilon work.                                                                                                                                                                                                                                                                                   |
| <b>B6 idle tick</b>                         | A real clock tick after the complete system has reached the B5/global fixed point. It performs no epsilon-prefix/suffix work and leaves non-clock state unchanged; it is treated as epsilon-stutter only by the observable trace-matching convention.                                                                                                                                                                                                                                                                                                                                   |
| <b>Observable unit</b>                      | A singleton $\Sigma$ event or an explicitly atomic same-cycle group such as a rendezvous Push/Pop, paired synchronization transaction, R2C fixed-point unit, signal-anchor group, or an explicitly selected proof-internal RESET unit. RESET may be carried by the Core.17 observable-unit machinery without being added to $\Sigma$ .                                                                                                                                                                                                                                                  |
| <b>Commit</b>                               | Core.8 defines a boundary commit as the selected proof model's legal E4 channel-commit transition, producing the committed Push/Pop unit and abstract channel-state update. For RTL_B, boundary mapping/B-faithfulness relates that abstract event to the clock cycle in which mapped valid and ready are simultaneously satisfied and the payload is observed. Commit(q) is the designated process-facing completion/accept/release occurrence for source operation q; CommitCycle(q) is its cycle index. Commit/release is distinct from L4/Core.RegSeq returned-result availability. |
| <b>Issue</b>                                | Core.9: Issue(q) is the first cycle in which the calling process begins requesting the transfer for q at its selected explicit boundary endpoint. It is the cycle index of that operational issue event; witness/scheduler records may carry the value but cannot choose a different timestamp. Core.10 separately identifies the operation-specific process-owned BoundaryReq. Issue is not generally reconstructible from committed labels alone and remains outside KObs history fields.                                                                                             |
| <b>ReturnedAtCycle</b>                      | The cycle boundary at which a message-call result becomes process-visible. For a successful blocking or non-blocking call it is the transfer-commit cycle; for a failed PushNB/PopNB it is the cycle in which the failure status returns even though no message commits. It is an operational timestamp, not a KObs field.                                                                                                                                                                                                                                                              |
| <b>Sys_ref</b>                              | The prescribed source-reference transition system: Sys_B in the ordinary back-annotated case, or Sys_B <sup>elab</sup> when explicit staged, bundle, join, guard, coupler, or epoch-gate boundaries are required. Raw rendezvous Sys remains Appendix G's conceptual presentation; the G-K proof stack uses Sys_ref, with B(c)=0 recovering rendezvous semantics.                                                                                                                                                                                                                       |

| Term                                       | Engineer-oriented meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Horizon</b>                             | The selected cycle/decision horizon used by J to group construction actions and enforce registered result availability and L1/L2/L3 phase ordering.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Finite ideal</b>                        | A finite down-closed set of observable units: if it contains a unit, it also contains every unit required to precede it. Appendix J constructs the source witness through a chain of these legal partial-order prefixes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>KObs</b>                                | The compact Core.18c cutpoint record $\langle E, w, H, O \rangle$ . It contains the elaboration/witness binding, canonical committed history, and current residual message offers. It deliberately does not contain historical Issue times, a Core.16 activation certificate, an in-flight-path witness, or future action-opportunity facts. Core.FlightStateRec and the J.KObs reconstruction lemmas derive the current proof-model state/action relation from KObs plus the fixed qualified schemas.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Residual offer</b>                      | A typed current outstanding source-level request at an explicit abstract boundary, including process, frontier item, dynamic call, direction, and reset epoch.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>ReleaseOwnerSets / WaitOwners / WFG</b> | For each disabled message frontier item $x$ of $P$ , the selected elaboration supplies $\text{ReleaseOwnerSets\_E}(x, \sigma)$ : a finite antichain of minimal nonempty modeled-process release supports. Members of one support are conjunctive; different supports are alternatives. $\text{WaitOwners\_E}$ is their union, and Core.WFG adds $P \rightarrow Q$ for each internal $Q$ in that union. $\text{K.Dead}$ does not flatten the family: it uses exact release closure over $\text{ReleaseOwnerSets\_E}$ plus induced-WFG connectivity. At an ordinary primitive point-to-point boundary the family is exactly $\{\{Q\}\}$ for the unique complementary process owner. Stateless couplers, FIFO stages, and other elaboration-local nodes are not WFG process vertices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>SettlePoint / BoundaryEvalPoint</b>     | $\text{SettlePoint}$ is the first fixed point reached by the complete source L2 epsilon-settling relation after L1 issue closes and requests freeze; Definition Core.Settle itself is conditional on that endpoint existing. $\text{BoundaryEvalPoint}$ is the settled explicit-boundary request/enablement interface. In the standard scope Core.QSync-Ref proves source $\text{SettlePoint}$ existence and uniqueness.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Core.QSync</b>                          | The qualified synchronous cycle/settling architecture has three RTL levels. $\text{QualifiedSafetyCarrierModel\_RTL\_B}$ is the safety/carrier foundation: QS1 single-clock scope + QS2 registered/frame separation and Sigma-silence + QS2a-D total/per-frame-unique designation over the independently defined Core.RTLFrameEpochReach domain. It deliberately does not require RTL QS3-QS5. QS2 is nevertheless conservatively retained in the safety carrier; possible QS2 minimization to a narrower registered-frame/origin discipline is an explicitly deferred audit item, not a current theorem weakening. $\text{QualifiedSynchronousCarrierModel\_RTL\_B}$ extends that foundation with QS3 relation-to-network realization/adequacy, deterministic local evaluation, QS4 finite acyclic settling, and QS5 selected-boundary completeness; Core.QSync-Post consumes this layer to obtain the unique same-cycle post $\text{KCut}$ and $\text{Core.KCutClosure\_post}$ . $\text{QualifiedSynchronousModel\_RTL\_B}$ extends the synchronous-carrier layer with QS2a-R/Core.RTLKCutSettlingWindow only for routes that begin from a reachable post $\text{KCut}$ and must recover its current carrier origin. The single named $\text{Core.QualifiedSynchronousCarrierModel-SafetyProjection}$ lemma carries stronger qualification to the safety interface, including the full-to-synchronous-to-safety implication stated in that lemma. Settling determinism is unique result for fixed entry/drives, not one global $\text{Sys\_ref}$ or RTL execution. |

| Term                                        | Engineer-oriented meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RTLFrameEpochReach /<br>RTLKCSettlingWindow | RTLFrameEpochReach is the pre-designation RTL cycle-state/reset-epoch reachability domain, defined directly from legal real-cycle reachability and RW1-RW5 without using RTLDesignatedOriginFrame, RTLCycleStep, or KOriginCarrier. QS2a-D then designates exactly one origin for every admitted pair. RTLKCSettlingWindow is the separate selected-witness SW1-SW3 placement predicate whose universal closure is QS2a-R; it directly identifies the current/greatest designated-origin prefix and excludes intervening non-settling steps, without depending on a later carrier theorem or making historical epoch a function of the bare KCut state.                                                                                                                                                                                                                                                                             |
| Core.KCut                                   | The K-facing comparison domain: a reachable state that is model-specific epsilon-quiescent and whose explicit BoundaryReq/ChannelEnabled/ChannelDisabled network is at its selected fixed point. Source SettlePoints are KCuts; RTL has its own KCut domain and no source L0-L4 phase tag.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Core.KCutClosure_post                       | The downstream RTL KCut-existence/non-vacuity interface. Under QualifiedSynchronousCarrierModel_RTL_B, Core.QSync-Post derives it from origin-indexed forward settling; QS2a-R is not needed for this closure result. The lighter QualifiedSafetyCarrierModel_RTL_B is sufficient for the cycle/carrier safety induction but does not by itself license canonical post settling. When a theorem starts from an arbitrary reachable post KCut, full QualifiedSynchronousModel_RTL_B supplies Core.RTLKCSettlingWindow and Core.QSync-KCutCanonical/Core.KOriginCarrier-KCut recover the corresponding carrier-relative origin/NormPost representation. K.CertCov remains a distinct package-totality interface, but the preferred standard route derives it from J.QSyncCertInd/J.CertExist-QSync/K.CertCov-QSync after that stronger KCut-placement qualification.                                                                  |
| Core.SyncFence                              | Constitutive Sys_ref synchronization-completion legality: a synchronization may commit only after every earlier blocking Push/Pop in pref_S(s) has process-facing committed, possibly in the same cycle. On RTL, the same rule is a conformance obligation; equal-cycle “before interval advance” accounting is certificate attribution, not observable physical sub-cycle ordering.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Core.MemSyncOrder                           | Local schedule/mapping-layer qualification for array or memory accesses source-ordered with synchronization calls: mapped storage commits before s occur no later than s; commits after s occur strictly later. It is recorded in M.7a but does not add memory events to $\Sigma$ or become an Appendix I/Theorem J.1 replay premise; J.Mem remains separate for cross-process shared-memory value flow.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Core.ReachPrep /<br>Core.IC                 | ReachPrep permits finite dependency-independent L1 preparation/reach of a source-later operation before an earlier blocking result returns, without erasing source order or implying Issue. Core.9 makes Issue the actual first request cycle. Core.10 makes BoundaryReq the fixed-mapping, operation-specific process-owned request rather than raw channel/coupler availability. Core.IC then gives the constitutive blocking lifecycle: every asserted blocking BoundaryReq has exactly one live owner and remains nonwithdrawably attributed until process-facing commit or affecting RW2 RESET; for a back-to-back next same-direction blocking q1, continued process-owned request ownership begins in t+1 with Issue(q1)=t+1, with no ownerless interval. DeclaredTerminal may end execution while a request remains pending. RTL lifecycle/attribution and owner decoding are separately checked M.7a conformance evidence. |
| Core.CycleState /                           | Core.CycleState is the model-indexed cycle-entry theorem state carrying the registered,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Term                                                                      | Engineer-oriented meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core.CycleResult /<br/>Core.CycleClosure</b>                           | environment, clock, reset-unit, and cycle-valued monitor state at the cycle origin. Core.CycleResult is the typed real-cycle outcome carrying the full next state, committed observable units, and scoped RESET units. Core.RTLCycleCarrier is a narrow post-side lemma: under QualifiedSafetyCarrierModel_RTL_B plus Core.RTLPartialCycleClosure, RC1 requires that the finite prefix has already entered real cycle t and RC2 requires that no distinct later designated origin lies at or before that prefix; the Closure entered-cycle clause supplies nonterminality, so there is no separate nonterminal theorem premise. The lemma then carries that same exact-history cycle through its registered update to the successor K-origin. Core.KOriginCarrier-Extend is different: once an explicit typed RTLCycleStep and nonterminal final carrier node are supplied, it appends the step with no ambient QSync/safety qualification predicate. Core.KOriginCarrier-AtOrigin/-Decomp consume the safety-carrier layer to recover finite ancestry. Core.CycleClosure separately qualifies legal nonterminal source cycles and the partial-cycle extension used by Core.CycleChoiceExt; when K.2b.E builds a fair continuation, Core.FairPick/FairCycleDovetail supply the constructive selector/dovetail bridge. |
| <b>Core.IssueFair /<br/>Core.DeclaredTerminal /<br/>Core.TerminalIdle</b> | IssueFair is weak fairness for a reached operation that remains IssueEnabled at every applicable L1 opportunity; merely intermittent legality does not trigger the antecedent, and it is separate from B2/B3 boundary progress. Core.DeclaredTerminal is the modeled machine/test termination predicate; a host watchdog or simulator stop is not terminal unless it is part of the selected theorem-instance semantics. Core.TerminalIdle is the stronger B6-eligible fixed/no-future-enabling condition for machine plus environment; it may be established by a stable declared terminal state or by a permanently idle nonterminating service model. An ordinary temporarily silent cycle is a Core.SilentCycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Core.SelectLatitude</b>                                                | Constitutive finite-prefix source selection rule: under Policy L or Policy S, otherwise legal L1 issue or L3 boundary actions need not be selected maximally at a particular opportunity. Nonselection is ordinary legal source-cycle behavior, does not imply Core.16 activation, and cannot override Core.16 or another legality rule. Core.IssueFair and B2 are separate complete-execution eventual-selection obligations; a unit-empty legal cycle may be a Core.SilentCycle with ordinary full-state next-cycle evolution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Core.MemFaithful /<br/>Core.MemProfile /<br/>J.Mem</b>                 | Core.MemFaithful constrains the existing $\Sigma$ -boundary selection freedom for mapped memory against the declared untransformed logical mapping semantics: hidden cache/merge/split/reorder behavior must preserve the selected outer events/order/values/atomicity and, for K, the represented blocking/WFG facts. J.Mem separately requires complete lowering of every qualifying cross-process memory dependency. Core.MemProfile packages the qualified reusable logical boundary above transaction-changing behavior, including MemFaithful, J.Mem, MemSyncOrder, K disposition, A6 arbitration/ownership, Q.1 where elaborated, and RW reset treatment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>FlightStateRec /<br/>InFlightStepRec</b>                               | KObs-pure reconstruction of the finite current proof-model state and one-step already-admitted action/effect relation. Actions are admitted because their work is represented now in requests, explicit storage/token state, or a ready synchronization carrier; the relation has no candidate-D or ownership filter. Operational future opportunity is not part of this relation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>WaitRoots /<br/>InFlightEscape /<br/>DrainClosedNow</b>                | WaitRoots_D is a process-indexed family of current frontier, enablement/disablement, and exact release-support data for P in D. InFlightEscape holds when a finite qualified InFlightStepRec path changes that family. DrainClosedNow is its negation. This replaces the former offer-only snapshot intuition; D selects which roots are observed, not which actions may occur.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Term                                                                             | Engineer-oriented meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SyncCarrier /<br/>SyncFire /<br/>EpochGateAdvance</b>                         | Core.7b-Core.7e give synchronization a separate typed carrier: endpoint/transaction identity, participants, pending/ready/blocked state, exact internal SyncReleaseOwnerSets, and the real source SyncFire action. A lowering-only EpochGateAdvance changes explicit Sys_K gate state and projects to source stutter. None of these definitions widens MsgOf, message BoundaryReq, or literal Dead_K/WFG.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>LowAdminNormalize<br/>/ LowWaitRoots /<br/>K.Low.ModalQual</b>                | Stage-2 lowering correspondence infrastructure. LowAdminNormalize finitely retires enabled lowering-only guard/epoch residue; LowWaitRoots is an extended source root view used only inside K.Low; ModalQual binds the local lowering cases, normalization/rank, root correspondence, bidirectional action projection/lifting with finite-suffix root reflection, separate Sys_K/Sys_ref K.InFlightQual, and certificate/schema versions.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| ExecutionCarrier /<br>CycleResultOn /<br>ResetEpochKey                           | Exact-history/occurrence provenance for history-sensitive cycle reasoning. Core.ExecutionCarrier identifies one legal finite history; CycleResultOn binds a real-cycle result to the exact occurrence on that history; ResetEpochKey/ResetEpochAdvance give canonical matched-RESET epoch provenance. Equal bare state or equal KObs does not authorize cross-history successor substitution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| RTLPartialCycleClosure / RTLPartialCycleExt                                      | Core.RTLPartialCycleClosure is the narrow finite-completion qualification used when an RTL prefix has already entered a real cycle and produced a complete selected unit. Its entered-cycle clause also establishes that the exact cycle-entry state is nonterminal. The closure accepts either the already-present occurrence-relative CycleResultOn on the same history or a finite exact-history extension that completes only that entered cycle. Core.RTLPartialCycleExt is a lemma derived from Closure that selects the appropriate one of those two witnesses; it adds no independent qualification or history-preservation premise. RTLCycleCarrier and J.CycleIdealCover/J.1-Safe-QSync consume Closure on the all-finite-prefix route. This interface is not QSync, fairness, generic RTL successor totality, or a complete/fair extension theorem. |
| Core.L3CloseView /<br>K.NormStable                                               | K-facing sibling branch of an extendible J.GWR source checkpoint. L3CloseView closes the selected L3 action set without creating a new scheduling action; J.L3CloseView-H preserves canonical history. K.NormStable NS0-NS6 then qualifies the observation-only KObsNormRef suffix. The closed branch is recreated for each K-facing use and is never the prefix extended by the next safety step.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Core.KObsSettleQualified /<br/>Core.KObsSettle-Ref<br/>/ Core.KObsNormRef</b> | K-facing-only post-L3 source observation settling. After Core.L3CloseView has irrevocably closed the selected L3 action set, Core.KObsSettleQualified_M(I) checks KSQ1-KSQ4: the declared boundary equations/topology remain valid under the actual post-L3 roots; only observation epsilon steps are phase-legal; the observation network is finite, deterministic, acyclic and complete; and its outputs cannot feed back into the already-closed L3 selection. Core.KObsSettle-Ref then supplies the finite fixed point used by Core.KObsNormRef and the source KCut. This is not a premise of Theorem J.1, J.QSyncTraceInd, CycleIdealCover, or the standard safety-only M.7b profile.                                                                                                                                                                     |
| B1 /<br>Core.B1ObsPrefix                                                         | B1 is observable RTL projection determinism at a selected alphabet, not unique internal execution. Its finite-prefix reading is cycle-bucketed: completed buckets agree, a prefix ending inside its current cycle may contain only a legal unsplit partial realization of that common bucket, atomic units remain indivisible, and a canonical selected occurrence may not migrate to another real cycle. Core.B1ObsPrefix is the directional helper relation used by the B1-run proof. Plain J.1, J.TraceCertCov, and the standard QSync safety chain do not require B1 merely for pairwise correspondence.                                                                                                                                                                                                                                                   |
| Core.B1RunMax /<br>BRM1 / BRM2                                                   | The selected-terminal outer-occurrence maximum used only by J.1-Safe-B1Run/M.7c. BRM1 requires a finite reachable cycle-closed prefix ending in modeled Core.DeclaredTerminal; cycle-closed means the endpoint is a full real-cycle boundary, with the zero-cycle initial-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Term                                                     | Engineer-oriented meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | terminal case permitted. BRM2 says the selected run contains every complete canonical selected $\Sigma\_DUT$ occurrence from every reachable RTL prefix. Under B1+BRM1, BRM2 is equivalently the absence of any reachable complete selected occurrence in a real cycle later than the selected run's final completed cycle (or no selected occurrence at all in the zero-cycle case). Later silent/erased cycles with no new retained occurrence are permitted.                                      |
| Core.OuterResetSupport /<br>Core.ResetScheduleCongruence | B1-run cross-run reset qualification. RSC0/ORS1 proves the chosen reset-scope/epoch-coordinate support is conservative for every reset that can affect retained outer identity/grouping or a contracted predecessor path. RSC1 compares supported completed-cycle reset signatures, RSC2 handles the partial-current-cycle reset frontier through already-realized outer groups, and RSC3 supplies reset-owned predecessor-domain closure. Full RTL-state equality is neither required nor intended. |
| Q.OuterProjectionCongruence                              | B1-run cross-run elaboration/projection qualification. For B1ObsPrefix-related RTL prefixes, QPC1 preserves outer-group restriction, QPC2 preserves old-old contracted required order, and QPC3 ensures predecessor-domain closure for any already-realized successor. Ordinary non-elaborated singleton projection is vacuous; qualified elaboration templates should prove the interface structurally and reuse it across instances.                                                               |

**DETERMINISM TAXONOMY - do not conflate these notions**

| Notion                                          | Logical status                                         | Engineer-oriented meaning                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Same-cycle settling determinism</b>          | Model-class restriction                                | R2C/Core.QSync: fixed registered state, fixed cycle-t inputs, and fixed explicit request/issue drives yield one settled boundary valuation.                                                                                                                                                                                                                                               |
| <b>Conditional synchronous-step determinism</b> | Functional cycle view after explicit choices are fixed | Once the currently legal witness, issue/scheduling, arbitration, and environment choices are fixed, the cycle can be viewed functionally. This does not remove Policy-L latitude or WC-sensitive source nondeterminism.                                                                                                                                                                   |
| <b>B1 observable RTL trace determinism</b>      | Theorem-instance assumption                            | Under fixed theorem-instance initial state/stimulus and selected alphabet, every reachable RTL behavior has the same cycle-bucketed selected observable projection. Internal state, epsilon paths, carrier provenance, and irrelevant same-cycle micro-order may still differ. A fully deterministic RTL transition semantics is one strong reusable discharge, not the definition of B1. |
| <b>I.DR / I.DR'</b>                             | Derived theorem                                        | Replay-checkpoint equality proved from the Appendix I premises. It is not an independent assumption to enter in the M.7a discharge record.                                                                                                                                                                                                                                                |
| <b>K.CV</b>                                     | Appendix K premise                                     | Authoritative Appendix-K design-level causal/value determinism and provenance premise. K.CV binds the common OpKey universe, the                                                                                                                                                                                                                                                          |



|                                           |                                  |                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           |                                  | K.CVPred/K.CV-WF typed fact-provenance certificate, serial-route comparison-availability, and observation determinism/isolation conditions under the fixed theorem instance. K.CVPred/K.CV-WF supply the low-level CV2 certificate structure and do not assume an ambient K.CV discharge. The broad Appendix-G causal graph and rendezvous mutual enablement are not automatically provenance predecessors. |
| <b>SDet</b>                               | Selected-instance fixing/premise | Fixes the Policy-S simulator, scheduler, arbitration, normalization, and related choices so the selected instance has no second incompatible committed history.                                                                                                                                                                                                                                             |
| <b>Normalization selection/congruence</b> | Selected proof data plus theorem | Records which replay normalization is selected and why the replay quotient is preserved. Core.QSync canonicalizes only K-facing same-cycle settling; Appendix I/J replay normalization remains separate.                                                                                                                                                                                                    |

For theorem premises, Appendix M.7a supplies the evidence axis: inapplicable to the selected theorem scope, satisfied by a syntactic/design restriction, discharged by tool/RTL evidence, or discharged by a source/RTL witness construction. A required condition may not be marked inapplicable; derived theorems such as I.DR do not become independent premises merely because their proof artifacts are recorded.

#### CUTPOINT BOUNDARY: KObs is not full-state equivalence

KObs is the typed Core.18 cutpoint record  $\langle E, w, H, O \rangle$ . Core defines the source record, its pure reconstruction functions, and the cross-model equality notion; J.OfferReach/OfferConf/KObsConf establish one realized common record for the selected source/RTL package. J.KObs-Proc/Chan/Offers/WFG reconstruct process replay, explicit channel state, message frontier/request roots, settled handshake, ReleaseOwnerSets\_E/WaitOwners\_E, and Core.WFG. Stage 2 adds J.KObs-SyncCarrier/J.KObs-SyncInFlight and J.KObs-InFlight, which reconstruct the current synchronization carrier and finite Q.2 in-flight action/effect relation without adding fields to KObs. Core.FlightStateRec explicitly excludes activation certificates and future opportunity claims. J.InFlightPathSound is a separate model-indexed operational realization theorem: it consumes Q.2 effect soundness plus Core.InFlightCycleExt/CycleClosure-equivalent evidence. Equal KObs is therefore typed current-observation equality, not full-state equivalence, historical Issue equality, or a proof that every reconstructed future edge is realizable.

## 4.1 Safety direction: why Post -> Ref is the normal signoff argument

For ordinary safety signoff, the useful deductive direction is: every covered reachable RTL prefix has a matching reachable Sys\_ref prefix with the same canonical observable history and Core.17  $\Sigma$ Match events/values. Therefore, if all relevant Sys\_ref histories satisfy a J.HCanon-invariant safety property meeting Definition J.IdealPrefixClosed, the covered RTL histories do too. Here “prefix-closed” means downward closure over finite required-order ideals, not merely deleting a suffix from one chosen linearization of independent events. When one complete source run is used as the verification representative, J.RefProjCover must cover the relevant source observations by finite canonical ideals/down-closed observable prefixes of that run. In the standard QSync route, J.CycleReach\_post/J.QSyncTraceInd/J.TraceCertCov-QSyncCycle first establish the cycle-aligned Post -> Ref packages; Core.RTLCycleCarrier and J.CycleIdealCover then cover every reachable finite outer observable history by a reachable cycle-aligned history, and J.1-Safe-QSync uses J.IdealPrefixClosed to obtain the all-finite-prefix safety conclusion without creating a separate J.GWR/J.TraceCertCov package for each intra-cycle ideal. Assertion-style no-bad-event invariants, per-channel FIFO/value legality, and explicitly downward-closed protocol invariants are natural uses. A cross-interface scoreboard whose check depends on an otherwise droppable explanatory event may fail J.IdealPrefixClosed and should instead use direct J.TraceCertCov on the required arbitrary-prefix domain (or refine the retained observation/order).

The reverse direction is intentionally a selected-pair compatibility result, not a constructive-existence theorem. J.RTLConsistentRefPrefix is typed over a reachable source prefix already equipped with a selected reachable RTL prefix and compatible annotations. Their complete selected observable-unit projections must match under Core.17 - neither side may contain an extra unmatched selected unit - and any cutpoint use must carry the matching J.OfferReach/J.OfferConf/J.KObsConf package. Because the RTL witness is already supplied, the Ref -> Post clause does not depend on a generic fair-extension theorem. Appendix L can select such a source witness when latency-sensitive choices require RTL-aligned epsilon outcomes.

## 5. Normative Formal Core and Appendix G - shared semantics and observable contract

### PURPOSE

**Define the shared source-reference semantics and the observable scheduling contract that the post-HLS implementation must preserve.**

**Engineer takeaway:** The Core is the normative semantic skeleton for G-K; Appendix G constrains that skeleton through R1-R5 and E1-E5. Neither layer by itself constructs the matching source execution - Appendices I and J do that work.

### 5.1 Source-side rules R1-R5

| Rule                              | Role in the architecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>R1 - synchronization order</b> | Synchronization calls of each process remain in dynamic source order and commit in strictly increasing cycles.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>R2 - signal anchoring</b>      | For a sequential process, a signal Read is logically anchored to the closest preceding synchronization and a signal Write to the closest succeeding synchronization. This anchor also determines the signal event's formal ordering for frontier reasoning: lexical placement around intervening message operations does not by itself impose message order. If several source Writes by the same process to the same signal share one E2 anchor, the last executed source Write supplies the canonical visible value and earlier same-anchor Writes are epsilon-internal. |

| Rule                                   | Role in the architecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>R2C - combinational fixed point</b> | <p>Combinational observations are defined from the cycle-entry valuation and the unique settled result of the qualified component/network. Internal epsilon/stutter re-evaluation is permitted so long as it converges to that unique result. R2C fixed-point units are matched by component identity and local emitted-occurrence ordinal rather than by a global cycle bucket; Sys_ref and RTL_B need not share an internal combinational model or absolute cycle numbering.</p> <p>ComponentParticipatesInCycle defines each side's observation domain. A cycle-locked component emits exactly one complete fixed-point unit in every participating cycle and none outside it; per-side participation/completeness and the implementation/certificate justification are part of the R2C conformance evidence.</p>                                             |
| <b>R3 - sync isolation</b>             | <p>A synchronization boundary separates the message operations in the immediately preceding and succeeding source intervals, and its Core.SyncFence completion clause requires every earlier blocking Push/Pop in pref_S(s) to process-facing commit no later than the synchronization commit. Same-cycle completion is permitted.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>R4 - safe issue order</b>           | <p>Dynamic message operations preserve prefix-closed source issue order. Core.9 Issue is the actual first process-owned request cycle, not a witness-selected annotation. For distinct blocking operations on the same scalar explicit boundary/direction, Core.IC(4) gives strict issue order; the ordinary same-interface special case is discharged from the fixed Core.10 process-owned BoundaryReq mapping, Core.IC live-owner/nonwithdrawal/back-to-back attribution, endpoint serialization, and the dynamic source sequence. If q0 commits at t and a continuing process-owned request at t+1 belongs to the next same-direction blocking q1, q1 owns it beginning at t+1 and Issue(q1)=t+1; the request may not be ownerless in between. Distinct-interface no-reverse and cross-interface prefix closure remain substantive implementation checks.</p> |
| <b>R5 - channel capacity semantics</b> | <p>The source reference uses rendezvous or finite-capacity channel semantics at the selected explicit abstract boundaries; elaboration may introduce proof-internal channels but does not reorder ordinary source calls.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

#### DEFINITION: Core.RegSeq - registered sequential-interface semantics

For a sequential process, request and Push-data outputs in cycle t depend only on registered state and operands fixed before L2 settle. If ReturnedAtCycle(q,t) holds - for a success or failed non-blocking return - any dependent evaluation, control, endpoint/address, payload, or issue has a later Horizon. ReturnedAtCycle is defined normatively in Appendix C; Core.RegSeq states the dependent-use rule.

- **Why it matters.** It eliminates same-Horizon decision-to-dependent-request edges that would be inconsistent with clocked hardware.
- **Where it is used.** Appendix I local preparation, J.FoundationRank/J.DepRank, J.RegPhaseNF, and the Appendix K serial-reduction assumptions.
- **DV implication.** Checked source/tool evidence must establish the registered-cycle rule. Zero-time dependent NB cascades must be identified and then rejected, excluded, isolated, remodeled atomically, or aligned through Appendix L; silent scope exclusion is forbidden.

#### CLARIFICATION: Core.OrdFrontSingleton - ordinary-source frontier uniqueness

Within an ordinary non-elaborated sequential source interval, dynamic message calls remain ordered by dynamic program order, including distinct-interface calls, repeated loop iterations, and ordered unrolled instances. Therefore Front\_P has at most one member. A genuine message-operation multi-frontier requires explicitly modeled incomparable Sys\_B^elab frontier items with the corresponding order, boundary, projection, and attribution evidence.

## 5.2 Normative Core operational semantics and the cycle-level hardware model

The Normative Formal Core treats Sys\_ref as an operational transition system with explicit process state, channel state, theorem-instance dynamic operations, requests, environment/reset state and epsilon micro-state. The current architecture adds exact-history provenance wherever equal bare state is not enough: Core.ExecutionCarrier records the legal history occurrence; Core.ResetEpochKey/Core.ResetEpochAdvance make reset epochs occurrence/scope based; occurrence-relative CycleResultOn/CycleClosure/CycleChoiceExt prevent history-sensitive successor splicing. On RTL, RTLFrameEpochReachOn/RTLFrameEpochReach is defined below designation; QS2a-D then supplies designated origins. Core.RTLPartialCycleClosure independently supplies the finite already-entered-cycle completion needed by all-finite-prefix safety, while Core.RTLPartialCycleExt is derived from that qualification. The RTL safety carrier itself is only QS1 + QS2 + QS2a-D; same-cycle canonical post settling belongs to the stronger QS3-QS5 synchronous-carrier layer, and selected-witness queried-KCut recovery adds QS2a-R on top. These interfaces govern finite cycle/settling structure, not generic time progress. A separate current-state layer reconstructs K-facing in-flight state/action relations from KObjs; operational future realizability remains a later qualification.

| Policy                                | Meaning and use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core.PolicyL /<br/>Core.LLegal</b> | Core.PolicyL is the permissive issue policy used by the safety witness and A5-L. Core.LLegal fixes exact step legality. Core.ReachPrep may allow a later dependency-independent operation to become reached before an earlier same-interval blocking call commits; reach is not Issue, source order remains, and Post -> Ref coverage must ensure Sys_ref is permissive enough for every preparation segment used by the accepted J construction. Core.SelectLatitude then permits an otherwise legal reached issue to remain unselected at a finite L1 opportunity; QSync does not remove either preparation or selection latitude.                                                                                        |
| <b>Core.Phase /<br/>Core.LAdm</b>     | Core.Phase defines L0-L4 and the pre-settle/settled-frontier boundary. Core.SelectLatitude applies to the selected L1 issue segment and L3 boundary-action set; they are finite and need not be maximal, and the L1 issue selection may be empty when no operational rule requires an action. Core.LAdm qualifies finite Policy-L prefixes and complete executions by adding the drain/settle-to-KCut discipline; SelectLatitude-permitted nonselection and any resulting legal CompleteCycle/SilentCycle remain Policy-L-legal. When K.2b.E constructs a complete fair continuation it additionally consumes Core.CycleClosure/Core.CycleChoiceExt and Core.IssueFair/B2/B3; a B6 suffix still requires Core.TerminalIdle. |
| <b>Core.PolicyS</b>                   | The conservative serial-blocking issue policy: a later same-interval blocking operation may issue only after every earlier blocking operation has process-facing committed in a strictly earlier cycle. Policy S inherits Core.SelectLatitude: the earlier legal operation need not issue at the first opportunity merely because it is legal. SDet later fixes the concrete selected Policy-S simulator/run used by the primary Appendix K route; Core.IssueFair/B2 govern eventual selection only where a complete fair execution is required.                                                                                                                                                                            |
| <b>Core.SelectLatitude</b>            | One no-maximal-progress rule shared by L1 issue selection and L3 boundary-action selection under both issue policies. It is finite-prefix operational legality, not fairness. It never licenses a Core.16-forbidden action, and absence of issue/commit does not by itself prove disablement or Core.16 activation.                                                                                                                                                                                                                                                                                                                                                                                                         |

**DEFINITIONS: Core.Phase, Core.Settle, Core.KCut, Core.16, Core.Drain, and Core.LAdm**

Core.Phase gives each source cycle L1 preparation/issue, L2 settling, L3 boundary actions and L4 registered result availability. Core.SelectLatitude permits finite non-maximal L1/L3 selection but is not a bridge-existence theorem. Core.Settle names the first L2 fixed point when reached; Core.QSync-Ref proves finite uniqueness in the standard source scope and Core.SettleIsKCut relates it to K-facing evaluation. Core.L3CloseView is a different phase operation: after a J.GWR construction checkpoint has exported all selected L3 work, it forms a fresh closed K-facing sibling with no reopened selection; J.L3CloseView-H and K.NormStable qualify the observation suffix. Core.16/Drain govern no-replenishment. For complete continuations, occurrence-relative Core.CycleResultOn/CycleClosure/CycleChoiceExt and FairPick/FairCycleDovetail thread the exact source history; an equal state reached by another history is not a replacement. On RTL, Core.RTLPartialCycleClosure is the separate finite already-entered-cycle qualification used by RTLCycleCarrier/CycleIdealCover; RTLPartialCycleExt is derived from it rather than bound as separate evidence.

**ARCHITECT MENTAL MODEL - one ordinary synchronous cycle**

| Cycle-level view                                                                                                                    | Formal correspondence                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Registered state and selected cycle-t exogenous inputs are fixed.</b>                                                         | QS1/QS2; ClockScope audit; L2/QSync frame.                                                                                                                                                                                                                                                                                                                                                   |
| <b>2. Sequential process request/payload outputs are determined from registered state and operands already fixed for the cycle.</b> | Core.RegSeq. Ordinary combinational logic from registers is allowed; a newly returned cycle-t result cannot feed a dependent same-cycle sequential output.                                                                                                                                                                                                                                   |
| <b>3. The complete system combinational/ready-valid/guard/join/coupler network settles to one deterministic valuation.</b>          | Delta-settling; R2C; QS3/QS4; Appendix Q for Sys_B^elab. Pure combinational loops are outside the standard scope; feedback crosses explicit state.                                                                                                                                                                                                                                           |
| <b>4. The settled boundary snapshot is read.</b>                                                                                    | BoundaryEvalPoint; source SettlePoint; Core.KCut when the K-facing proof consumes the snapshot. These are proof views of the ordinary settled cycle state, not additional hardware states.                                                                                                                                                                                                   |
| <b>5. Boundary decisions and commits occur.</b>                                                                                     | Core.Phase L3; E4; R3/E5; Core.SyncFence.                                                                                                                                                                                                                                                                                                                                                    |
| <b>6. The next registered state/result availability is established.</b>                                                             | Core.Phase L4 / next cycle. RTL_B has its own registered update rather than source L0-L4 tags.                                                                                                                                                                                                                                                                                               |
| <b>Selection latitude (not a hardware stage)</b>                                                                                    | Core.SelectLatitude. At the source proof level, the legal L1 issue segment and L3 boundary-action set may be non-maximal or empty when no other rule requires an action. A no-selected-Σ-commit result can be a real Core.SilentCycle whose registered/environment/monitor state still evolves. Core.IssueFair/B2 are the later fairness counterparts, not part of finite-prefix J legality. |

The functional cycle picture is conditional on explicit choices, not a claim that Sys\_ref has only one execution. WC-sensitive outcomes, Policy-L issue/scheduling choices, arbitration, and reactive-environment choices remain part of the admissible execution space. Likewise R2C/Core.QSync determinism means a unique settled result for a fixed cycle-entry valuation and fixed drives/choices; internal epsilon/stutter re-evaluation is allowed, and Sys\_ref and RTL\_B need not share the same internal combinational model or absolute cycle buckets.

### 5.3 Post-HLS compatibility obligations E1-E5

| Obligation | What the RTL/tool flow must establish                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E1         | Preserve each process's synchronization-event order.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| E2         | Preserve anchored signal values/visibility for sequential processes and fixed-point signal units for combinational processes. For sequential processes, canonicalization records at most one Read and at most one Write per (process, signal) at each E2 anchor; when several source Writes share an anchor, the last executed source Write supplies the matched value and earlier same-anchor Writes are epsilon-internal.                                                                                                                                                                        |
| E3         | Preserve safe message Issue order and attribution. Core.9 fixes Issue as the operational first-request cycle and Core.10 fixes the process-owned BoundaryReq mapping. For the ordinary scalar same-interface blocking case, strict order is derived from qualified Core.IC persistence/back-to-back ownership, scalar endpoint serialization, and dynamic source order rather than chosen Issue annotations. Distinct-interface no-reverse ordering and the prefix-closed issued-set condition remain substantive E3 checks; Core.17/ΣMatch separately checks committed occurrence/value matching. |
| E4         | Realize legal rendezvous or finite-capacity FIFO behavior at each selected explicit abstract channel boundary, including payload/order/occupancy behavior.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| E5         | Do not move a message operation across its immediate synchronization boundary: predecessor operations issue no later than the sync; successor operations issue strictly later; and every earlier blocking Push/Pop in pref_S(s) must process-facing commit no later than the sync. Same-cycle message/sync completion is allowed. On the post side, the "accounted before interval advance" convention is certificate/trace attribution, not a physical sub-cycle RTL ordering requirement.                                                                                                        |

**BOUNDARY CONDITIONS: B-faithfulness/Core.ElabProj and mapped-memory  
Core.MemFaithful/Core.MemProfile**



E4 only has meaning at a chosen abstract storage/rendezvous boundary. At an ordinary primitive Sys\_B boundary, B-faithfulness checks selected transfers and  $B(c)/occ_c$ , and primitive E4 legality also determines the complementary handshake. In  $Sys\_B^{elab}$ , operational storage/rendezvous channels use the explicit  $B\_E(d)/occ_d$  state declared by E; Q.1(a)-(c) separately uses outer Sys\_B  $B(c)/occ_c/E4$  history as the refinement target. Core.ElabProj coordinates  $\Pi\_elab$ , outer trace map  $\pi_{\Sigma,E}$ , finite token projection  $\Pi\_token,E$ , signed  $A\_E$  state accounting, the settled handshake network, and the process-dependency projection. At  $\epsilon$ -quiescent cycle-boundary states  $A\_E$  must agree with independently reconstructed outer history; its legal nonnegative range is then derived from outer E4 rather than certified as a separate obligation. Internal staged/bundle/join transfers with outer effects have a unique declared token attribution, while transfers with no outer effect may have none; one token may account for several outer Push/Pop effects only without invention, omission, duplication, or double counting. ReleaseOwnerSets\_E/WaitOwners\_E contracts stateless couplers to modeled processes. A stateless coupler may sit at a process-facing boundary but is not a process, residual offer, or stored channel state. Evidence reuse is explicit in M.7a: once the underlying E4 boundary legality or Core.IC request-lifecycle/attribution obligation has been authoritatively discharged for the selected boundary and instance, downstream B-faithfulness, J.CCert, correspondence, and certificate consumers may cite that same evidence instead of re-proving the underlying legality; each consumer still discharges its distinct mapping, snapshot, abstraction, or correspondence obligations.

Mapped memory uses the same selected-boundary discipline but a separate qualification. Core.MemFaithful names the untransformed logical mapping-layer semantics as the reference and permits transformed physical memory traffic to remain below the selected boundary only when cache/merge/split/reorder behavior preserves the selected Core.17/E1-E5 events/order/values/atomicity. A selected message or anchored-signal event cannot be selectively hidden. J.Mem still requires complete lowering of qualifying cross-process memory dependencies. If transaction-changing behavior or persistent backpressure needs a reusable boundary above the physical stream, Core.MemProfile supplies the qualified logical interface; its K-facing boundary must faithfully represent capacity/backpressure/release dependencies or the transformation is outside the covered K instance.

## 5.4 Trace compatibility is observational, not temporal equivalence

The relation approximately-equal ( $\approx$ ) is Core.17  $\Sigma Match$  plus E1-E5.  $\Sigma Match$  explicitly matches the complete selected occurrence sequences and value labels - including per-channel Push/Pop ordinals, canonicalized sequential signal observations, R2C fixed-point units matched by component/local emitted-occurrence ordinal, synchronization occurrences, and any explicitly selected proof-internal RESET units carried by the observable-unit machinery (without adding RESET to  $\Sigma$ ). Independent matched observations may occur in different cycles or be grouped differently where the ordering and atomicity rules permit. Core.SelectLatitude is the source-side operational reason a matching finite witness may defer otherwise legal issue/boundary actions without making that delay itself observable. Appendix J may insert only the finite invariant-preserving unit-empty cycle bridge required by selected cycle-sensitive evidence; J.HCanon does not equate the bridge length to the raw RTL Horizon gap. Appendix G expands this matching convention; it does not define a second competing trace relation.

Appendix G's explanatory causal-dependency graph is not a universal well-founded proof order. It may contain legal same-cycle atomic cycles. Appendix J instead generates its own construction relation: D1-D4 are semantic prerequisite edges covered by J.DepSound, while D5 is Horizon proof stratification and is justified separately by J.HorizonOrderSafe commutation evidence. Appendix K uses yet another relation under the authoritative K.CV premise: Definition K.CVPred/K.CV-WF supplies the well-founded fact-provenance certificate used by K.CV and must not inherit cycles merely because two operations are causally or temporally adjacent. K.CVPred/K.CV-WF define that certificate structure without assuming K.CV as a prior premise.

**READING RULE: What to take from Appendix G**

For architects: the Core fixes the operational model and G defines what implementation freedom remains invisible to a latency-insensitive environment. For DV: together they identify the observable checker boundary and conformance obligations. For formal implementers: the Core supplies the shared transition-system/KObs/WFG/drain vocabulary, while G supplies the R/E contract consumed by I and J.

## 6. Appendix I - per-process replay and preparation

### PURPOSE

Show that each process can locally replay the selected RTL observable prefix and prepare for its next participation using a finite source-owned script.

**Engineer takeaway:** Appendix I is intentionally local. It identifies the right dynamic operation and reaches its request/decision point, but it does not assert that the complementary endpoint is present or that all process scripts can be merged globally.

### 6.1 Inputs and preconditions

| Input                            | Why I needs it                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fixed initial state and stimulus | The process is replayed under the same global Sigma-causal environment behavior, not an arbitrary P-local stimulus.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| WC witness                       | Every epsilon-modeled choice that can affect later observable control, values, identities, requests, or frontiers is fixed consistently.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| I.A0 / E3/E5                     | The RTL message-issue discipline is supplied as an implementation assumption, including prefix-closed issue attribution and the E5/Core.SyncFence rule that no earlier blocking request survives a synchronization commit into the next interval.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Core.RegSeq                      | Dependent next-cycle requests cannot be created from same-cycle returned results.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| B4 / finite pipeline depth       | Internal micro-progress has a finite epsilon bound when the process is not externally stalled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| B2/B3 when progress is invoked   | These are theorem-applicability fairness/progress assumptions, not consequences of source code and not generically “environment assumptions.” B2 requires eventual selection only for a boundary action that remains continuously ChannelEnabled; B3 supplies complementary channel discharge under its persistent-request antecedent. Their discharge owner may be source/scheduler semantics, HLS-generated or hand-written RTL arbitration/back-pressure, channel-library behavior, an external master/environment, or a qualified reusable class bound to the concrete theorem instance. Intermittent enablement does not satisfy the weak-fairness antecedent by design. |
| Source well-formedness           | Signal anchors, dynamic operation identities, direct-input contracts, shared-memory lowering, and reset behavior are well formed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

### 6.2 The replay checkpoint: I.ReplayObs, I.CutpointOfferObs, I.ReplayObs-Congruence, I.DR, and I.DR'

I.ReplayObs is the logical P-local replay checkpoint determined by the committed observable-unit history, fixed stimulus, and witness. It contains the logical replay control position, replay-relevant values, witness-selected outcomes, replay-fixed dynamic attribution, completed-item status, candidate NextFront\_P, anchored values, returned results, reset epoch, and normal-completion status. It is not the complete raw operational state: certified outcome-independent L1 preparation may advance the operational cursor or compute temporaries without changing the logical checkpoint.

I.CutpointOfferObs is the separate operational slice for current uncommitted blocking offers: Pending\_P, Front\_P, BoundaryReq, owning dynamic calls, boundary/direction/epoch facts, and stable Push payload obligations. The K-facing cutpoint therefore combines replay history H with residual offers O. Deterministic replay does not manufacture O from H; Appendix J constructs and audits O through J.OfferReach and J.OfferConf.

**LEMMA: I.ReplayObs-Congruence**

From equal logical replay checkpoints, the same next complete observable unit produces equal canonical immediate post-unit checkpoints. An exact selected finite normalization may then mix replay-inert preparation segments, which leave each checkpoint unchanged, with witness-fixed epsilon transitions that may update replay fields. The normalization convention must pair those replay-changing transitions so that equal checkpoints remain equal; the theorem does not assume per-path neutrality, global confluence, or uniqueness of all legal raw endpoints.

**LEMMA: I.DR and I.DR' - deterministic replay**

I.DR' establishes equality of the logical ReplayObs checkpoint at the canonical immediate post-unit state, before optional following L1 preparation or epsilon-normalization. I.DR establishes equality at the selected epsilon-quiescent cutpoints by applying paired congruence to the exact proof-relevant normalization paths. Appendix J uses the immediate form when constructing the next Horizon and records normalized witnesses explicitly when a quiescent cutpoint is required.

## 6.3 Deferred non-blocking decisions

A process-local proof cannot decide a non-blocking poll from process state alone because success or failure depends on the common incident-channel snapshot and possible complementary requests. Stopping the local replay at every poll would be too restrictive: a registered process may execute independent same-Horizon work that does not depend on the poll result.

**DEFINITION: I.DeferredNBHole**

A deferred hole records the dynamic non-blocking call, explicit boundary, snapshot key, Horizon, result destinations, reset epoch, witness-selected expected outcome, and NoDependentUse evidence. The process-local preparation script can carry the unresolved hole through independent work while forbidding any dependent use of its result.

## 6.4 The local preparation theorem

**LEMMA: I.3 - finite process-local preparation script**

Given the replay state, exact dynamic operation identity, Core.RegSeq, WC, E3/I.A0, source well-formedness, and typed footprints, I.3 constructs a finite process-owned epsilon/request path to the selected operation, its persistent issued request, or a carried deferred non-blocking decision hole. The script does not commit an unmatched observable unit and does not assume global enablement.

- **Output.** The information later packaged in J.PCert: the logical ReplayObs checkpoint, exact dynamic identity, finite per-Horizon preparation actions, separate operational issue/pending attribution, stable payload/value facts, field-sensitive footprints, and deferred-hole records.
- **Non-result.** No complementary endpoint, common channel snapshot, globally legal merge, or reachable system extension is proved here.
- **Why finite.** The source-control segment is finite and internal progress is bounded by B4/FPD when not externally stalled.

## 6.5 Engineer example: independent work after a poll

Suppose a process performs a PopNB, updates an independent diagnostic counter, prepares a request on another endpoint whose control and payload do not depend on the PopNB result, and only later branches on the poll status. Appendix I may carry the poll as a DeferredNBHole through the independent counter/request work. Appendix J later freezes the complete channel request snapshot at L2, decides the poll once globally, and prevents the branch from occurring until the result is available under Core.RegSeq.

### ARCHITECTURAL SEAM: Why Appendix I cannot be the whole safety proof

Local scripts may each be valid when viewed against read-only summaries, yet still conflict through shared channel state, atomic rendezvous pairing, a common non-blocking snapshot, reset scope, or overlapping field accesses. Appendix J proves that the scripts coexist and derives a legal global order; it does not merely conjoin them.

## 7. Appendix J - local-to-global safety construction

### PURPOSE

Construct one reachable Sys\_ref execution from local process/channel/atomic certificates and prove observable compatibility with a selected RTL prefix.

Engineer takeaway: Appendix J has a B1-free pairwise correspondence engine and multiple coverage wrappers. Standard QSync supplies the general all-finite-prefix safety route; the alternate B1-run wrapper can replace RTL carrier/package induction for terminating B1-qualified instances.

### 7.1 Theorem J.1: the top-level safety result

#### THEOREM: J.1 - execution-level / trace-set compositional compatibility

For every covered reachable RTL\_B prefix, the normal compositional route uses Definition J.LocalGWR, source-side QualifiedSynchronousModel\_Sys\_ref(I)/Core.QSync-Ref, Core.SelectLatitude, and Theorem J.GWR-Comp to construct a reachable Sys\_ref prefix and a global witness package whose annotated projections satisfy E1-E5. A directly validated J.GWR package is an alternate route and may carry equivalent checked per-Horizon L2-settling/bridge evidence instead of global source QSync. Plain finite-prefix J.1 does not require Definition B1 and does not require post-side RTL QSync merely to construct the selected pair. Any B1 specialization is separate from the pairwise theorem. The reverse Ref -> Post clause remains selected-pair compatibility only through J.RTLConsistentRefPrefix; it does not construct RTL behavior from an arbitrary source run.

Four details are important for engineering use:

- **Finite reachability, not only fair executions.** The generic Appendix I/J domains are Reach\_post and Reach\_ref finite reachable prefixes. Theorem J.1 does not require generic Exec\_post/Exec\_ref objects or a J.FE fair-extension lemma. Finite postponement of an otherwise legal L1 issue or L3 boundary action is ordinary Core.SelectLatitude legality; Core.IssueFair and B2/B3 are not imported into this finite-prefix construction. Complete/fair source continuation is introduced only by later liveness machinery such as Core.LAdm/Exec\_L^adm and K.2b.E.
- **Required-prefix induction.** The proof orders observable units by the constraints that must be preserved, rather than by raw RTL cycle buckets or full source order across independent endpoints.
- **Annotated carrier.** Compatibility includes dynamic identities, witness choices, Issue/Commit attribution, request facts, and channel histories. It is stronger than matching an unannotated list of labels, but weaker than full-state equivalence.
- **Source-side settling and exact bridge route.** On the normal J.LocalGWR -> J.GWR-Comp route, source QSync supplies the finite L2 SettlePoint needed by RegPhaseNF/UnitExt. When selected source/environment/reset evidence requires a later source cycle-entry state before the next nonempty construction bucket, J.SourceBridgeClosure provides an exact-history finite bridge and finite-arrival proof; J.SourceBridgeExt provides literal extension/restriction of that bridge for incremental construction. Core.SelectLatitude may justify a permitted nonselection inside a certified bridge cycle but does not create the bridge or prove arrival. A directly validated J.GWR package may instead carry equivalent checked settling/bridge evidence.

### 7.2 The local certificate package

| Component | What it certifies                                                                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| J.PCert   | Per-process logical replay checkpoint plus separate finite preparation/issue evidence, exact dynamic identity, field-sensitive footprints, and deferred non-blocking holes produced from Appendix I. |
| J.CCert   | Per-explicit-channel boundary identity, capacity, E4 history, occupancy/head, payload, request snapshot, enablement, reset, and B-faithfulness evidence.                                             |

| Component                      | What it certifies                                                                                                                                 |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>J.ACert</b>                 | Agreement for explicitly atomic units: rendezvous pairs, paired synchronization, E2/R2C groups, RESET, and other named multi-participant actions. |
| <b>J.EnvCert / J.ResetCert</b> | Environment and reset-unit identity, scope, action, and locality evidence.                                                                        |
| <b>Typed footprints</b>        | Field-sensitive ReadSet/WriteSet evidence for every construction action, including component-local elaboration actions.                           |
| <b>LocalAgree</b>              | Equality, uniqueness, ownership, coverage, and duplicate-field agreement across local records.                                                    |
| <b>Generated dependencies</b>  | Locally justified Dep_J edges and enough evidence to prove that omitted pairs commute; the package does not supply a convenient global rank.      |

**DEFINITION: J.LocalGWR**

The complete local package for one RTL prefix: PCert for every process, CCert for every explicit boundary, ACert for every atomic unit, applicable environment/reset records, complete action/deferred-hole inventory, field-sensitive footprints, Core.RegSeq and source conformance, generated dependencies, LocalAgree, and coverage. It must not assume that these records already form a realizable global replay.

## 7.3 How J derives a global construction order

J converts the local certificate package into a finite graph of primitive ConstructionAction occurrences. Every genuine A5 action carries a checked PhaseRole\_A5 (PreL1, L3Boundary, or ResetBoundary); ordinary L2 settling and the plain next-cycle update remain phase/cycle machinery rather than invented graph vertices. Proof-unit order can be richer than executable action order: J.ProofUnitConstructionSupport marks each c-order/VCO proof unit as action-backed or order-only, and maximal c-order paths through order-only units contract to COrdActPred/Dep\_J action constraints when they connect distinct supports. D1-D4 encode semantic prerequisites; D5 is proof stratification and requires HorizonOrderSafe when it adds an order not already semantically required. J.FoundationRank/DepRank prove acyclicity, while J.SupportInsertionExists derives the simultaneous support-consistent placement of order-only units rather than requiring every LocalGWR package to supply an existential cut.

| Definition / lemma                     | Role                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>J.DepJ</b>                          | Generated dependency relation over actual ConstructionAction occurrences. D1-D4 are semantic prerequisites; D5 is Horizon proof stratification. PhaseRole_A5 types genuine A5 actions. Q.1(c-order) proof-unit paths that contain order-only vertices are reflected through J.ProofUnitConstructionSupport and contracted to COrdActPred/Dep_J constraints where applicable. |
| <b>J.DepSound / J.HorizonOrderSafe</b> | J.DepSound proves only D1-D4 semantic edges are real operational prerequisites. For a D5-only edge with no semantic path, J.HorizonOrderSafe requires a commutation certificate showing the actions may be exchanged without changing legality, complete observable units, terminal post-settle semantic state, or J.HCanon.                                                 |
| <b>J.DepComplete</b>                   | All noncommuting action pairs are ordered. The completeness proof also consumes the c-order/ProofUnitConstructionSupport reflection so an outer-required proof-unit order cannot be lost merely because an intermediate unit is order-only.                                                                                                                                  |
| <b>J.FoundationRank</b>                | A graph-independent lexicographic semantic rank based on reset epoch, Horizon, semantic stage, local ordinal, and component tie data.                                                                                                                                                                                                                                        |
| <b>J.DepRank</b>                       | Every generated edge strictly advances the foundation rank.                                                                                                                                                                                                                                                                                                                  |



| Definition / lemma     | Role                                                                                                                                                                                                                                                                                          |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>J.DepAcyclic</b>    | A directed cycle would strictly increase the well-founded rank back to its start, which is impossible.                                                                                                                                                                                        |
| <b>J.CanonicalRank</b> | Derived deterministic action order. J.SupportInsertionExists then proves that all order-only proof units admit simultaneous support-consistent placement respecting required unit order, reset epoch, Horizon/phase anchoring and atomicity; cut existence is not a LocalGWR package premise. |

#### LOAD-BEARING RESULT: J.DepComplete + J.DepRank

The flow cannot simply provide a global rank that is assumed correct. It must generate every non-commuting prerequisite from local rules, prove that unordered actions really commute, and prove each edge advances an independent semantic foundation rank. This is what makes the construction auditable and prevents a hidden global replay assumption.

## 7.4 Registered phase normalization

Within each Horizon, the proof must respect actual clocked interface semantics. It cannot decide a non-blocking result and then create a dependent request in the same cycle, and it cannot evaluate different polls against inconsistent partial channel snapshots.

#### LEMMA: J.RegPhaseNF - registered phase normal form

Every finite same-Horizon construction fragment can be permuted, without changing its canonical observable history, into: L1 process evaluation and request/attempt writes; one finite L2 settle producing the common snapshot; and L3 non-blocking decisions plus complete boundary commits. The lemma needs the local fact that the frozen L2 state admits a finite  $\rightarrow_{\varepsilon, L2}$  path to a SettlePoint. On the normal compositional route Core.QSync-Ref supplies that fact uniformly; a direct J.GWR route may carry the equivalent checked per-Horizon evidence. Registered results become available only in L4/next cycle.

#### LEMMA: J.NBDecision

Once all L1 predecessors have executed and J.CCcert supplies the common settled snapshot, each deferred non-blocking call has exactly one legal source result. A failed poll cannot coexist with a matching enabled rendezvous request, and a successful poll cannot be invented in a full/empty state that forbids it.

## 7.5 From one Horizon to the complete source witness

| Lemma / theorem         | Function in the construction                                                                                                        |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>J.Frame</b>          | A certified transition changes only its WriteSet; disjoint state and guards are preserved.                                          |
| <b>J.LocalLift</b>      | Embeds one Appendix I process-local script into a reachable global state when owned/shared locations match its certified summaries. |
| <b>J.PrepareCommute</b> | Moves independent preparation actions into the common L1 window while respecting snapshots and true dependencies.                   |

| Lemma / theorem     | Function in the construction                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>J.UnitEnable</b> | After L1/L2, the requests, payloads, channel state, atomic participants, and holes make each selected L3 unit genuinely enabled.                                                                                                                                                                                                                                                                                                                            |
| <b>J.UnitExt</b>    | Uses J.SourceBridgeClosure only when the next nonempty construction bucket requires later exact source arrival; the bridge is occurrence-bound, history/unit-empty for the J construction, invariant-preserving and finitely arriving. Then UnitExt executes the bucket, updates replay/channel/atomic state, resolves holes and preserves the construction invariant. Core.SelectLatitude is ordinary legality inside the bridge, not its existence proof. |
| <b>J.GWR-Comp</b>   | Outer induction over nonempty construction buckets constructs the reachable Sys_ref witness chain. J.GWR-Comp consumes PhaseRole_A5, ProofUnitConstructionSupport/SupportInsertionExists and exact SourceBridgeClosure. J.GWR-CompExt reuses the exported old tau_ref^G chain literally and appends only operationally legal work under LocalGWRExtCert's no-retcon frontier; it never extends an old K-facing L3CloseView branch.                          |

J ranks only actual construction actions in nonempty Horizon buckets. Empty RTL Horizons do not create synthetic CanonicalRank elements. If the next construction bucket requires a later source cycle-entry state, the proof invokes J.SourceBridgeClosure on the exact current source ExecutionCarrier; every bridge cycle is a typed occurrence-relative full cycle, and the bridge certifies finite arrival at the exact next required state while preserving the exported construction/history invariant. Persistent request levels are carried by Core.IC; they do not automatically manufacture a fresh A2 retention action in every Horizon.

#### THEOREM: J.GWR-Comp - local certificates construct global witness realizability

From J.LocalGWR plus QualifiedSynchronousModel\_Sys\_ref(I), Core.QSync-Ref supplies a finite source L2 SettlePoint for every reachable constructed nonempty Horizon. J.GWR-Comp validates the dependency graph and construction order, phase-normalizes each nonempty Horizon, and applies Horizon-batched J.UnitExt. Where the selected source witness, environment, reset-state contract, or other cycle-sensitive evidence requires a later source cycle-entry state, J.UnitExt first uses Core.SelectLatitude to take the finite evidence-determined invariant-preserving unit-empty CompleteCycle bridge needed to reach that entry. A no-Σ-commit member is Core.SilentCycle, but the proof does not assume its full state is unchanged: the typed cycle evolution and existing certificates preserve or re-establish the replay/channel/request/witness/reset/environment invariant. Empty RTL Horizons never become J.CanonicalRank buckets, so Height/FR/DepRank/DepAcyclic remain over Act\_L( $\tau_{\text{post}}$ ), and bridge length is not inferred merely from the raw RTL Horizon-index gap.

Local certificates

↓

Generated dependency graph

↓

J.DepRank / J.DepAcyclic

↓

Validated order with finite-ideal prefixes

↓

Core.SelectLatitude gap bridge where required + Core.QSync-Ref + J.RegPhaseNF + Horizon-batched J.UnitExt

↓

J.GWR + J.RegPhaseReach

↓

Proposition J.0

↓

Theorem J.1

Non-circularity. LocalGWR supplies no global order, no phase-normal replay, no assumed global extension, and no fairness premise. Core.SelectLatitude is constitutive source legality rather than B2/Core.IssueFair.

Stable interface. J.GWR can also be supplied directly by a translation validator or symbolic replay tool; that direct route must independently validate the same reachability, ordering, ideal-prefix, finite per-Horizon L2-settling, and any required invariant-preserving bridge interface.

Reset handling. RESET is an atomic observable unit; RW1-RW5 and Lemma J.RS splice a new matched reset epoch while framing unaffected components.

Finite-ideal invariant. Every flattened prefix of the validated order is down-closed: if it contains a unit, it contains all required predecessors. Unit-empty bridge cycles do not change that ideal because they add no selected J unit. Tie ordering among independent units is bookkeeping; required predecessors are not skipped and atomic units are never split.

The exact source bridge may contain real silent cycles, so registered, environment, reset or cycle-valued state can evolve according to the fixed theorem-instance choices. What is invariant is the selected construction/history package specified by SourceBridgeClosure, not the entire machine state. In incremental J.GWR-CompExt, LocalGWRExtCert is bound to the actual old  $W/\tau_{\text{ref}}^G$  construction and enforces a no-retcon exported checkpoint: new same-Horizon work must be operationally appendable after the old checkpoint, including the correct PhaseRole\_A5 and reset ordering. Old bridge/history choices restrict literally under SourceBridgeExt. Bridge length is never inferred from the raw RTL Horizon gap.

## 7.6 Proposition J.0 is deliberately separate

### PROPOSITION: J.0 - pairwise composition lemma

Given an already selected compatible source/RTL trace pair, compatible per-process annotated projections, and per-channel E4 well-formedness, J.0 lifts E1-E5 to the system level. It does not construct the source or RTL trace, prove B-faithfulness, or establish J.GWR. On J.1's normal Post  $\rightarrow$  Ref route, J.GWR-Comp constructs the reachable source/RTL pair and Theorem J.1 instantiates J.0 with that constructed pair. In the restricted Ref  $\rightarrow$  Post clause, RTLConsistentRefPrefix instead supplies the already-selected pair and J.0 is the non-circular composition step.

## 7.7 Safety transfer for the verification flow

### DEFINITION: J.TraceCertCov + COROLLARY: J.1-Safe

J.TraceCertCov\_I(D) is the named Post  $\rightarrow$  Ref safety-certificate coverage premise. For every RTL prefix in D it records either the normal J.LocalGWR/J.GWR-Comp package with source Core.QSync-Ref/Core.SelectLatitude or a directly validated J.GWR package with equivalent checked per-Horizon settling/bridge evidence, plus the remaining Post  $\rightarrow$  Ref premises. Corollary J.1-Safe consumes that route-sensitive coverage for a J.HCanon-invariant predicate satisfying J.IdealPrefixClosed. If one complete pre-HLS run is used as representative, J.RefProjCover must cover every relevant reachable source observation as a finite canonical ideal/down-closed prefix of that run.

Core.SelectLatitude enlarges admissible source latency realizations; pure unit-empty delay that leaves CanonHist unchanged is already absorbed, but if legal selection delay can change the canonical  $\Sigma_{\text{DUT}}$  history then the one-run RefProjCover discharge needs a separate latency-invariance/coverage argument. For  $D = \text{CycleReach\_post}(I)$ , the standard QSync route derives package coverage from J.QSyncTraceInd. These safety-side coverage predicates do not imply K.CertCov.

**QSYNC SAFETY PROFILE: J.CycleReach\_post / J.QSyncTraceInd / J.CycleIdealCover / J.1-Safe-QSync**

J.CycleReach\_post is the exact reachable RTL-prefix domain ending at K-origins, before same-cycle QSync settling. J.QSyncTraceState retains only the exact-prefix LocalGWR/J.GWR-Comp witness and J.HCanon needed by safety. J.QSyncTraceInd proves a base case and one-cycle preservation for every reachable successor (not just one observed path); it explicitly does not consume J.OfferReach, J.OfferConf, J.KObsConf, K.DrainReach, or K.RLAdmReach. J.TraceCertCov-QSyncCycle and J.1-Safe-QSyncCycle establish cycle-aligned package coverage/safety. Core.RTLCycleCarrier supplies the execution-carrier fact for an already-entered RTL real cycle; J.CycleIdealCover then proves every reachable finite outer canonical history is a finite ideal of a reachable cycle-aligned history, and J.1-Safe-QSync uses J.IdealPrefixClosed to transfer safety to every reachable finite RTL prefix without asserting a separate J.TraceCertCov package for each intra-cycle ideal. If the richer K-facing J.QSyncCertInd is already established, J.QSyncCertInd-Trace supplies the safety induction automatically. Cross-interface scoreboards that fail J.IdealPrefixClosed still require direct arbitrary-prefix J.TraceCertCov coverage.

**ALTERNATE B1-RUN SAFETY PROFILE: Core.B1RunMax / J.B1PostProjCover-RunMax / J.1-Safe-B1Run**

Fix one selected finite RTL prefix  $p\_post^0$ . B1 at outer  $\Sigma\_DUT$  plus Core.B1RunMax and the triggered premises of J.B1OuterHistCongruence derive J.B1PostProjCover: every reachable RTL canonical outer history is a finite ideal of CanonHist\_ $\Sigma\_DUT(p\_post^0)$ . The proof trims any outer-empty suffix of a queried prefix and uses J.OuterCanon-Ideal equality for that erased-only extension; no RTLPartialCycleClosure or fair extension is used. The singleton J.TraceCertCov( $\{p\_post^0\}$ ) package then constructs the selected source witness  $p\_ref^0$  by plain B1-free J.1. If  $\Phi$  is extensional on CanonHist\_ $\Sigma\_DUT$ , J.IdealPrefixClosed, and holds on that selected  $p\_ref^0$ , Corollary J.1-Safe-B1Run proves  $\Phi$  for every Reach\_post prefix. This route does not consume J.RefProjCover, universal TraceCertCov, RTL QSync carrier induction, LocalGWRExtCert/GWR-CompExt, or K-only evidence.

## 7.8 Alternate B1-run universal safety route

The B1-run route is a coverage shortcut around the RTL-side QSync carrier/package induction, not a different J.1 semantics. It is available only when the fixed-stimulus RTL instance has one finite reachable cycle-closed DeclaredTerminal run  $p\_post^0$  satisfying Core.B1RunMax and the claimed property is extensional on CanonHist\_ $\Sigma\_DUT$  and J.IdealPrefixClosed. B1 is discharged only at the outer  $\Sigma\_DUT$  alphabet used by the route; plain J.1 and its singleton TraceCertCov package add no wider proof-alphabet B1 premise.

The RTL-only coverage proof has three layers. First, Core.B1RunMax-Cover combines BRM2 with B1 no-cycle-migration to place every reachable selected occurrence into  $p\_post^0$  at the same real cycle. Second, J.B1OuterHistCongruence turns Core.B1ObsPrefix-related prefixes into finite ideals of one another. The built-in strict-generator cases are explicit: (i) ordinary Core.17/E1-E5 required generators are handled structurally; (ii) RESET/RW reset-epoch generators use RSC0-RSC3 over conservative Core.OuterResetSupport; and (iii) qualified Q.1(c-order) proof/elaboration generators use QPC1-QPC3. Any additional strict generator admitted through the open extension class must separately supply the same three-part cross-run contract: common occurrence/group restriction, old-old required-order restriction stability, and predecessor-domain closure for a successor already realized on the smaller carrier. Third, J.B1PostProjCover-RunMax trims any later outer-empty suffix, defines  $t^*$  from the maximum real cycle containing a retained selected occurrence in the trim, and derives universal outer-history ideal coverage without completing an arbitrary queried RTL cycle.

Operationally, BRM2 is universal over Reach\_post, so observing one terminating simulation is not enough. Under B1 and BRM1, however, the obligation has a sharp cycle-horizon form: for a nonzero selected run with final completed cycle  $t_f$ , BRM2 holds exactly when no reachable complete selected  $\Sigma\_DUT$  occurrence occurs in a real cycle later than  $t_f$ . A strong translation/semantics theorem may make this a reusable design/tool theorem with lightweight per-stimulus binding; otherwise it remains a global reachability/safety check. The final source property check must be on the actual selected  $p\_ref^0$  or on another source run whose CanonHist\_ $\Sigma\_DUT$  is separately proved equal. NB-CF alone does not justify substituting an arbitrary source regression run.

## 7.9 The J-to-K cutpoint bridge

Trace matching alone is insufficient for deadlock reasoning. Core.18 defines one common typed KObs seam, but J keeps the extendible safety checkpoint and the K-facing branch distinct. From  $\tau_{ref}^G$ , a K-facing consumer first forms a fresh  $\tau_{ref}^{cl}$  through Core.L3CloseView; J.L3CloseView-H preserves CanonHist and request roots. On the standard normal-compositional K-facing source route, Core.KObsSettleQualified\_Sys\_ref(I) is the separate post-L3 observation-only settling qualification, and Lemma Core.KObsSettle-Ref supplies the finite fixed point used by Core.KObsNormRef. K.NormStable NS0-NS6 then checks the exact close-view/normalization branch before OfferReach/KObsConf. Core.KObsSettleQualified is not a premise of Theorem J.1 or the standard safety-only QSync profile. That closed branch is never extended by the next QSyncTraceStep/GWR-CompExt. J reconstructs current message, synchronization and in-flight state from KObs, then deliberately stops: future path realizability is supplied by the separate occurrence-relative J.InFlightPathSound theorem and consumer-specific profile coverage/search evidence.

| Bridge element                                                                                        | Purpose                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>J.HCanon</b>                                                                                       | Proves that the selected source and RTL prefixes have the same canonical replay history H modulo the permitted epsilon/history quotient.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>J.RegPhaseReach / Core.L3CloseView</b>                                                             | J.RegPhaseReach records the exact extendible $\tau_{ref}^G$ construction under the registered phase discipline. A K-facing use then forms a fresh Core.L3CloseView $\tau_{ref}^{cl}$ from that same checkpoint; the close view is observation/K-facing infrastructure and is not the next safety-construction prefix.                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>J.L3CloseView-H / Core.KObsSettleQualified / Core.KObsSettle-Ref / K.NormStable / J.OfferReach</b> | J.L3CloseView-H preserves canonical history and request/root state across the phase-close view. On the standard normal-compositional K-facing source route, Core.KObsSettleQualified_Sys_ref(I) discharges the KSQ1-KSQ4 post-L3 observation-only settling qualification and Core.KObsSettle-Ref supplies the finite deterministic fixed point used by Core.KObsNormRef. K.NormStable NS0-NS6 validates that the resulting suffix is post-L3 observation-only and introduces no new selected unit, issue/attribution, witness-significant NB decision, reset or L4 update. J.OfferReach then realizes $\langle E, w, H, O \rangle$ at the exact resulting source KCut. This qualification is K-facing only and is not part of the safety-only QSync profile. |
| <b>J.OfferConf</b>                                                                                    | At the selected RTL KCut, bidirectionally and uniquely matches every residual offer in O to a relevant asserted RTL request and every relevant RTL request back to exactly one offer. Core.SyncFence is load-bearing for identifying unqualified residual offers with the current-interval Pending_P set.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>J.KObsConf</b>                                                                                     | Binds one exact source/RTL package with explicit NormRef/NormPost KCut endpoints, common I/E/w/H/O, complete request attribution, and boundary/reset/epsilon-opacity evidence. It is the downstream theorem obligation that establishes the Core.18 cross-model equality; the Core itself fixes only the common type/equality notion.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Corollary J.1</b>                                                                                  | Defines KObsPost_J, proves its equality with the realized source KObs, and exports the current-state reconstruction facts. Message-frontier/WFG reconstruction remains J.KObs-Proc/Chan/Offer/WFG; synchronization and generalized drain additionally use J.KObs-SyncCarrier/J.KObs-SyncInFlight/J.KObs-InFlight. The corollary does not itself assert K.Dead or future reconstructed-path realizability.                                                                                                                                                                                                                                                                                                                                                    |
| <b>J.KObs-SyncCarrier / J.KObs-SyncInFlight / J.KObs-InFlight</b>                                     | Reconstruct the typed current synchronization carrier, local SyncFire action/effect, and the complete selected Sys_ref FlightStateRec/InFlightStepRec action relation from the same KObs. No message offer is synthesized for synchronization and no future path opportunity is inferred.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Bridge element             | Purpose                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>J.InFlightPathSound</b> | Universal exact-path realization from every admitted profile start. Each reconstructed edge consumes the exact operational prefix/carrier occurrence produced by the previous edge; reset/terminal/activation may stop early only at the path endpoint or with a separately checked no-further-declared-edge fact. Profile membership/coverage is supplied separately by the consuming K theorem. |

#### AUDIT CRITICAL: J.OfferConf request-to-offer completeness

Every relevant asserted RTL boundary request must be attributed to exactly one source residual offer at the selected explicit boundary. Omitting a hidden asserted request, duplicating attribution, or using an unrelated source cutpoint invalidates the KObs bridge. This is the implementation abstraction boundary on which Appendix K relies.

## 8. Appendix K - conditional deadlock preservation

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PURPOSE</b> | Preserve absence of reachable generalized-drain-closed, release-closed internal message-passing wait cycles from the selected source-reference instance to covered RTL KCuts. Engineer takeaway: K is not a generic progress theorem. Current-state reconstruction, operational path qualification, consumer ProfileCover, and exact negative-search evidence are separate layers. The serial A5-L reduction likewise separates SDet selection, Policy-S progress, universal candidate applicability, explicit reset scope, prospective response domains, and post-SF response extraction. G2 lowering adds real-cycle stutter/reset-sensitive normalization qualifications rather than silently transporting Sys_K evidence to Sys_ref. |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 8.1 The deadlock predicate

Appendix K evaluates exact K-facing occurrences, not arbitrary epsilon-quiescent bare states. On source, the K-facing package begins from an extendible J.GWR checkpoint, takes Core.L3CloseView, then KObsNormRef/K.NormStable NS0-NS6 to the selected source KCut. Closing that L3 view cannot manufacture disablement/deadlock merely from a nonselected action. On RTL, Core.ExecutionCarrier/RTLFrameEpochReachOn record exact history/reset provenance and QS2a-D supplies forward designated origins under QualifiedSafetyCarrierModel\_RTL\_B. The QS3-QS5 QualifiedSynchronousCarrierModel\_RTL\_B extension is what licenses Core.QSync-Post and hence forward KCut closure. A theorem that begins from a queried reachable KCut adds full QS2a-R/RTLKCUTSettlingWindow on the selected witness and only then uses KOriginCarrier-KCut. ReleaseOwnerSets\_E remains the exact release-closure data, WaitOwners\_E the conservative WFG projection, and literal Dead\_K remains message-rooted. Generalized drain is evaluated per proof model under its bound action/path/profile/search qualifications rather than by comparing arbitrary source/RTL microstates.

| Object                 | Meaning                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Front_P</b>         | The minimal pending blocking message-operation items of process P whose endpoint requests are currently asserted. Merely being the next source call is not enough; the request must have issued. Core.OrdFrontSingleton proves $ Front\_P  \leq 1$ in an ordinary non-elaborated source interval; multiple members require explicit Sys_B^elab partial-order structure. |
| <b>Blocked process</b> | Front_P is nonempty and every frontier item is ChannelDisabled at the selected explicit boundary. A disabled edge for one item does not by itself make a multi-frontier process blocked.                                                                                                                                                                                |

| Object                                                                                                                 | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WFG edge <math>P \rightarrow Q</math></b>                                                                           | For each disabled frontier item $x$ of $P$ , $\text{Core.WFG}$ contains $P \rightarrow Q$ for every internal modeled process $Q$ in $\text{WaitOwners\_E}(x, \sigma)$ , the union of $x$ 's minimal $\text{ReleaseOwnerSets\_E}$ supports. This graph is intentionally conservative: one support may require several processes jointly, while different supports are alternatives. Multi-frontier processes may also have outgoing edges from several disabled items. $Q.1(j)$ makes the settled handshake and state-instantiated family reconstructible from $E$ plus the reconstructed current roots, and $J.\text{KObsConf}$ separately checks RTL faithfulness to that declared schema. $Q.1(k)$ requires family-level support soundness (no spurious rescue) and completeness (no missed rescue) relative to the qualified template before $\text{WaitOwners}$ is formed. The blocked-process condition remains an all-items-disabled test; exact deadlock closure is checked against the unflattened release families. |
| <b>DrainClosedNow(D)</b>                                                                                               | Generalized form: no finite nonempty qualified $\text{Core.InFlightStepRec}$ path from the current $\text{FlightStateRec}$ changes $\text{WaitRoots\_D}$ . $D$ scopes the observed process-root family only; it does not filter action ownership or causal scope. $\text{DrainCandidates\_D}$ is retained only as an ordinary primitive diagnostic/fast path, with $K.\text{PrimitiveDrainExact}$ giving the restricted equivalence case.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Definition <math>K.\text{Dead}</math> / <math>K.\text{DeadW}</math> / <math>K.\text{ReleaseSCCReduction}</math></b> | $K.\text{Dead}$ is the single authoritative reachable drain-closed internal wait-cycle predicate. A candidate component is nonempty, every member is blocked, every internal disabled frontier release support intersects the component (exact family-level release closure), and the induced coarse WFG is strongly connected and contains a genuine internal process-dependency cycle. For $ D =1$ , this requires an actual $P \rightarrow P$ dependency; the vacuous one-vertex SCC convention is not enough. $\text{Dead\_K}^W$ adds the declared $\text{KObs-closed}$ waiver exclusion. $K.\text{ReleaseSCCReduction}$ shows that connectivity can be reduced to a terminal SCC without loss when the release-closed set has a nonempty internal release family at each member; in the singleton terminal-SCC case its within-set outgoing-edge argument supplies the required genuine self-loop. Optional $\text{Dead\_KS}$ remains diagnostic only; later assumptions do not redefine $K.\text{Dead}$ .              |
| <b>WaitRoots_D</b>                                                                                                     | Process-indexed $P \mapsto \text{WaitRoot\_P}$ family containing each $P$ frontier, its enablement/disablement facts, and exact $\text{ReleaseOwnerSetsRec\_E}$ data. Process indexing makes restriction to $T \subseteq D$ definitional, enabling $K.\text{InFlightEscapeMono}/K.\text{DrainClosedHeredity}$ and the repaired $K.\text{ReleaseSCCReduction}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>FlightStateRec / InFlightStepRec / InFlightEscape</b>                                                               | Finite current-state reconstruction used by generalized drain. The step relation contains ordinary already-issued commits and qualified proof-internal staged/bundle/join actions; the synchronization profile also admits ready $\text{SyncFire}$ and, on lowered $\text{Sys\_K}$ , explicit lowering-admin actions such as $\text{EpochGateAdvance}$ . Multi-step paths matter: the two-stage non-fall-through artifact has no immediate root change on its first action but still provides a later escape.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

#### SCOPE: Reachable closed internal wait-cycle cutpoint

The operative predicate is stronger than “the system is currently slow” and different from a temporal permanent-deadlock definition. It identifies a reachable message-rooted internal wait cycle whose exact release alternatives are trapped and for which the qualified finite already-admitted transition system has no escape that changes the candidate's  $\text{WaitRoots}$ . The directed WFG remains a conservative  $\text{WaitOwners}$  union;  $K.\text{Dead}$  keeps the exact  $\text{ReleaseOwnerSets\_E}$  family. Synchronization carriers, NB-aligned guard causes, and lowering administration may participate in the proof of drain/lowering correspondence, but they do not become literal message WFG roots. Environment-only rescue, barrier mismatch, starvation, and other excluded stalls retain their separate scope rules.

## 8.2 Reconstructing the wait state and generalized in-flight relation from $J.\text{KObs}$

The drain argument is explicitly modal while keeping  $\text{KObs}$  compact. From equal  $\text{KObs}$ ,  $J$  reconstructs the current message-root state and the current finite proof-model transition system.  $\text{Core.FlightStateRec}$  contains only reconstructible current state;  $\text{Core.InFlightStepRec}$  gives one complete already-admitted action/effect step. Neither object contains historical Issue times, activation certificates, or a promise that a locally plausible action will be selectable in a future cycle.



K defines `WaitRoots_D` as a process-indexed family and `InFlightEscape` as a finite nonempty reconstructed path that changes it. The relation is D-independent, so outside-D admitted work remains relevant. Sound use requires four distinctions. First, `K.InFlightQual` fixes the Q.2 schema, finite drain, occurrence-relative opportunity/path soundness and record-level `K.InFlightRelationPrecision`. Second, `K.InFlightProfileCover` proves that each exact occurrence consumed by the theorem is an admitted profile start. Third, every negative drain claim binds an `occurrenceBound` or checked `uniformFiber` `K.InFlightSearchQuery` and an accepted `K.InFlightSearchStatus`; `ExhaustiveExact` carries `K.InFlightRelationComplete`, `OmittedRootIrrelevant` proves the omitted edges cannot affect this exact root query, and `UnknownIncomplete` cannot establish `DrainClosedNow`. Fourth, these consumer-specific objects are not fields of a route-neutral `K.CertifiedCutpoint`.

Two proof levels should therefore not be conflated. `Core.Drain/K.DrainEvidence/Core.SettleKBridge` qualify the exact reached source prefix and its no-replenishment history. Generalized `DrainClosedNow` is a current-state modal negative whose soundness additionally depends on relation precision, exact occurrence/profile coverage and accepted query status. In G1 the serial route's `Core.CycleClosure` can discharge only the IQ3-class continuation part of SR3-G; it does not establish Q.2 soundness, profile cover or IQ5 search evidence.

| Result                                                                             | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>J.KObs-Proc / Chan / Offers / WFG</b>                                           | Appendix J reconstructs process replay, explicit channel state, residual message offers, frontier/request roots, settled handshake, <code>ReleaseOwnerSets_E/WaitOwners_E</code> , <code>ChannelEnabled/ChannelDisabled</code> , and <code>Core.WFG</code> from the common KObs record. These are current message-root facts; generalized in-flight reconstruction is a separate J.KObs-InFlight layer.                                                                                                                                                                                                                                         |
| <b>K.EnvTerminalAdequate / K.KObsDerived</b>                                       | Appendix K first supplies the environment-terminal adequacy premise and gathers the K-facing derived functions needed before deadlock factoring.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>K.KObs-Dead / K.KObs-Ext</b>                                                    | K-owned factoring and equal-KObs transfer of release-aware deadlock/waiver predicates using the same reconstructed <code>FlightStateRec/InFlightStepRec/WaitRoots</code> relation. The generalized action/path schema identity is bound separately through <code>K.InFlightQual/K.CertHdr</code> ; equal KObs does not manufacture that operational evidence.                                                                                                                                                                                                                                                                                   |
| <b>K.0 / K.DeadChar</b>                                                            | K.0 reconstructs the operational minimal pending blocking frontier from KObs-derived inputs. <code>K.DeadChar</code> is the source-side/lowered-KCut channel-disabled characterization on the domain where <code>K.Dead</code> is defined; its buffered/rendezvous cases cite <code>Core.12/Core.13</code> directly rather than asserting an RTL microstate theorem.                                                                                                                                                                                                                                                                            |
| <b>K.2 / K.2a</b>                                                                  | Use equal reconstructed state to prove extensionality of the exact release/WFG data and generalized drain representation. K.2a compares the same <code>FlightStateRec/InFlightStepRec/WaitRoots</code> reachability structure, not a legacy offer-only selector.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>J.KObs-SyncCarrier / SyncInFlight / KObs-InFlight / J.InFlightPathSound</b>     | Current-state reconstruction and future-path realization are separate. J.KObs reconstruction recovers the exact selected Q.2 relation; <code>J.InFlightPathSound</code> realizes every finite path edge-by-edge on the same occurrence chain. Each consuming theorem separately proves that its exact occurrence is an admitted profile start through <code>K.InFlightProfileCover</code> .                                                                                                                                                                                                                                                     |
| <b>K.WaitRoots / K.InFlightEscape / K.DrainClosedHeredity / K.InFlightRelocate</b> | <code>WaitRoots</code> is process-indexed; an escape is any finite qualified path changing that family. Subset projection yields escape monotonicity and drain-closure heredity, which repairs <code>ReleaseSCCReduction</code> . If a candidate is not drain-closed, <code>InFlightRelocate</code> realizes an escape, stops at the first K-facing root change, and redispaches all candidate sets; it assumes neither SCC preservation nor monotone candidate identity.                                                                                                                                                                       |
| <b>K.InFlightQual / Q.2 / Core.InFlightCycleExt</b>                                | Qualification layer for generalized drain. Q.2 supplies exact action/effect schemas and finite choice; <code>Core.InFlightDrainWF</code> supplies a decreasing measure; <code>Core.InFlightCycleExt/J.InFlightPathSound</code> supply occurrence-relative operational realization. <code>K.InFlightRelationPrecision</code> is record-level. Every negative uses a bound <code>K.InFlightSearchQuery</code> plus <code>K.InFlightSearchStatus</code> : <code>ExhaustiveExact</code> requires <code>K.InFlightRelationComplete</code> ; <code>OmittedRootIrrelevant</code> is query-specific; <code>UnknownIncomplete</code> is diagnostic only. |

| Result                                                     | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>K.Low.ModalQual / InFlightPathCorr / DeadRootBridge</b> | G2 lowering stack. TimeStutterQual is required when administrative lowering may consume real cycles. ModalQual MQ1-MQ6 includes instance-wide MQ2: every relevant normalization window must be able to finish before a projection-changing affecting RESET; LT2 reset mapping alone is insufficient. MQ5 binds separate Sys_K/Sys_ref InFlightQual plus ProfileCover for every exact paired start consumed by the modal path/escape/drain correspondence. ResetStutterMatch/ResetKey-Stutter and fresh target-model negative-evidence transport are used when applicable. |

### 8.3 One certified cutpoint versus universal coverage

#### DEFINITION: K.CertifiedCutpoint

A K-ready package for one reachable RTL KCut rho\_post, together with at least one explicitly recorded reachable prefix tau\_post and finite normalization witness nu\_post satisfying NormPost(tau\_post, nu\_post, rho\_post). It binds the theorem instance and contains the J.1/KObs correspondence, exact source KCut witness, J.RegPhaseReach, K.DrainReach including Core.SettleKBridge, the A6 source-ownership/elaborated-dependency-projection facts for every evaluated KObs/WFG boundary, waiver closure, and the remaining K-facing assumptions for that one package. On the normal compositional route the package may cite K.DrainReach-Comp for the exact K.DrainReach field once the bound DR1-DR4 instance evidence is established. It does not imply K.CertCov. When generalized drain is selected, the common K.CertHdr also binds the exact Q.2 action/effect schema and uniform Sys\_ref K.InFlightQual used by K.1A; G2 serial/lowering packages additionally bind K.Low.ModalQual and both model-indexed InFlightQual records upstream.

#### THEOREM: K.1A - certified-cutpoint preservation from A5-L

Assume one certified RTL cutpoint satisfies Dead\_K^W. Equal KObs transfers the same reconstructed message-root data and generalized DeadKRec structure to the exact reachable Sys\_ref witness. The K-facing package supplies J.RegPhaseReach/K.DrainReach so the exact source prefix is Core.LAdm-admissible, while the theorem application separately binds the applicable Sys\_ref K.InFlightQual that makes generalized DrainClosedNow operationally sound. This contradicts source A5-L. K.1A is route-neutral: it consumes source A5-L and source K.InFlightQual, not a G1/no-lowering premise; a G2 serial route may have obtained A5-L upstream through K.Low.ModalQual/K.Low.A5.

#### COROLLARY: K.1A-Total - universal result under K.CertCov and KCut closure

K.1A is pointwise over an already selected certified KCut. A universal claim additionally consumes Core.KCutClosure\_post and K.CertCov. In the standard QSync scope, QualifiedSynchronousCarrierModel\_RTL\_B - the safety foundation plus RTL QS3-QS5 - invokes Core.QSync-Post and derives KCut closure without QS2a-R. The lighter QualifiedSafetyCarrierModel\_RTL\_B remains sufficient for the Appendix-J cycle/carrier safety route. The preferred package-totality route then adds full QualifiedSynchronousModel\_RTL\_B with QS2a-R/Core.RTLKCutSettlingWindow so each queried reachable KCut can be placed in its selected witness and related to a carrier position by Core.KOriginCarrier-KCut. J.QSyncCertInd proves a base case plus one-cycle preservation over every reachable RTL successor; J.CertExist-QSync lifts that induction to J.QSyncCertCov and K.CertCov-QSync binds the tuples through K.CertHdr to derive K.CertCov. A single observed simulation path or sampled KCut set is insufficient unless a separately proved determinism/exhaustiveness result collapses the successor set. K.CertCov remains total only on KCutReach\_post(l), not every arbitrary RTL prefix or normalization witness.

**STANDARD QSYNC PACKAGE-TOTALITY ROUTE**

J.QSyncCertState contains the substantive exact-prefix J/K package: canonical NormPost endpoint, accepted J.LocalGWR/J.GWR-Comp witness, J.HCanon/OfferReach/OfferConf/KObsConf, theorem-derived J.RegPhaseReach, and QC5 K.DrainReach/Core.SettleKBridge. J.QSyncCertInd is anti-circular: CI0 establishes the initial tuple; CI1 must extend it through any reachable one-cycle RTL successor; CI2 requires exhaustive successor coverage and may not assume the successor K.CertifiedCutpoint or K.CertCov. On the standard normal route, CI1 preserves applicability of the independently established DR1-DR4 K.DrainReach-Comp evidence, rebuilds the successor J.GWR-Comp/I.A0 attribution, history, and QSync settling facts, and then re-applies K.DrainReach-Comp to obtain successor QC5; there is no independent candidate-specific DR5 history input. J.CertExist-QSync plus K.CertCov-QSync then supplies K.CertCov on demand over the canonical KCut domain. The same J.QSyncCertInd proof projects to the cycle-aligned J safety proof.

## 8.4 The central source-liveness premise A5-L

**PREMISE: A5-L - scheduler-robust liberal source liveness**

A5-L abbreviates A5L\_Sys\_ref: no finite Policy-L prefix admissible under Core.LAdm for the selected Sys\_ref instance reaches a source K-facing cutpoint satisfying Dead\_K^W. Drain-closedness is part of Dead\_K itself. Under generalized drain, the theorem application also binds the applicable Sys\_ref K.InFlightQual; proving A5-L alone does not manufacture path-opportunity evidence. On the primary G2 serial route, the bounded-response reduction first establishes A5L\_Sys\_K on the normalized lowered model and generalized K.Low.A5 transports it to this source A5-L. Because Policy-S-shaped serial prefixes are included among admissible Policy-L prefixes, a reachable serial wait cycle still makes A5-L false.

A5-L is intentionally global. Internal wait cycles depend on interactions among processes, capacities, environment behavior, and the selected issue policy. Unlike the J safety construction, it cannot generally be reduced to independent per-process liveness theorems.

## 8.5 The primary Policy-S discharge route

The primary user-facing activity is still one selected complete instrumented Policy-S execution, but the current route has an explicit producer/consumer stack. SDet fixes one functional artifact; K.PolicySProgress separately binds Core.IssueFair\_{Policy S} and applicable B2/B3 on that artifact; K.CompleteS packages maximal/complete execution. K.CVPred now supplies exact-prefix Avail\_pi, predecessor-before-use, evaluator applicability, typed ObsKey/version identity, finite ResetSupport\_CV/K.CVResetSupportSound and CVP-M historical persistence; SegWork recurses directly on WellFounded(prec\_CV). SR3-G binds Sys\_K K.InFlightQual, exact counterexample-domain ProfileCover and candidate negative-search evidence. K.SerialCandidateApplicability is universal over every non-waived A5L\_Sys\_K counterexample candidate. Nested Continue selects a valid DomResetScope and prospectively fixes PolicySCompareDomains before E/P0/R/P/P-Frozen/SF/Q. K.RespCov/K.CompleteS remain direct K.2b inputs; A5-S supplies those plus K.RespClean for K.2c/K.1.

Lowering is proof transport, not a new deadlock predicate. K.Low.C transports finite admissible prefixes/cutpoints; complete-execution correspondence additionally requires separately model-indexed fairness/progress preservation. When lowering can insert/delete real cycles, K.Low.TimeStutterQual explicitly covers environment, reset/terminal timing, cycle-valued semantics and admissible-prefix stutter. K.Low.ModalQual MQ2 is instance-wide over every relevant normalization start and MQ5 includes paired occurrence ProfileCover on both Sys\_K and Sys\_ref. LT2-mapped resets use K.Low.ResetStutterMatch/ResetKey-Stutter, the closed permitted cross-history ResetKey pairing. Cross-model negative drain evidence is retyped through K.Low.InFlightNegativeEvidenceTransport; a Sys\_K search query is never reused as a Sys\_ref query. K.Low.A5 then transports A5L\_Sys\_K to the source A5-L consumed by K.1A.

| Element                                                               | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>K.Low / K.Low.A5 / Sys_K</b>                                       | Witness-indexed lowering remains the route for synchronization waits and aligned non-NB-CF dependencies. When lowering-only administration may consume real cycles, K.Low.TimeStutterQual is required before finite-prefix K.Low.C/K.Low.A5 transport. ModalQual MQ1-MQ6 binds the exact lowering/action schemas; instance-wide MQ2 normalization existence/rank plus proof that every relevant normalization window can complete before any projection-changing affecting RESET; MQ3/MQ4 state/root and bidirectional path/reflection; MQ5 separate Sys_K/Sys_ref InFlightQual plus paired-occurrence InFlightProfileCover; MQ6 schema/checker binding. LT2-mapped resets use ResetStutterMatch/ResetKey-Stutter and cross-model negative drain evidence is reconstructed as a fresh target-model query/status.                                                 |
| <b>K.EnvMF</b>                                                        | For StandardEnv/B1-available, a candidate satisfying K.JointFreeze-Ord's ordinary/single-release-support premises gets the no-release fact theorem-wise. Every other admitted candidate - including a single frontier item with a non-singleton or multi-alternative ReleaseOwnerSets_E family and every explicit multi-frontier - supplies K.JointFreeze JF1/JF2. Timed/reactive or otherwise nondefault environments require exactly one selected internal frontier per process plus the K.JointFreeze-Ord single-singleton-release-support premises. K.EnvMF-Uniform is the standard instance-level sufficient discharge of that timed/nondefault branch; in the common ordinary primitive case its structural part reduces to Core.OrdFrontSingleton plus the Core.WFG {{Q}} release-support fact and Dead_K membership.                                     |
| <b>K.JointFreeze / K.CompareAdvanceLegal</b>                          | K.JointFreeze-Ord is theorem-derived for the qualifying ordinary/comparability-certified single-frontier/single-singleton-release-support case; K.JointFreeze is required for other admitted StandardEnv structures. Generalized drain narrows one part of JF2: at the activation cutpoint, the candidate's DrainClosedNowRec already excludes every finite currently admitted InFlightStepRec path that changes D's roots. JF2 still has genuine continuation work: phase-legal activity not already contained in that initial closure, including outside-D progress and subsequently admitted work, must preserve the selected frozen operations/release closure for as long as they remain pending. K.CompareAdvanceLegal remains the separate P0 preparation/R Issue legality condition, with the existing JointFreeze/CompareAdvanceLegal consistency rule. |
| <b>SDet / K.PolicySProgress / K.CompleteS</b>                         | SDet fixes one functional Policy-S artifact and every history-affecting choice. It does not imply progress. K.PolicySProgress separately binds IssueFair_{Policy S} plus applicable B2/B3 on that exact artifact and with the selected reset/terminal semantics. K.CompleteS packages the resulting maximal/complete run. These are deliberately separate from response cleanliness.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>K.DomResetScope / K.InterruptionGate / K.PolicySCompareDomains</b> | For each nested Continue candidate, bind one explicit valid S_dom (or the applicable uniform constructor) and evaluate early Reset then Terminal. On Continue-followed-by-Continue, prospectively fix K.PolicySCompareDomains=<SFControl,Handoff,Response> before SF/Q. R-L/T-L protect the liberal continuation; R-S/T-S protect only SFControl through the canonical monitor trigger. Post-handoff affecting reset/terminal are Response dispositions. DirectFail remains independently mapped K.RespFail. Scope selection and domains may not be changed after observing SF.                                                                                                                                                                                                                                                                                  |
| <b>K.RespCov / K.RespSerialBlockerClassSet / K.SFOutputEnvelope</b>   | K.RespCov is default-total over dynamically reachable internal process-facing blockers and constructs one canonical logical K.RespMon for literal or static-derived dispositions. Before SF, K.SFOutputEnvelope defines the non-predictive owner/frontier/provenance/key envelope of possible outputs. K.RespSerialBlockerClassSet conservatively covers every possible SF response class; a class may be pruned only by K.RespSerialClassExclusionSound from a closed pre-SF evidence base and the retained set must be waiver-disjoint.                                                                                                                                                                                                                                                                                                                        |

| Element                                                    | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CD3-C / CD4 / K.RespClean                                  | CD3-C is the prospective universal Handoff-completeness fact: for every relevant dynamic instance in SFOutputEnvelope whose canonical monitor triggers on the fixed SDet artifact, Handoff already represents the exact canonical first-trigger record. After SF, K.2b.Q derives CD4 actual membership by instantiating CD3-C. Failure invalidates the certificate/application; it cannot be repaired by changing Handoff/domains or early-DirectFail rerouting. K.RespClean excludes every non-waived K.RespFail on the successful complete run, including concrete early DirectFail.                                                                |
| K.PolicySProgress / response phase                         | K.2b.SF/Q may use Policy-S eventual selection/response only through the exact-artifact K.PolicySProgress record. The SF result is handoff-stable/non-advancing only through the certified handoff and thereafter until the first applicable response disposition - not permanently. K.2b.Q Step 3 consumes Response only and orders outcomes: affecting RESET/RW2 cancelled-pending, then DeclaredTerminal maximal-pending while the same monitor remains pending, then timeout.                                                                                                                                                                      |
| K.SerialRoutePremises                                      | Authoritative K.2b applicability record. SR1-SR3 carry the fixed Sys_K operational/lifecycle/CV/cycle premises; SR3-G separately binds current Sys_K InFlightQual, exact counterexample-domain InFlightProfileCover and accepted per-candidate IQ5 search evidence. SR4/4-U bind EnvMF, CompareAdvanceLegal, SegmentPlan/Adm and universal K.SerialCandidateApplicability. SR5 binds the selected reset/terminal gate plus prospective PolicySCompareDomains package, not an actual SF blocker. SR6 fixes theorem-instance choices; SR7 binds NB-Align and K.Low/ModalQual where triggered. The record does not contain A5-L or response cleanliness. |
| A5-S / CF-S                                                | The complete response certificate over the same SDet artifact. For K.2c/K.1 it supplies K.CompleteS + total K.RespCov + K.RespClean and references PolicySProgress, SerialCandidateApplicability, class/domain and checker identities through the common header/ledger. It does not duplicate K.2b's applicability proof.                                                                                                                                                                                                                                                                                                                             |
| K.CV / K.OpKey / K.CVPred / K.CV-WF / K.2b.P0-P1           | K.CV remains the authoritative design-level causal/value-determinism premise. K.CVPred supplies exact-prefix Avail_pi, predecessor-before-use closure, evaluator applicability, typed historical ObsKey occurrence/version identity, finite ResetSupport_CV with K.CVResetSupportSound, and CVP-M persistence. CompareScopeFootprint includes those supports. The predecessor relation prec_CV is directly WellFounded; a natural CVRank may be an implementation convenience but is not a normative premise. K.CompareFact-Mono invokes CVP-M rather than inferring historical fact persistence from DomEpoch alone.                                 |
| Core.CycleClosure / Core.FairPick / Core.FairCycleDovetail | K.2b.E constructs the frozen liberal continuation through occurrence-relative Core.ExecutionCarrier/CycleResultOn/CycleClosure/CycleChoiceExt successors. FairPick/FairCycleDovetail thread the same exact history through dependent choice. Equal bare states from a different history cannot supply a C2/C3 successor. B5 remains separate.                                                                                                                                                                                                                                                                                                         |
| K.InFlightQual / Q.2 / J.InFlightPathSound                 | Model-indexed generalized-drain qualification. K.InFlightQual binds Q.2 soundness, finite drain, occurrence-relative continuation/path soundness and record-level InFlightRelationPrecision. Every consumer separately binds InFlightProfileCover. Each negative binds an OccurrenceBound or checked UniformFiber query plus InFlightSearchStatus; ExactRelation negatives additionally use proof-carrying InFlightRelationComplete. SR3-G is the serial consumer; G1 CycleClosure can supply only IQ3, not the remaining fields.                                                                                                                     |

**LEMMA: K.2b - deterministic serial-source dominance with response-failure extraction**

Assume one exact non-waived A5L\_Sys\_K counterexample occurrence under K.SerialRoutePremises, the SDet artifact's K.PolicySProgress/K.CompleteS, and K.RespCov. SR3-G supplies the generalized-drain/profile/query qualification; K.SerialCandidateApplicability supplies environment plus reset-first/terminal-second applicability and the prospectively fixed domains. The E/P0/R/P/P-Frozen construction yields the common frozen Policy-L continuation; SF identifies each actual process-facing Policy-S serial frontier while staying inside SFControl through its canonical monitor trigger. Step 2 proves the actual blocker lies in K.SFOutputEnvelope and its class is in the conservative class set, then instantiates pre-SF CD3-C to obtain CD4 membership in the already-fixed Handoff. Step 3 consumes Response and yields a finite non-waived K.RespFail through reset-first cancelled-pending, terminal maximal-pending, or timeout. The theorem does not require a permanently non-advancing Policy-S frontier.

**COROLLARY: K.2c - the selected deterministic Policy-S execution suffices**

If the selected Policy-S execution satisfies A5-S, its K.CompleteS/K.RespCov/K.RespClean package plus the exact-artifact K.PolicySProgress and K.SerialRoutePremises/SerialCandidateApplicability rules out every liberal Dead\_K^W counterexample. In G1 this yields A5-L on Sys\_ref. In G2 it yields A5L\_Sys\_K and the separately qualified finite-prefix/time-stutter/modal K.Low.A5 transport yields source A5-L. A clean run cannot coexist with any concrete early DirectFail or post-handoff response failure produced by K.2b.

**STATUS: q→K paper architecture closed for G1 and qualified G2; Lean mechanization, reusable checker implementation, and concrete template/flow qualification outstanding**

The paper architecture includes the exact-history safety repairs, occurrence-relative generalized drain, explicit relation/search/profile coverage, the strengthened G2 time-stutter/ModalQual interfaces, exact-prefix K.CV provenance, PolicySProgress, universal SerialCandidateApplicability, selected DomResetScope, and prospective SFControl/Handoff/Response response extraction. These are paper-level interfaces and certificate/checker obligations; mechanization and concrete template/flow qualification remain implementation work.

**PRACTICAL WARNING: One run must be complete and flow-qualified**

For a terminating design, the selected run must reach the declared maximal terminal state and resolve every covered monitor. For a nonterminating design, a finite trace is only evidence; an invariant, sound model-checking result, recurrence/lasso proof, or equivalent argument must establish completeness and continued response cleanliness. In either case, the flow must also establish SDet, route applicability, total finite-bound response coverage, and certificate validity; a clean trace alone is not the theorem premise.

## 8.6 Alternate direct discharges of A5-L

| Route                                     | When it can discharge A5-L                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>K.Str-1 - acyclic dependency graph</b> | If the full static modeled-process dependency graph induced by the possible WaitOwners_E projections is acyclic, no reachable coarse WFG cycle can occur and therefore no K.Dead component can satisfy its connectivity clause.                                                                                                        |
| <b>K.Str-2 - ranked dependency</b>        | A well-founded rank strictly decreases along every realizable coarse WFG edge induced by WaitOwners_E. Therefore a WFG cycle would force an impossible descent, thereby excluding the internal cycle/connectivity required by K.Dead. This is a conservative sufficient route; K.Dead itself retains exact ReleaseOwnerSets_E closure. |



| Route                                        | When it can discharge A5-L                                                                                                                                                                                                                        |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>K.Str-3 - credit/token cycle breaking</b> | Every static dependency cycle contains an edge proved unrealizable at reachable cutpoints, often by occupancy, initial-token, or credit invariants.                                                                                               |
| <b>CF-3 / direct invariant</b>               | An inductive abstraction proves the prohibited Policy-L cutpoint unreachable. When exposed through the common certificate API, package the result as model-indexed K.DirectA5Proof with exact A5L_M and the applicable InFlightQual/ProfileCover. |
| <b>CF-4 / exploration</b>                    | Sound exhaustive or bounded model checking over the selected transition system. The certificate is model-indexed K.DirectA5Proof; an artifact over Sys_K requires explicit K.Low.A5 transport before a Sys_ref consumer may use it.               |
| <b>CF-5 / tool certificate</b>               | A tool-emitted certificate using knowledge of the scheduled control graph, storage, arbitration, joins, and automatic flush.                                                                                                                      |

These are alternate proofs of exactly the same A5-L property. They cannot rescue a design for which an admissible serial-shaped Policy-L prefix already reaches Dead\_K^W.

Route-dependent continuation, reset, termination, and direct-A5 typing. Alternate A5-L routes prove the same abstract source premise but no longer use an untyped catch-all: K.DirectA5Proof carries the exact model M, exact A5L\_M target, and applicable model-indexed InFlightQual/ProfileCover. A Sys\_K direct proof reaches a Sys\_ref consumer only through K.Low.A5. The Policy-S route separately uses occurrence-relative CycleResult/CycleClosure/FairPick, PolicySProgress, SR3-G and SerialCandidateApplicability. K.DomResetScope is explicitly selected, with Min/leastness/join and only the stated transfer rules. Early Reset/Terminal DirectFail is independent mapped K.RespFail evidence; nested Continue leads to the prospectively fixed SFControl/Handoff/Response package. A reset or terminal after handoff is handled on Response rather than making the earlier comparison domain retroactively invalid.

## 8.7 Standing assumptions A1-A11 in engineer terms

| Assumption | Plain-language role                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A1</b>  | The source and RTL satisfy the previously defined R-rules and B-rules at the selected boundaries.                                                                                                                                                                                                                                                                                                                                                                              |
| <b>A2</b>  | Finite internal epsilon depth / finite stutter when not externally stalled.                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>A3</b>  | Weak fairness: a continuously enabled action is eventually selected.                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>A4</b>  | Channel progress: persistent complementary requests eventually discharge full/empty/rendezvous blocking.                                                                                                                                                                                                                                                                                                                                                                       |
| <b>A5</b>  | For the user-facing Policy-S route, a complete bounded-response-clean selected serial run; K.1A itself consumes A5-L directly.                                                                                                                                                                                                                                                                                                                                                 |
| <b>A6</b>  | Point-to-point source ownership plus elaborated dependency projection: each source-level logical channel has one Push-owning process and one Pop-owning process; shared source resources are explicitly arbitrated. At an elaborated boundary a stateless coupler may occupy one side without becoming a process. The qualified elaboration supplies the ReleaseOwnerSets_E/WaitOwners_E process projection, including soundness/completeness and no spurious self-dependency. |
| <b>A7</b>  | Selected witness/equal-KObs correspondence and K-facing admissibility. A7 does not consume Definition B1. The selected source witness is fixed by the exact J.GWR/WC or Appendix-L package, and equal KObs comes from J.KObsConf; any independent B1 consumer must name its own alphabet and trigger.                                                                                                                                                                          |
| <b>A8</b>  | Barrier participation is well formed; barrier mismatches are outside the internal message-passing deadlock theorem.                                                                                                                                                                                                                                                                                                                                                            |



| Assumption | Plain-language role                                                                                                                                                                                                                                                                                                                      |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A9</b>  | Instance assumption referring back to the Core scalar endpoint/cardinality convention: at most one Push and one Pop commit per channel per cycle unless an explicit multi-lane/burst model is selected. It does not redefine the Core rule.                                                                                              |
| <b>A10</b> | Instance assumption referring back to the Core reset-empty buffered-channel convention: every selected buffered channel is logically empty at each applicable reset boundary unless preload is explicitly modeled. It does not create a competing reset-state definition.                                                                |
| <b>A11</b> | Ke: explicit normative WFG-inert epsilon condition. After the WFG-relevant request/attribution cutpoint is fixed, an epsilon-only normalization segment with no selected observable/request-changing event may not create, remove, redirect, or otherwise change the K-relevant frontier, enablement, release-support, or WFG structure. |

#### THEOREM: K.1 - user-facing deterministic Policy-S preservation

Combine the uniform K.1A premises with the selected Policy-S route. Bind the applicable generalized-drain qualifications in addition to K.SerialRoutePremises and A5-S. In G1, Sys\_K=Sys\_ref and the same valid K.InFlightQual serves the serial and source theorem. In G2, the selected Sys\_K run binds its own K.InFlightQual and K.Low.ModalQual binds a separate Sys\_ref K.InFlightQual plus the global lowering stuttering/root correspondence; K.Low.A5 transports the derived A5L\_Sys\_K result to the source A5-L used by K.1A. The universal RTL layer remains Core.QSync-Post/QSync-KCutCanonical plus J.QSyncCertInd/J.CertExist-QSync/K.CertCov-QSync and Core.KCutClosure\_post. The conclusion is still absence of the defined non-waived reachable internal message-passing Dead\_K cutpoint class, not “RTL never hangs.”  $q \rightarrow K$  is paper-closed; concrete Q.2/path/lowering qualifications and mechanized checkers remain required.

## 8.8 What Appendix K does not cover

- Barrier mismatches or conditional SyncChannel participation remain outside the literal internal message-passing Dead\_K theorem and are handled by A8/separate protocol verification. The Stage-2 SyncCarrier/SyncFire machinery exists only so a well-formed synchronization obstruction can be represented faithfully in in-flight/lowering correspondence; it does not add synchronization to MsgOf or the WFG.
- Waiting indefinitely for an external environment input, output readiness, starvation/perpetual under-supply, termination, EOF, cancellation, or reset behavior not represented by the defined closed internal WFG/Dead\_K predicate. The persistent-deadlock conclusion is deliberately limited to that internal wait-cycle class.
- Independent nonterminal withdrawal, cancellation, retry, or reattribution of an already-issued blocking request; Definition Core.IC excludes those lifecycles from Sys\_ref unless a separate operational/correspondence/progress extension is defined, and the RTL side must separately demonstrate Core.IC conformance.
- Cycle-accurate RTL response bounds; K.Resp evidence is a source-side discharge of A5-L, not an RTL timing contract.
- Designs whose correctness requires favorable eager/same-cycle issue when a serial-shaped admissible prefix deadlocks.

## 9. Appendix L - witness selection and snooping

### PURPOSE

**Conditionally realize a WC-compatible source witness for latency-sensitive epsilon choices while leaving the unique free-running RTL behavior untouched.**

**Engineer takeaway:** Snooping selects one admissible source witness only when the required WC/L.Dir witness exists. It must not feed pre-HLS readiness back into RTL, relabel payloads, drop/reorder observations, or treat an unrealizable witness as a silent stall.

### 9.1 Why snooping is needed

A non-blocking arbiter, interrupt controller, or retry/timeout block may make a later observable choice based on which request arrived first or how many polls failed. HLS changes internal latency, so the raw pre-HLS and post-HLS simulations can select different epsilon outcomes even though both satisfy the broad scheduling contract. When those outcomes affect later Sigma-visible behavior, Appendix I needs a specific WC-compatible witness.

If NB-CF holds - failed non-blocking outcomes do not affect later observable control, values, frontier, or request attribution - the witness is trivial and snooping is unnecessary. Cadence-sensitive polling is different: it is a localized latency-sensitive protocol and must be isolated or explicitly snoop-aligned. Appendix L is conditional: if no admissible source execution can consume the RTL event sequence in ordinal order with matching kind and payload, WC/L.Dir is not discharged for that comparison.

#### DEFINITION: L.1 - witness selector / snooping

At each relevant epsilon-quiescent source witness-choice cutpoint, the selector returns the outcome chosen by the matched RTL prefix using only causally available history. This is intentionally a generic quiescent witness-selection interface, not a Core.KCut requirement: KCut adds a K-facing boundary-evaluation condition that an arbitrary snooped epsilon choice need not satisfy. The selector cannot depend on future RTL events or invent a retry/timeout outcome inconsistent with the isolated protocol.

#### LEMMA: L.2 - conditional snooping discharge of WC

If the snooped set covers every epsilon choice that can affect later observable behavior, the wrapper changes only epsilon-choice selection, selected outcomes are causal and RTL-consistent, successful and failed non-blocking calls retain the required Sigma/epsilon interpretation, cadence-sensitive blocks are isolated, and the selected ordinal event stream is realizable, then the selected source execution satisfies WC for Appendices I and J. This is a conditional witness-construction theorem, not a guarantee that every RTL event stream has a matching source witness at the chosen boundary.

### 9.2 One-directional architecture

#### CONDITION: L.Dir - one-directional alignment / RTL-side non-interference

The RTL\_B execution is free-running under the fixed stimulus. The pre-HLS snoop sequence must consume the recorded RTL snoop sequence in ordinal order, with matching kind and payload, and no pre-HLS backpressure/readiness may stall or alter RTL. Benign cycle skew is permitted. Order, kind, or payload divergence means that the required WC witness does not exist at the chosen boundary; the run has no theorem-backed guarantee and the harness must abort and triage the failure rather than silently gating or hanging.

## 9.3 Wrapper realization requirements W1-W5

| Requirement                              | Engineering meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>W1 - recorded ordinal observation</b> | Capture each RTL snooped event by dynamic ordinal and hold it until the pre-HLS side consumes the same ordinal. Do not rely on overlapping live assertion windows.                                                                                                                                                                                                                                                                                                                                                           |
| <b>W2 - gate only commit enable</b>      | Gate only the mirror signal that enables the pre-HLS commit: observed valid for a pre-HLS Pop or observed ready for a pre-HLS Push. Never modify the source endpoint-owned request.                                                                                                                                                                                                                                                                                                                                          |
| <b>W3 - registered sampling</b>          | Register the RTL observation before it enters the source-side gate to avoid delta-cycle races; the extra source latency is part of witness selection.                                                                                                                                                                                                                                                                                                                                                                        |
| <b>W4 - compare and watchdog</b>         | At each source snoop commit, compare ordinal, kind, and payload against the recorded RTL event. Never substitute the RTL payload into the source, because that would hide value divergence. Detect missing, extra, reordered, or never-consumed observations. A watchdog must convert non-existence or failure to realize the witness into a reported result; if the watchdog bound is not proved sufficient, a bound-exceeded abort is an inconclusive witness-realization failure, not by itself a proved L.Dir violation. |
| <b>W5 - independent coverage proof</b>   | The wrapper mechanism is sound only for the points it records. A separate analysis must prove that all epsilon choices capable of affecting later observables are covered.                                                                                                                                                                                                                                                                                                                                                   |

### IMPLEMENTATION PITFALL: Why a live AND is unsound

ANDing the current pre-HLS and current RTL handshake levels can drop an RTL event when the RTL assertion retires before the source arrives. It can also reorder or duplicate matching under multiple outstanding events. The normative rule is an AND against a recorded per-ordinal observation retained until source consumption.

## 9.4 Harness soundness and shared testbenches

### LEMMA: L.5 - online realization of the witness selector

A co-simulation harness realizes the abstract selector only when a matching witness exists, the harness preserves the free-running RTL trace, implements W1-W5, satisfies L.Dir, and produces the same source-side witness choices assumed by the formal comparison. The formal theorem is about the fixed RTL trace and selected source witness; harness correctness, coverage, and successful witness realization are additional implementation obligations.

A shared reactive testbench becomes problematic once one model can lag significantly, because reactions may be driven by whichever model the testbench observes. The flow should replicate the stimulus/testbench per model, or structure a shared environment around a common causal transaction stream rather than raw cycle timing.

## 9.5 What snooping means formally

Snooping does not use RTL to change the source specification. The source model is intentionally under-specified with respect to permitted latency and epsilon choice. Core.SelectLatitude supplies the ordinary finite L1/L3 non-maximal-selection component of that latency freedom; WC/Appendix L handles witness-sensitive epsilon choices whose outcomes affect later observable behavior. When a WC/L.Dir witness exists, the RTL observation selects one admissible execution of that specification for comparison. If the event order, kind, or payload cannot be realized, the comparison is outside theorem scope and must be triaged among source-model ill-formedness, incomplete snooping coverage/harness behavior, concrete source-carrier nonconformance to Core.SelectLatitude/WC, and tool/RTL non-compliance.

## 9.6 Latency-robustness stress testing

Appendix L also recommends randomized stalls, varied channel latency/capacity, and adversarial weakly fair scheduling in the pre-HLS model. This is not a substitute for the formal premises, but it is a practical way to detect designs that accidentally rely on incidental ordering of causally independent events. A failure under such stress usually indicates that the source specification itself is outside the intended latency-insensitive model or that an explicit snoop/isolation boundary is missing.

## 10. How the appendices work together in a verification flow

### PURPOSE

**Translate the theorem architecture into a concrete signoff workflow.**

**Engineer takeaway:** Safety-only use stops after the J.1/J.1-Safe package. Appendix K is added only when the claim includes the defined internal wait-cycle freedom, and it requires substantially more global evidence.

### 10.1 Step-by-step safety flow

| Step                                                     | Deliverable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Fix the theorem instance                              | First screen the design patterns against Appendix F of the scheduling-rules document. Then select one single-global-clock theorem instance: Sys_ref = Sys_B or Sys_B^elab, explicit boundaries and capacities, and, where elaborated, the Core.ElabProj package ( $\Pi_{\text{elab}}$ , $\pi_{\Sigma}$ , E, $\Pi_{\text{token}}$ , E, signed A_E plus the declared handshake/dependency interface); observable alphabet; reset interpretation; fixed stimulus/environment; witness provenance; tool/RTL identities; and any mapped-memory logical reference/boundary/profile. Multi-clock designs require a separate semantic layer or per-domain verified boundaries.                                                                                                                                                        |
| 2. Establish Core/G and select the safety-coverage route | Check the common Core/G semantics: Commit/Issue/BoundaryReq/Core.IC; R/E rules; source-side Core.QSync on the normal J.LocalGWR/J.GWR-Comp route (or equivalent checked per-Horizon settling for direct J.GWR); boundaries/capacities/elaboration; reset/memory qualifications; and current consumed-interface identities. Then select the universal safety route. Standard QSync additionally binds RTL ExecutionCarrier/reset provenance, RTLFrameEpochReach and QualifiedSafetyCarrierModel_RTL_B/QS2a-D, plus Core.RTLPartialCycleClosure only for the all-finite-prefix entered-cycle completion step. Alternate M.7c instead binds B1 at outer $\Sigma_{\text{DUT}}$ , one Core.B1RunMax run, and any triggered RSC0-RSC3 / QPC1-QPC3; it does not require the RTL carrier/QSync induction machinery merely for safety. |
| 3. Discharge WC                                          | Use NB-CF, an Appendix L snooping/L.2 construction with a realized WC/L.Dir witness, direct witness construction, or a complete mixed per-site coverage argument.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4. Produce local evidence                                | Generate J.PCert from Appendix I and per-channel/atomic/environment/reset certificates plus typed footprints and local dependencies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5. Run J.GWR-Comp                                        | Generate actual ConstructionAction occurrences with PhaseRole_A5, D1-D4/D5 dependencies, Q.1(c-order)/ProofUnitConstructionSupport reflection and validated action order. Derive SupportInsertionExists. Use SourceBridgeClosure/Ext for any required later source cycle arrival; SelectLatitude is not the bridge-existence theorem. For a single selected J.1 package, run J.GWR-Comp on the extendible $\tau_{\text{ref}}^{\wedge}G$ branch (or validate direct J.GWR). GWR-CompExt/LocalGWRExtCert are needed only when the selected coverage route actually performs incremental cross-prefix construction, such as standard QSync induction; M.7c's singleton route does not require them.                                                                                                                              |
| 6. Apply Proposition J.0 and plain Theorem J.1           | Use the accepted J.GWR package to instantiate the B1-free pairwise Post $\rightarrow$ Ref theorem. Do not add B1 simply because a later coverage route may consume it. On the normal route the source package carries source-side Core.QSync; the direct route carries equivalent checked settling. K-facing close-view/Offer/KObs/drain facts remain separate.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Step                                        | Deliverable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7. Bind the certificate-backed safety claim | When the selected J.1, J.TraceCertCov, or safety-corollary evidence is generated or manually assembled as a certificate package, bind it through K.CertHdr/K.CertPkg. Record the theorem/document/schema version, source/Sys_ref/RTL artifact identities, selected elaboration/boundaries/capacities, fixed stimulus/environment/reset interpretation, witness and WC provenance, the authoritative consumed-interface manifest, and any incompatible-schema revalidation or K.InterfaceSchemaMigration status. Include only subordinate evidence whose M.7a triggers fire: a safety-only package does not fabricate K-only waiver, KObs, A5, response, lowering, or liveness fields. K.CertPkg binds checked evidence to the exact theorem instance and trust boundary; packaging does not itself create or discharge a theorem premise.                                                                                                         |
| 8. Transfer the safety property             | Use J.TraceCertCov and a safety predicate extensional on J.HCanon/CanonHist and satisfying J.IdealPrefixClosed. Standard QSync: derive cycle-aligned TraceCertCov from J.QSyncTraceInd, then use Core.RTLPartialCycleClosure/J.CycleIdealCover when arbitrary finite prefixes may end inside an entered cycle; properties that fail ideal-prefix closure still require direct broader-domain J.TraceCertCov. Alternate B1-run: prove B1 at $\Sigma\_DUT$ , Core.B1RunMax, triggered J.B1OuterHistCongruence premises (RSC0-RSC3 and/or QPC1-QPC3 as applicable), and singleton J.TraceCertCov( $\{p\_post^0\}$ ); J.B1PostProjCover-RunMax + J.1-Safe-B1Run then cover all Reach_post. Evaluate $\Phi$ on the selected $p\_ref^0$ or a separately proved equal CanonHist, and bind checker extensionality when a noncanonical linearization is used. Maintain M.7a as the definitive discharge ledger; M.7b and M.7c are packaging overlays only. |

## 10.2 Additional liveness flow

| Step                                     | Deliverable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8. Establish the KCut observation domain | Establish the K-facing observation domain without merging it with the safety checkpoint. From the exact source $\tau\_ref^G$ form Core.L3CloseView $\tau\_ref^{cl}$ . On the standard normal-compositional source route, bind Core.KObsSettleQualified_Sys_ref(I) and use Core.KObsSettle-Ref to obtain the finite post-L3 observation fixed point represented by Core.KObsNormRef; alternate routes must validate the same close-view/observation interface directly. Apply K.NormStable NS0-NS6 and then Offer/KObs correspondence. Core.KObsSettleQualified is not a safety-profile premise. On RTL, QualifiedSynchronousCarrierModel_RTL_B adds QS3-QS5 to the safety foundation and Core.QSync-Post gives forward KCut closure; if starting from a queried reachable KCut, full QualifiedSynchronousModel_RTL_B adds QS2a-R/RTLKCutSettlingWindow for selected-witness recovery. |
| 9. Build the KObs bridge                 | Build the KObs bridge and current reconstruction. Bind J.InFlightPathSound on exact occurrence chains, the applicable K.InFlightQual, and consumer-specific K.InFlightProfileCover. Every negative drain result must bind an accepted occurrence-bound or uniform-fiber query/status; ExactRelation negatives require K.InFlightRelationComplete.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Step                                                     | Deliverable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>10. Qualify the source witness</b>                    | Qualify the exact source branch through the four-layer RLAdmReach: L3CloseView, RegPhaseReach on tau_ref^G, NormStable on the closed branch, and DrainReach on tau_ref^K. DrainReach-Comp uses DR1-DR4; no DR5 is added.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>11. Certify the cutpoint</b>                          | Assemble the route-neutral K.CertifiedCutpoint correspondence package. Do not insert downstream consumer-specific ProfileCover/IQ5 search evidence into the cutpoint merely because K.1A/K.1 will later need it; bind that evidence at the consuming theorem/application under the same instance identity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>12. Run and qualify the A5-L discharge</b>            | For the primary A5-L route, fix the SDet artifact, establish K.PolicySProgress and K.CompleteS, K.CVPred exact-prefix/ResetSupport_CV provenance, SR3-G, and universal K.SerialCandidateApplicability. For each nested Continue candidate select valid S_dom and prospectively fix SFControl/Handoff/Response plus class coverage/CD3-C. Run SF, derive CD4 by instantiation, then run Response reset-first/terminal/timeout. In G2 additionally bind TimeStutterQual, ModalQual MQ2/MQ5, reset/negative evidence transport and K.Low.A5.                                                                                                                                                                                                                                                                                                                             |
| <b>13. Establish universal KCut closure and coverage</b> | Apply K.1A directly to one selected certified KCut. For K.1A-Total or universal K.1, consume Core.KCutClosure_post and K.CertCov. In the preferred standard QSync route, QSync-Post/QSync-KCutCanonical derive the complete canonical KCut domain, while J.QSyncCertInd proves the full per-cutpoint correspondence package by base/one-cycle/exhaustive-successor induction. On the normal route CI1 preserves applicability of the already-bound DR1-DR4 evidence, reconstructs the successor J/QSync facts, and re-runs K.DrainReach-Comp to supply successor QC5; it does not assume an arbitrary successor DrainReach package. J.CertExist-QSync then gives J.QSyncCertCov and K.CertCov-QSync derives K.CertCov. Alternate independently checked K.CertCov routes remain permitted. A single observed run or sampled KCut set is not package-totality evidence. |

### 10.3 Recommended reading order in the scheduling-rules document

- Start with the front Abstract and Formal Guarantees at a Glance, then read Appendix F for the architect-facing design constraints. Use Appendix M for the formal conclusions, scope, and division of responsibilities. Continue with the unlettered Guide to the Formal Appendices, the Normative Formal Core for the shared semantic skeleton, and Appendix G for the R/E observable contract.
- In the Normative Formal Core, focus on Sys\_ref; Commit/Issue/BoundaryReq/Core.IC; ReachPrep/RegSeq/SelectLatitude/PolicyL/PolicyS; Core.ExecutionCarrier and ResetEpochKey/ResetEpochAdvance; occurrence-relative CycleResultOn/CycleClosure/CycleChoiceExt; source Settle plus Core.L3CloseView; RTLFrameEpochReachOn/RTLFrameEpochReach, RTLKOriginFrame/RTLDesignatedOriginFrame/RTLKCutSettlingWindow, the safety-carrier/synchronous-carrier/full QSync hierarchy and its single Core.QualifiedSynchronousCarrierModel-SafetyProjection lemma, Core.RTLPartialCycleClosure with theorem-derived Core.RTLPartialCycleExt, RTLCycleCarrier/KOriginCarrier; FairPick/FairCycleDovetail; SyncFence, memory and ElabProj/WFG/release-support semantics; Core.16/Drain/SettleKBridge; generalized in-flight Core; Core.17 SigmaMatch; and Core.18 KObs. Read the interfaces in dependency order: exact history/reset provenance before designation/carrier; safety qualification before optional post-settling/full extensions; partial-cycle completion separately from QSync; and L3CloseView only as a K-facing sibling of an extendible safety construction checkpoint.
- Read Appendix I to understand exactly what is local and why deferred non-blocking holes are needed.



- Read Appendix J as the central safety construction. Start with LocalGWR, ConstructionAction/PhaseRole\_A5, J.ProofUnitConstructionSupport, the D1-D4/D5 distinction, COrdActPred, dependency soundness/completeness/rank, SourceBridgeClosure/Ext, RegPhaseNF, GWR-Comp, J.0, and plain B1-free J.1. Then choose the coverage layer: J.QSyncTraceInd/CycleIdealCover for the standard QSync route, or Core.B1ObsPrefix/Core.B1RunMax, Core.ResetScheduleCongruence, Q.OuterProjectionCongruence, J.B1OuterHistCongruence/J.B1PostProjCover-RunMax and J.1-Safe-B1Run for the alternate terminating B1-run route. Only after safety coverage is understood should a K-facing reader continue through L3CloseView/KObs normalization and Appendix K.
- Read Appendix L before K when witness-selected non-blocking or latency-sensitive interfaces are present.
- Read Appendix K for exact K-facing reconstruction and generalized drain first: DrainReach/RLAdmReach with L3CloseView/NormStable NS0-NS6; InFlightRelationComplete/InFlightRelationPrecision, InFlightSearchQuery/Status, InFlightProfileCover, J.InFlightPathSound and DeadStructCandidate/Cover. Then read A5-L and the Policy-S route: K.CVPred exact-prefix/ObsKey/ResetSupport\_CV/CVP-M and WellFounded(prec\_CV), K.PolicySProgress, SR3-G, K.SerialCandidateApplicability, explicit selected K.DomResetScope, K.SFOutputEnvelope, K.RespSerialClassExclusionSound/K.RespSerialBlockerClassSet, prospective K.PolicySCompareDomains, and K.2b SF/Q with post-SF CD4 and reset-first Response dispositions. For G2, read TimeStutterQual, ResetStutterMatch/ResetKey-Stutter, ModalQual MQ1-MQ6 including instance-wide MQ2 and paired MQ5 ProfileCover, InFlightNegativeEvidenceTransport, and K.Low.A5. Finally read K.InterfaceSchemaMigration/K.CertHdr/M.7 for artifact-version binding.

## 11. Assumptions and evidence matrix

### PURPOSE

**Summarize the named assumptions that most often determine whether a theorem instance is usable.**

**Engineer takeaway:** The important audit question is not merely “is the assumption plausible?” but “what artifact establishes it for this exact source, RTL, capacity/elaboration, stimulus, witness, reset, and environment instance?”

Architect-facing entry point. Appendix F remains the design-pattern screen, but the current evidence boundary is route-sensitive. The standard QSync safety route binds exact RTL ExecutionCarrier/reset/carrier provenance and cycle induction; the alternate M.7c route replaces those RTL-side coverage artifacts with B1, Core.B1RunMax and the triggered reset/elaboration congruence interfaces while retaining the same B1-free selected J.1 correspondence machinery. M.7a is definitive in both cases.

**How to read this matrix.** The entries below are not a checklist of new manual tasks for every architect or design. Many design- and implementation-conformance entries formalize conditions that were already explicit or implicit in HLS-generated and manual RTL flows; the formalism exposes them so a flow can determine applicability, assign ownership, and retain auditable evidence. The safety assurance layer also requires proof-construction, provenance, and coverage artifacts - such as global witness construction, exact prefix-extension evidence, carrier/origin provenance, and implementation-wide cycle-induction coverage - that conventional RTL flows do not ordinarily produce. The intended discharge architecture is reusable tool/library/template qualification, translation validation, or automatically generated and independently checked certificates where available; the corresponding mechanization and concrete flow qualification are not yet complete. Depending on the selected claim, an entry may be inapplicable, satisfied by a genuine theorem-scope design restriction, theorem-derived, discharged once through reusable qualification, or established automatically for the particular implementation. Appendix F remains authoritative for the design restrictions and alternate-route conditions. The deadlock/liveness claim adds further provenance, release-family, drain, lowering, and response evidence beyond the safety stack.

### Premise / evidence

### Where it matters and typical discharge

| Premise / evidence                         | Where it matters and typical discharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B1 - observable RTL projection determinism | Triggered only where a named theorem/profile explicitly consumes B1. Plain Theorem J.1, Corollary J.1, J.TraceCertCov, A7/equal-KObs, and the standard QSync safety chain do not require Definition B1 merely for pairwise Post -> Ref compatibility. Current normative consumers include the Appendix-G specialization and the alternate B1-run outer-coverage route. Discharge uniqueness of the selected Core.3 cycle-bucketed projection at the named alphabet; internal state/epsilon/carrier nondeterminism may remain. Evidence may be a reusable RTL semantics/refinement theorem uniformly quantified over a stimulus class plus lightweight per-instance binding, or an instance-specific observable-projection uniqueness proof. Historical Appendix-K “B1-avail” is a different StandardEnv availability convention.                                                                                                                                  |
| B1-RunMax / Core.B1RunMax                  | Triggered only by Corollary J.1-Safe-B1Run / M.7c. Bind one finite reachable cycle-closed RTL prefix $p\_post^0$ ending in modeled Core.DeclaredTerminal and prove BRM2: it contains every complete canonical selected $\Sigma\_DUT$ occurrence from every reachable RTL prefix. Under B1 + BRM1, BRM2 has the equivalent cycle-horizon form: for a nonzero selected run with final completed cycle $t\_f$ , no reachable complete selected occurrence may occur later than $t\_f$ ; the zero-cycle case requires no reachable selected occurrence. A simulation trace alone does not discharge this universal fact unless backed by a completeness/translation/reachability theorem.                                                                                                                                                                                                                                                                             |
| B2 - weak fairness                         | Consumed only where a complete/fair execution or eventual boundary selection is required, notably later liveness/fair-continuation reasoning. B2 is weak fairness: a boundary action that remains continuously ChannelEnabled from some point onward must eventually be selected; merely intermittent enablement does not satisfy the antecedent. It is not a premise of either finite-prefix direction of Theorem J.1 and is not the source of ordinary finite deferral; Core.SelectLatitude supplies that legality. Discharge ownership is architectural rather than categorically “environmental”: it may lie with source/scheduler semantics, HLS-generated or hand-written RTL resource/memory-port/interface/channel arbitration/back-pressure, channel-library behavior, external masters, or reusable qualified arbiter classes bound to the concrete instance. Evidence: scheduler/arbiter fairness proof, assertions, or Appendix N/M.7a qualification. |
| B3 - channel progress                      | Consumed by I.2 and K. For Push at a full buffered channel, B3 guarantees a complementary Pop discharge that frees space; for Pop at empty, it guarantees a complementary Push discharge that supplies data. This is not by itself a conclusion that the original blocked operation commits. If that request remains pending and continuously enabled afterward, E4 plus B2 govern its eventual commit. Evidence may be a per-boundary progress certificate or the stated persistent two-endpoint-request premises.                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Premise / evidence                                                            | Where it matters and typical discharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>B4 - finite epsilon-step depth</b>                                         | <p>Triggered wherever Lemma I.1, finite replay/preparation normalization, B5 charging, or the generic source-cycle finiteness argument relies on FPD/B4. D_P (or an equivalent well-founded measure) bounds consecutive P-owned epsilon-work while P remains internally advancing toward its next observable action and is not externally stalled; stable waiting, peer/environment delay and back-pressure are outside the charge. A certified Appendix-I preparation segment may cite its local certificate. Generic Core.CycleClosure use requires the corresponding uncapped per-cycle source-control finiteness evidence for its source-cycle-existence clause. Evidence: finite source-control/pipeline/FSM-depth proof, control-graph ranking, simulator-semantics theorem, or tool-generated progress certificate. B4 is not the QSync same-cycle settle theorem. Packaging: M.7b bundle 2 owns the replay/preparation-normalization use; bundle 3 owns the B5-charging and generic Core.CycleClosure source-cycle-existence uses. This split is packaging only: M.7a retains one B4/FPD trigger and discharge, and each triggered use references that same authoritative ledger evidence. In M.7c, B4/FPD is carried only if the selected singleton replay/preparation or quiescence construction actually triggers the same M.7a obligation; the alternate coverage route does not trigger B4 merely because it uses B1/B1RunMax.</p> |
| <b>B5 / B6-Terminal</b>                                                       | <p>B5 supplies bounded global stabilization where invoked. A B6 idle suffix is permitted only after Core.TerminalIdle proves both a genuine B5/global fixed point and permanent no-future-enabling machine/environment behavior (Core.EnvTerminal supplies only the environment component). A currently quiet nonterminal cycle is Core.SilentCycle, not B6 stutter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>A8/A9/A10 - barrier, endpoint-cardinality, reset-empty well-formedness</b> | <p>Triggered independently where applicable. A8 applies when SyncChannels/barriers are omitted from the internal Push/Pop WFG and requires matched barrier participation/no permanent bypass under the fixed stimulus. A9 applies to scalar per-cycle endpoint semantics and requires at most one Push and at most one Pop commit per channel per cycle unless an explicit multi-lane/burst model is used. A10 applies to buffered channels at initial/reset boundaries and requires zero logical occupancy at each selected reset boundary or an explicitly modeled preload. Typical evidence is source/elaboration checks, barrier-count/participation proof, endpoint-cardinality assertions, and reset-state/channel initialization checks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>WC</b>                                                                     | <p>Consumed by I and J; provenance may be NB-CF, L.2 snooping, direct witness construction, or mixed per-site coverage.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Core.RegSeq</b>                                                            | <p>Consumed by I/J/K. Evidence: sequential interface structure; request/data outputs are functions of registered state and operands fixed before L2; ordinary combinational derivation from those registers is allowed, but no same-cycle newly returned interface result may feed a dependent sequential output.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Premise / evidence                                           | Where it matters and typical discharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core.QSync / QSync-Ref / QSync-Post</b>                   | Source-side Core.QSync is required on the normal J.LocalGWR -> J.GWR-Comp -> finite-prefix J.1 route because each constructed Horizon needs a finite L2-settled source snapshot; direct J.GWR may instead carry independently checked per-Horizon settling. On RTL_B, QualifiedSafetyCarrierModel_RTL_B = QS1 + QS2 + QS2a-D belongs to the standard cycle/carrier safety route, not to M.7c merely for safety coverage. The standard all-finite-prefix route additionally consumes Core.RTLPartialCycleClosure; canonical post settling/KCut closure adds QS3-QS5; queried-KCut recovery adds QS2a-R. The alternate B1-run route still uses source-side QSync on the normal J route but does not consume the RTL carrier hierarchy unless another selected consumer independently triggers it.                                                                               |
| <b>Core.RTLPartialCycleClosure / Core.RTLPartialCycleExt</b> | Triggered only for the all-finite-prefix ideal-cover route when a reachable RTL prefix has already entered a real cycle and must be completed within that same exact-history occurrence. It is not a premise of J.QSyncTraceInd/J.1-Safe-QSyncCycle. The semantic qualification may be proved directly for the instance or reused from an independently checked RTL operational-semantics/translation class qualification, with the instance checker validating class/version identity and side conditions rather than repeating exhaustive state exploration. Core.RTLPartialCycleExt is theorem-derived from Closure and is not a second evidence obligation.                                                                                                                                                                                                               |
| <b>Core.KObsSettleQualified / Core.KObsSettle-Ref</b>        | Triggered only when a Corollary J.1 / Appendix K source K-facing package obtains sigma_ref from an extendible J.GWR checkpoint by first forming Core.L3CloseView and then Core.KObsNormRef. Bind the exact source model/elaboration and discharge KSQ1-KSQ4: valid post-L3 boundary equations/topology, observation-only phase legality, finite deterministic acyclic complete settling, and no feedback into the closed L3 action selection/E4 decision. Core.KObsSettle-Ref then supplies the finite observation fixed point on the standard normal-compositional route; an alternate route must validate the same close-view/observation interface directly. This is not a premise of Theorem J.1, J.QSyncTraceInd, J.TraceCertCov-QSyncCycle, J.CycleIdealCover, or standard safety-only M.7b.                                                                            |
| <b>J.QSyncTraceInd / J.QSyncCertInd</b>                      | QSyncTraceInd is the safety-only base/one-cycle/exhaustive-successor certificate over CycleReach_post and carries only the exact LocalGWR/J.GWR-Comp + J.HCanon state needed by safety. QSyncCertInd is the richer K-facing induction carrying OfferReach/OfferConf/KObsConf, theorem-derived RegPhaseReach, and QC5 K.DrainReach. On the standard normal route, CI1 preserves the bound DR1-DR4 K.DrainReach-Comp applicability, re-establishes the successor attribution/history/QSync settling facts, and re-applies the lemma to obtain QC5; no separate DR5 history certificate or arbitrary successor DrainReach assumption is required. J.QSyncCertInd-Trace projects the richer induction to the safety form. For either induction, one observed simulation path is insufficient unless independent determinism/exhaustiveness collapses the reachable successor set. |
| <b>J.LocalGWR</b>                                            | Consumed by the normal J.GWR-Comp route. Evidence: complete local process/channel/atomic/env/reset records, typed footprints, dependencies, agreement, and coverage. The optional J.GWR-Direct route may replace the local construction for a selected prefix only with a directly validated J.GWR package and J.ValidatedConstructionOrder satisfying all V1-V6 obligations, including the phase-normal V6 condition; the direct route is not a shortcut around those checks.                                                                                                                                                                                                                                                                                                                                                                                                |

| Premise / evidence                                           | Where it matters and typical discharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>J.LocalGWRExtCert / J.GWR-CompExt</b>                     | Required whenever one selected extendible J.GWR source witness is carried across a larger RTL prefix and exact source-prefix nesting is needed, including every T11/CI1 safety-stage carrier edge in the standard QSync induction route. Bind the exact old LocalGWR package and selected old W/tau_ref^G to the new package through LocalPrefixAgree/LocalGWRExtCert, including the no-retcon frontier, stable old actions/dependencies/ranks/bridge choices, and legal suffix-only appendability. J.GWR-CompExt then reuses the old tau_ref^G chain literally and appends only permitted later work. It is not required for an isolated one-prefix J.1 application with no old/new witness relation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| B-faithfulness / Core.ElabProj / Q.OuterProjectionCongruence | Consumed by J/K at every explicit abstract boundary. Evidence: back-annotation, structural RTL checks, interface monitors, refinement checks, or tool certificates. Elaborated instances additionally need the coordinated Core.ElabProj package: $\Pi_{\text{elab}}$ channel fibers, outer trace $\pi_{\Sigma,E}$ , signed state accounting $A_E$ , finite $\text{ElabToken}(E)/\Pi_{\text{token},E}$ projection and functional transfer attribution, the complete stateless handshake equations, and proof-internal process-dependency visibility. Q.1(b) checks $A_E$ against independently reconstructed outer Sys_B history at $\epsilon$ -quiescent cycle boundaries; the resulting $0 \leq A_E \leq B(c)$ range is derived from outer E4 rather than a separate qualification item. Q.1(j) requires the settled handshake and state-instantiated ReleaseOwnerSets family to be reconstructible from E plus the allowed reconstructed roots and requests, with the same schema/signals validated by J.KObsConf against the implementation. Q.1(k) separately requires exact family-level support soundness/no-spurious-rescue and completeness/no-missed-rescue relative to the declared template semantics before WaitOwners_E is formed. A local coupler is not a process vertex or residual offer. For M.7c, an elaborated/multi-effect or erased-path outer quotient additionally carries Q.OuterProjectionCongruence QPC1-QPC3: common-group restriction, old-old contracted-order restriction, and predecessor-domain closure. Ordinary singleton projection is vacuous; qualified templates should prove this once and bind it to each E instance. |
| RW1-RW5 / Core.ResetScheduleCongruence                       | RW1-RW5 apply whenever dynamic reset is in the compared executions. For M.7c, if dynamic RESET can affect retained outer identity/grouping or a contracted required-predecessor path, additionally bind Core.ResetScheduleCongruence. RSC0/ORS1 proves Core.OuterResetSupport conservativeness; RSC1 compares supported completed-cycle reset signatures, RSC2 the partial-current-cycle frontier through already-realized groups, and RSC3 reset-owned predecessor-domain closure. No-dynamic-reset makes the cross-run condition trivial. Taking the complete theorem-instance reset-scope family is a conservative cheap RSC0 default but enlarges the RSC1-RSC3 comparison domain; narrowing support needs a checked omitted-reset irrelevance proof.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Premise / evidence                                                  | Where it matters and typical discharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q.1(l) handshake support / CouplerPersistence</b>                | Triggered when the K.2b/Policy-S serial route consumes a coupling-sensitive boundary action. The template identifies finite earlier channel/history support plus persistent source-attributed request roots sufficient to determine the target ChannelEnabled result, and proves CouplerPersistence over the fixed-DomEpoch base frozen continuation: every permitted whole action before the target commits preserves that support or directly preserves ChannelEnabled at the next applicable settled opportunity. Q.1(d)-(k) well-formedness does not by itself establish this for mutual-stall, shared-capacity, or arbitrated/non-monotone handshake topologies. If the selected template cannot supply the Q.1(l) proof, the candidate is outside K.PSF/CF-S and requires another A5-L route. Evidence may be a reusable template invariant/proof, such as the sole-producer join persistence argument, bound to the concrete E instance. |
| <b>J.TraceCertCov</b>                                               | Consumed by J.1-Safe over the selected RTL coverage domain. Coverage is route-sensitive. Standard QSync derives TraceCertCov on CycleReach_post from J.QSyncTraceInd and extends ideal-prefix-closed safety to arbitrary finite prefixes through CycleIdealCover. M.7c needs TraceCertCov only on the singleton {p_post^0}; B1 + B1RunMax + J.B1OuterHistCongruence provide universal RTL outer-history coverage instead. Plain TraceCertCov itself does not carry B1 merely as part of a route-complete J.1 package. Properties outside J.IdealPrefixClosed still need direct coverage at the required domain.                                                                                                                                                                                                                                                                                                                                 |
| <b>Core.ReachPrep / Core.16 / K.Drain / Core.SettleKBridge / Kε</b> | ReachPrep and Core.16/K.Drain qualify the actual source execution: once a fully disabled frontier settles, no source-later replenishing work may launch/Issue and only finite already-admitted work may progress. K.DrainReach/Core.SettleKBridge connect that reached activation to the selected K-facing cutpoint, with Kε covering only WFG-inert normalization. This is not the generalized modal drain theorem. Generalized DrainClosedNow is separately computed from FlightStateRec/InFlightStepRec and requires K.InFlightQual/J.InFlightPathSound. On the normal J.GWR-Comp route, DR1-DR4 and K.DrainReach-Comp supply the path-indexed reached-prefix facts; Q.2/InFlightQual supply the future already-admitted path abstraction.                                                                                                                                                                                                   |
| <b>J.OfferReach / J.OfferConf / J.KObsConf</b>                      | Consumed by Corollary J.1 and all K cutpoint reasoning. Evidence must bind the exact same source/RTL prefix, witness, E/w/H/O, and NormRef/NormPost endpoints. Core.18 fixes the common typed record/equality interface; J.KObsConf is the theorem-level proof that J's KObsPost_J representation equals the realized source KObs and that the bound E handshake/release-support schema reconstructs the same MirrorAvail, ReleaseOwnerSets_E/WaitOwners_E, and WFG facts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Premise / evidence                                          | Where it matters and typical discharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>K.CertCov</b>                                            | Consumed by K.1A-Total and universal K.1 together with Core.KCutClosure_post. In the preferred standard QSync route, QualifiedSynchronousCarrierModel_RTL_B/Core.QSync-Post derives closure from the safety foundation plus QS3-QS5, while full QualifiedSynchronousModel_RTL_B adds QS2a-R/Core.RTLKCtSettlingWindow so Core.KOriginCarrier-KCut can recover the carrier position of every queried reachable KCut. J.QSyncCertInd then proves the full per-cutpoint package by base/one-cycle/exhaustive-successor induction, J.CertExist-QSync gives J.QSyncCertCov, and K.CertCov-QSync derives K.CertCov. The normal route first establishes the independent DR1-DR4 evidence once for the bound instance; CI1 then re-applies K.DrainReach-Comp at each exact successor rather than requiring unrelated per-cutpoint drain certificates. Alternate total-extraction/refinement/exhaustive routes remain valid. One CF-0 package, sampled KCuts, or a clean simulation is insufficient.                                                           |
| <b>A5-L</b>                                                 | Consumed directly by K.1A. Evidence: Policy-S route through K.2c or a valid structural/direct/invariant/exploration/tool proof.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Selected Policy-S run / SDet / A5-S</b>                  | Consumed by K.2b/K.2c/K.1. SDet fixes one functional artifact but does not establish progress; K.PolicySProgress separately binds IssueFair_{Policy S} and applicable B2/B3 with the selected reset/terminal semantics, and K.CompleteS packages maximal/completeness evidence. K.SerialRoutePremises/SR3-G and universal SerialCandidateApplicability are independent applicability inputs. A5-S supplies K.CompleteS, total K.RespCov and K.RespClean for the same artifact. In G2, K.Low.A5 transports the resulting A5L_Sys_K only after TimeStutter/ModalQual qualification.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>K.EnvMF</b>                                              | Consumed by the serial route. Under StandardEnv/B1-available, candidates satisfying all K.JointFreeze-Ord ordinary/single-release-support premises use that theorem; every other admitted candidate - including a single frontier with a non-singleton or multi-alternative ReleaseOwnerSets_E family and every explicit multi-frontier - needs K.JointFreeze JF1/JF2. Under a timed/reactive or otherwise nondefault environment, every relevant process must have exactly the selected internal frontier and exactly one singleton release support together with the remaining K.JointFreeze-Ord premises; environment-facing co-frontiers, second internal frontier items, and non-singleton/multi-alternative release supports are outside the serial route. K.EnvMF-Uniform is the candidate-independent sufficient discharge when these structural conditions hold at every reachable K-facing cutpoint. In the common ordinary primitive case Core.OrdFrontSingleton, Dead_K membership, and Core.WFG's {{Q}} fact supply the structural part. |
| <b>W.Env - environment-rescue / response-monitor waiver</b> | Optional and exact-scope only. A KObs-side waiver records the specific component class/rescuing item, environment contract, and KObs-closure justification. A K.Resp waiver records the exact dynamic frontier class, the permitted timeout/reset-interruption reason, and the coverage-preservation proof showing that every possible Policy-S serial-blocker response class for the non-waived candidate lies outside the waived response set. Broad end-of-test or reset waivers are insufficient. Waivers are removed explicitly from Dead_K^W/diagnostic variants and K.RespFail^W and propagate through the affected K conclusions.                                                                                                                                                                                                                                                                                                                                                                                                             |



| Premise / evidence                                                                 | Where it matters and typical discharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>K.CV / K.OpKey / K.CVPred / K.CV-WF</b>                                         | Authoritative serial provenance. K.CVPred is exact-prefix: Avail_pi, predecessor-before-use, evaluator applicability, typed ObsKey occurrence/version identity, finite ResetSupport_CV/K.CVResetSupportSound and CVP-M persistence. CompareScopeFootprint includes consumed supports. The predecessor relation prec_CV is directly WellFounded; a numeric rank is optional certificate implementation data rather than a theorem premise.                                                                                                                                                                                                                                                                                                                     |
| <b>NB-Align / K.SerialRoutePremises</b>                                            | NB-Align remains separate from K.CV. SerialRoutePremises now includes SR3-G generalized-drain qualification/ProfileCover/query evidence over the exact counterexample occurrence domain and universal SerialCandidateApplicability. On nested Continue it binds an explicit selected valid DomResetScope and prospectively fixed PolicySCompareDomains package through the applicability record. Actual SF blocker/Handoff membership and K.RespFail remain theorem conclusions, not premises.                                                                                                                                                                                                                                                                |
| <b>K.DomResetScope / K.InterruptionGate / K.ResetEpoch / K.TerminalDisposition</b> | Consumed by the Policy-S route. K.DomResetScope is a predicate over an explicitly selected valid S_dom, with DomResetScopeMin/leastness/join and only the named transfer rules. Early Reset then Terminal produce independent DirectFail K.RespFail or Continue. On nested Continue, R-S/T-S constrain the already-fixed SFControl domain; post-handoff affecting reset/terminal are Response dispositions and do not justify retrospective scope/domain changes.                                                                                                                                                                                                                                                                                             |
| <b>L.Dir and W1-W5</b>                                                             | Consumed by the online snooping harness. Evidence: free-running RTL, ordinal recording/consumption, comparison/watchdog, noninterference, and complete choice-point coverage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>ClockScope - single global clock</b>                                            | Applies to the G-K theorem stack and directly discharges QS1 evidence. The selected theorem instance uses one global synchronous cycle relation; CDCs are excluded from that instance or represented through separately specified/verified crossing components. Appendix O is only an informative multi-clock sketch until a separate semantic layer is supplied.                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Core.SyncFence / R3/E5 completion fence</b>                                     | Core.SyncFence is constitutive synchronization-completion legality on Sys_ref and a separate RTL conformance obligation. Every blocking q in pref_S(s) has a designated process-facing commit no later than sync s. Same-cycle q/s completion is allowed; on RTL, ordering “before interval advance” is annotation/certificate accounting rather than physical sub-cycle timing. Typical evidence: R3/E5 trace checks, control-state assertions, boundary mapping, or formal refinement.                                                                                                                                                                                                                                                                      |
| <b>Core.MemSyncOrder</b>                                                           | Separate local scheduling/mapping qualification whenever an array/memory access is source-ordered with a selected synchronization call, whether storage is internal or external. Verify mapped storage commits before s no later than s and after s strictly later, including any cache/merge/split/reorder transformation evidence. A single merged storage commit may not represent accesses on opposite sides of the same synchronization boundary. It does not add memory events to $\Sigma$ or become an Appendix I/J.1 replay premise; J.Mem is separately required for qualifying cross-process shared-memory value flow, and Core.MemFaithful is additionally required when transformed physical traffic is hidden below the selected proof boundary. |
| <b>Core.KCutClosure_post / Core.KCutObserve</b>                                    | Consumed by K.1A-Total and universal K.1, separate from K.CertCov. In the standard user-facing scope, Core.KCutClosure_post is a theorem-derived interface: accepted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Premise / evidence                                                                                  | Where it matters and typical discharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                     | RTL-side Core.QSync evidence invokes Core.QSync-Post, giving a finite same-cycle normalization to a unique PostKCut before the next registered update. $K\epsilon$ plus Core.KCutObserve preserves any persistent relevant request/frontier/enablement configuration into that KCut. The former arbitrary invariant/checked-normalization routes are not independent standard-scope discharges; a non-QSync model needs a separately qualified extension.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Core.9/Core.10/Core.IC - Issue/BoundaryReq lifecycle conformance</b>                             | Core.9 makes Issue(q) the first cycle in which the calling process begins requesting q at the selected endpoint; an annotation may carry but may not choose that cycle. Core.10 makes BoundaryReq a fixed-mapping, operation- and boundary-specific process-owned request; raw valid/ready levels representing complementary channel/storage availability or coupler logic are not BoundaryReq merely because they are high. Core.IC is constitutive Sys_ref transition legality. On RTL_B, establish bidirectionally complete owner decoding; live-owner uniqueness; stable blocking request/payload persistence from Issue until process-facing Commit or affecting RW2 RESET; and back-to-back ownership: if q0 commits at t and the same process-owned request continues at t+1 as the next same-direction blocking q1, q1 owns it and Issue(q1)=t+1, with no ownerless interval. Genuine delay is only before q1 BoundaryReq begins. Evidence may be interface assertions/monitors, schedule/control-state evidence, request-attribution certificates, or formal refinement. K.BF consumes this persistence but does not own the conformance obligation. |
| <b>K.BF - blocking-only serial fragment</b>                                                         | Consumed by K.2b/K.2c/K.PSF/CF-S on the bounded-response serial route. In addition to citing established Core.IC/RW2 persistence, show every WFG-relevant wait is an ordinary blocking Push/Pop on a finite B-faithful point-to-point boundary, local computation cannot diverge before the process-facing frontier, and shared arbitration is explicit and deterministic or witness-fixed. This is a K-route applicability condition, not the definition of request persistence itself.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Core.CycleState / Core.CycleResult / Core.CycleClosure / Core.CycleChoiceExt / Core.FairPick</b> | Required by complete/fair source continuations, not finite-prefix J.1. Bind exact Core.ExecutionCarrier/CycleResultOn occurrences so both CycleChoiceExt completions and FairCycleDovetail extend the same history; do not splice a successor from another equal state. Policy-S use additionally binds K.PolicySProgress. B5 remains separate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Core.SelectLatitude</b>                                                                          | Finite-prefix source-model qualification used by Post -> Ref whenever an otherwise legal L1 issue or L3 boundary action may be postponed. The abstract Sys_ref discharges it definitionally; a concrete executable source/reference model used as the theorem carrier must prove it does not impose hidden maximal progress that excludes the accepted J construction. Nonselection is legal under both Policy L and Policy S, remains subordinate to Core.16 and all other legality, and is separate from Core.IssueFair/B2. The M.7a entry also warns that J.RefProjCover needs latency coverage only when different legal delays can change canonical $\Sigma_{DUT}$ history.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Core.MemFaithful</b>                                                                             | Required when mapped-memory channels/signals or transformed physical memory activity are hidden below the selected J/K boundary. Bind the declared untransformed logical mapping-layer semantics as reference and prove the transformed realization preserves the selected Core.17/E1-E5 event identities, values, required order, and atomicity. Selected message/signal events cannot be selectively hidden. For K, hidden realization must also preserve the represented process-facing blocking/frontier/enablement/release-support/WFG facts; persistent backpressure therefore needs a K-faithful logical boundary or is outside the transformed K scope.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Premise / evidence                                                                               | Where it matters and typical discharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>J.Mem</b>                                                                                     | Required for every shared-memory dependency through which one process's write can affect another process's control flow, payload/value expressions, or the identity/order of later operations. Discharge at dependency level, not per physical RAM access, but prove completeness: every qualifying path is lowered to covered proof-visible operations (including a qualified logical memory boundary) or the separate memory-state extension, and no other qualifying unlowered memory-carried path remains.                                                                                                                                                                                                                                                                                                                               |
| <b>Core.MemProfile</b>                                                                           | Used for a reusable logical memory boundary above transaction-changing cache/merge/split/reorder behavior, including transformed cross-process memory and persistently backpressuring memories whose raw transaction stream cannot be the E4/Core.17 boundary. The profile binds the logical boundary/alphabet and transformation set; discharges Core.MemFaithful, J.Mem, Core.MemSyncOrder, and the K WFG-inert/K-faithful disposition; enforces A6 explicit arbitration/point-to-point ownership, Appendix Q.1 where the exposed boundary is Sys_B^elab, and RW reset disposition; and binds the qualified profile to the instance through K.CertHdr. If a K-faithful transformation-invariant boundary cannot be supplied for a persistently backpressuring realization, disable the transformation or exclude that memory from K scope. |
| <b>Q.2 / Core.InFlightDrainWF / Core.InFlightCycleExt / K.InFlightQual / J.InFlightPathSound</b> | Required whenever generalized DrainClosedNow/Dead_K is consumed. InFlightQual binds the Q.2 schema/effect soundness, finite drain, occurrence-relative continuation/path soundness and record-level InFlightRelationPrecision. Every consumer separately proves InFlightProfileCover for its exact occurrence domain. Every negative binds a K.InFlightSearchQuery/Status: ExhaustiveExact requires proof-carrying InFlightRelationComplete; OmittedRootIrrelevant is query-specific; UnknownIncomplete is diagnostic only. J.InFlightPathSound threads exact operational successor occurrences edge by edge.                                                                                                                                                                                                                                |
| <b>Core.SyncCarrierWF / K.Low.ModalQual MQ1-MQ6</b>                                              | Triggered by G2 lowered generalized-drain claims. TimeStutterQual is required first when administrative work may consume real cycles. ModalQual MQ2 is instance-wide: every relevant real-cycle normalization window must be able to complete before any affecting RESET that changes the projection; LT2 reset mapping alone is insufficient. MQ5 binds separate Sys_K/Sys_ref InFlightQual plus ProfileCover for every exact paired start consumed by modal path/escape/drain correspondence. LT2-mapped resets use ResetStutterMatch/ResetKey-Stutter; cross-model negative evidence is freshly transported, never query-reused.                                                                                                                                                                                                          |
| <b>K.SFOutputEnvelope / K.RespSerialBlockerClassSet / K.PolicySCompareDomains</b>                | Policy-S K.2b prospective-response qualification. Before SF, bind the non-predictive SFOutputEnvelope, conservative all-possible-output response-class set with waiver disjointness, every class-pruning RespSerialClassExclusionSound proof, and one SFControl/Handoff/Response package satisfying CD1-CD3/CD5-CD6. CD3-C is universal over relevant dynamic monitor instances on the fixed functional SDet artifact. CD4 is produced only after SF by instantiating CD3-C at the actual blocker; failure rejects the package/application rather than permitting domain mutation or early-DirectFail rerouting.                                                                                                                                                                                                                             |

## 11.1 A practical certificate boundary

A useful implementation is a versioned package whose common `K.CertHdr` contains the theorem-instance identity, trust information, the authoritative consumed-interface schema manifest, and incompatible-schema revalidation status. `K.CertPkg` applies whenever such a package discharges a certificate-backed Appendix-J safety theorem/corollary/coverage result as well as the K-facing/liveness claims. Safety-only packages carry only the common binding fields and subordinate evidence whose `M.7a` triggers fire; K-only waiver, `KObs`, `A5`, response, lowering, and liveness fields are not fabricated merely to fill the common package. In addition to source/RTL IDs, boundaries, capacities, elaboration, stimulus/environment/reset, witness and `Core.18` package where applicable, record exact `ExecutionCarrier/CycleResultOn/reset-epoch` provenance; for a no-dynamic-reset instance, record the checked exclusion fact, prescribed initial epoch binding, and identity threading rather than a changing reset-key vector. Record `RTLFrameEpochReachOn/QS2a-D`; `Core.RTLPartialCycleClosure` where the all-finite-prefix route consumes it, either as a direct instance proof or as a binding to the qualified RTL operational-semantics/translation class; `QS3-QS5 QualifiedSynchronousCarrierModel_RTL_B` evidence when `Core.QSync-Post/canonical` post settling is consumed; source bridge and construction-support/c-order schema identities; for K-facing source packages on the standard normal-compositional route, the separately triggered `Core.KObsSettleQualified_Sys_ref` record and `Core.KObsSettle-Ref` theorem reference together with the `L3CloseView/Core.KObsNormRef/K.NormStable` branch; and any full-`QSync QS2a-R` selected-witness recovery evidence. `Core.RTLPartialCycleExt` is theorem-derived and need not be stored as a second qualification. Generalized-drain applications record model/profile `InFlightQual`, consumer-domain `ProfileCover`, relation precision/completeness identity and every accepted negative-search query/status. `Policy-S` records `SDet` plus `PolicySProgress`, `K.CVPred ObsKey/ResetSupport_CV/Avail_pi/CVP-M` and `WellFounded(prec_CV)` schema, `SR3-G`, universal `SerialCandidateApplicability`, each selected valid `S_dom`, prospective `PolicySCompareDomains` and class/exclusion evidence, `CD3-C` and the post-SF `CD4` checker, response configuration and cleanliness. `G2` records `TimeStutterQual`, instance-wide `MQ2` normalization-opportunity/rank evidence, both model `ProfileCover` identities, `ResetStutterMatch` pairs and any negative-evidence transport. Prior evidence establishing `QualifiedSynchronousCarrierModel_RTL_B` may discharge the current weaker safety interface through the named `Core.QualifiedSynchronousCarrierModel-SafetyProjection`, whose stated second implication also carries full qualification through the synchronous-carrier layer to safety, without a `MIG1-MIG5` migration solely for that premise weakening. If a stored artifact consumed an otherwise incompatibly changed interface, regenerate/recheck it or bind `K.InterfaceSchemaMigration MIG1-MIG5`; a raw theorem/version name is insufficient.

### SIGNOFF RULE: Do not mix witnesses or cutpoints

A `J.OfferReach` proof for one source state, a `J.OfferConf` proof for another RTL cycle, and a `K.Drain` proof for a different witness do not compose. The formal architecture repeatedly requires exact identity of the theorem instance, stimulus, elaboration, witness, `H`, `O`, source prefix, RTL prefix, and normalized cutpoints.

## 11.2 Standard QSync and alternate B1-run certificate profiles

Appendix `M.7a` is the definitive trigger/discharge ledger. Appendix `M.7b` provides the standard `QSync` four-bundle packaging used by the general safety/K-facing flow. Appendix `M.7c` is a second normative packaging profile for Corollary `J.1-Safe-B1Run` only. It does not alter `M.7a` requirements: it reorganizes the applicable safety evidence into one RTL maximum/congruence record, one selected-run correspondence record, one `WC` variant, and one source-safety/property record. The `B1-run` profile is not a K-facing certificate profile and is not a replacement for `M.7b` when Appendix `K` claims are required.

Logical bundle

What it owns in the architecture

| Logical bundle                             | What it owns in the architecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Instance and QSync qualification</b> | <p>The theorem-instance identity and structural/mapping qualifications, including exact-history provenance. The common RTL safety-carrier core is exactly QualifiedSafetyCarrierModel_RTL_B (QS1 + QS2 + QS2a-D), together with adjacent Core.ExecutionCarrier/reset-epoch schema, RTLFrameEpochReachOn/RTLFrameEpochReach, designated-boundary/carrier checks, source QSync, Core.RegSeq/ReachPrep/SelectLatitude, the applicable R/E contract, boundaries/capacities/elaboration, Q.1(c-order) group/order metadata, memory/reset/interface conformance, and A8/A9/A10 where their individual triggers fire. Direct-input obligations remain owned by Core.RegSeq, the applicable R/E entry, and Core.QSync/QS2 frame discipline rather than by a separate dangling package item. If no post-base dynamic reset is possible, the retained epoch coordinates use the fixed initial valuation and identity advancement. Core.RTLPartialCycleClosure is additional only when all-finite-prefix completion is claimed; it may be referenced through reusable independently checked RTL operational-semantics/translation qualification with per-instance class/version binding, and RTLPartialCycleExt is theorem-derived. If the selected route consumes Core.QSync-Post/canonical post settling, the bundle adds QualifiedSynchronousCarrierModel_RTL_B/QS3-QS5. KCut-starting recovery further adds full QualifiedSynchronousModel_RTL_B with QS2a-R/RTLKCUTSettlingWindow and pre-origin disposition. For the richer K-facing source-cutpoint profile, but not for the safety-only profile, bundle 1 additionally carries or references Core.KObsSettleQualified_Sys_ref(I) and may cite Core.KObsSettle-Ref for the post-L3 Core.KObsNormRef endpoint after Core.L3CloseView. Stored artifacts are tied to the authoritative current consumed-interface manifest, with the named stronger-to-safety projection recorded as a compatible premise projection rather than an incompatible schema migration.</p> |
| <b>2. J correspondence/refinement</b>      | <p>LocalGWR/GWR-Comp or validated direct GWR with PhaseRole_A5, ProofUnitConstructionSupport/COrdActPred/SupportInsertionExists, exact SourceBridgeClosure/Ext and Q.1(c-order)/OuterCanonHist order preservation. When their own M.7a triggers fire, bundle 2 owns the separately triggered B1 observable-projection-determinism and B4/FPD finite replay/preparation-normalization evidence; plain J.1 and the standard QSync safety chain do not themselves trigger Definition B1, and omission from this short summary never makes a triggered obligation inapplicable. The summary names the common replay/preparation B4 use only: when B5 charging or generic source-cycle finiteness/Core.CycleClosure consumes the same M.7a B4/FPD discharge, that triggered use is owned by bundle 3. Standard-route safety uses QSyncTraceInd and group-respecting ideal closure, including exact carrier-edge extension/coverage rather than independent per-prefix witness reconstruction. K-facing use additionally forms a fresh Core.L3CloseView and binds J.L3CloseView-H/K.NormStable NS0-NS6, OfferReach/OfferConf/KObsConf, RegPhaseReach and DrainReach on the exact branch.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Logical bundle                       | What it owns in the architecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3. K applicability/provenance        | K-specific applicability/provenance: request persistence/ownership/release families; DrainReach/K-epsilon; generalized Q.2/InFlightQual plus consumer-domain InFlightProfileCover and accepted query/status family; exact-history source continuation; the triggered M.7a B4/FPD discharge when B5 charging or the generic Core.CycleClosure source-cycle-existence clause consumes it; EnvMF/CompareAdvanceLegal; K.CVPred ObsKey/ResetSupport_CV/Avail_pi/CVP-M; PolicySProgress; SR3-G and universal SerialCandidateApplicability; explicit selected DomResetScope; applicable NB-Align/waivers/lowering. G2 includes TimeStutterQual and ModalQual MQ1-MQ6 with instance-wide MQ2 and paired MQ5 covers. |
| 4. Policy-S response                 | Primary bounded-response A5-L discharge over the fixed functional SDet artifact: K.CompleteS, total K.RespCov with canonical literal/static-derived monitors, K.RespClean, response waivers, conservative RespSerialBlockerClassSet and class-exclusion evidence, the prospective SFControl/Handoff/Response package/CD3-C, and the post-SF CD4 checker/response-failure evidence. This bundle references SerialCandidateApplicability rather than predicting an SF blocker in advance.                                                                                                                                                                                                                      |
| M.7c alternate B1-run safety profile | Safety-only packaging for one selected terminating RTL run. Record 1: outer B1 + Core.B1RunMax + ordinary Core.17/R-E + triggered RSC0-RSC3/QPC1-QPC3. Record 2: singleton J.TraceCertCov({p_post^0}) using normal LocalGWR/GWR-Comp + source QSync or direct J.GWR; no wider B1 is inherited from plain J.1. Record 3: WC Variant A (NB-CF/identity-style) or Variant B (Appendix-L snooped WC, with W4 failure leaving the singleton package undischarged). Record 4: $\Phi$ on the selected p_ref^0/equal CanonHist, J.IdealPrefixClosed, and checker extensionality when a noncanonical encoding is used.                                                                                                |

### 11.3 How the bundles compose by claim

| Claim                            | Bundle composition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standard QSync functional safety | Bundles 1 + safety form of 2. RTL safety uses exact ExecutionCarrier/reset provenance, QualifiedSafetyCarrierModel_RTL_B (QS1 + QS2 + QS2a-D), and QSyncTraceInd. Plain J.1 and the standard QSync safety chain do not themselves trigger Definition B1. Any B1, B4/FPD, A8/A9/A10, direct-input, or other M.7a item is included only when its own independent trigger fires. Core.RTLPartialCycleClosure is added only for all-finite-prefix CycleIdealCover after a prefix has already entered a real cycle; it may be directly proved or supplied through the applicable reusable RTL-semantics/translation qualification, and RTLPartialCycleExt is derived from Closure. J uses c-order group preservation and exact SourceBridgeClosure. RTL QS3-QS5, QS2a-R, Core.KObsSettleQualified/Core.KObsSettle-Ref, and K-facing close-view/drain/response evidence are not safety premises. |
| One selected K.1A cutpoint       | Applicable instance qualification plus one exact K-facing J/cutpoint package. On the standard normal-compositional source route, that package binds Core.KObsSettleQualified_Sys_ref(I), uses Core.KObsSettle-Ref after L3CloseView to obtain the Core.KObsNormRef endpoint, and then applies K.NormStable before Offer/KObs correspondence. It also binds the Sys_ref generalized-drain InFlightQual, ProfileCover/query evidence for the selected contradiction occurrence, and any valid source A5-L discharge. Universal K.CertCov is not required for one already selected certified KCut.                                                                                                                                                                                                                                                                                            |



| Claim                                               | Bundle composition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Universal Theorem K.1 via Policy-S</b>           | All four bundles. The K-facing theorem-derived route first uses QualifiedSynchronousCarrierModel_RTL_B/QS3-QS5 for Core.QSync-Post and KCut closure, then full QualifiedSynchronousModel_RTL_B/QS2a-R plus KOriginCarrier-KCut to recover queried carrier positions. QSyncCertInd supplies the canonical per-carrier package and J.CertExist-QSync/K.CertCov-QSync supply universal package totality; bundle 3 supplies occurrence/profile/query drain qualification and the current serial applicability stack; bundle 4 supplies the prospective response-domain/clean selected run contradiction. In G2, TimeStutterQual/ModalQual/Low.A5 transport the serial result to Sys_ref before K.1A-Total. |
| <b>Universal K result with alternate A5-L route</b> | Bundles 1-3 as triggered plus a typed model-indexed K.DirectA5Proof payload instead of the Policy-S response bundle. The direct artifact concludes exactly A5L_M and binds the applicable InFlightQual/ProfileCover; a Sys_K direct proof must be transported explicitly through K.Low.A5 before Sys_ref consumption. There is no untyped 'other permitted route' escape hatch.                                                                                                                                                                                                                                                                                                                        |
| <b>Alternate B1-run functional safety</b>           | M.7c only. B1 + B1RunMax + triggered RSC/QPC derive J.B1PostProjCover; one selected J.TraceCertCov package and J.1-Safe-B1Run transfer the ideal-prefix-closed property to every Reach_post prefix. Does not require QualifiedSafetyCarrierModel_RTL_B, ExecutionCarrier/RTLFrameEpochReach carrier induction, RTLPartialCycleClosure, LocalGWRExtCert/GWR-CompExt, universal TraceCertCov, J.RefProjCover, or K-facing evidence unless some other selected consumer independently triggers them.                                                                                                                                                                                                      |

This packaging is intentionally an API/user-interface simplification rather than a semantic simplification. A user-facing flow may show four top-level statuses with drill-down to M.7a, but an independent checker must still validate the referenced subordinate evidence and recompute theorem-derived or KObs-derived facts where required.

## 12. Common misunderstandings

| Misreading                                                            | Correct interpretation                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>“Trace compatible” means cycle accurate.</b>                       | No. It preserves selected labels, values, ordering, atomicity, channel semantics, and anchors. Independent events may move between cycles.                                                                                                                                                                                                                                                                                               |
| <b>The source pass transfers by Ref -&gt; Post.</b>                   | For ordinary safety, the deductive path is Post -> Ref plus J.1-Safe: RTL adds no covered canonical safety behavior absent from Sys_ref. Ref -> Post is restricted to annotated RTL-consistent or witness-selected source prefixes.                                                                                                                                                                                                      |
| <b>Per-process replay certificates already imply a system replay.</b> | No. Shared channel state, snapshots, atomic participants, resets, and field interference can conflict. J.GWR-Comp proves the merge.                                                                                                                                                                                                                                                                                                      |
| <b>The causal-dependency graph is the J induction order.</b>          | No. Appendix-G causal dependency explains information flow and may contain legal atomic cycles. J.DepSound applies only to D1-D4 semantic prerequisite edges; D5 is proof stratification and uses J.HorizonOrderSafe commutation evidence. J.FoundationRank/J.DepRank prove the generated J graph acyclic. Appendix K's K.CVPred/K.CV-WF is a third, separately well-founded provenance order and must not reuse the broad causal graph. |



| Misreading                                                                                             | Correct interpretation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>KObs is a normalized execution or full state.</b>                                                   | No. Core.18 makes the cross-model cutpoint seam equality of the typed record $\langle E, w, H, O \rangle$ . The source record is defined by Core.18c; Appendix J constructs KObsPost_J and J.KObsConf proves typed equality for the exact selected package. Core.18d/e reconstruct K-facing facts from that record; historical Issue timing and arbitrary RTL/source microstate are not equated.                                                                                                                                                                                                                                                                   |
| <b>No-reverse scheduling proves deadlock freedom.</b>                                                  | No. It prevents one deadlock-introduction mechanism. Timing overlap, finite capacity, pipeline drain, environment behavior, and serial-shaped source deadlocks require Appendix K.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>One clean Policy-S simulation is automatically a proof of A5-S.</b>                                 | No. The primary workflow uses one selected run, but the serial reduction also requires Definition K.SerialRoutePremises, which binds SDet, constructive fairness, K.BF/ownership/release projection, authoritative K.CV and its OpKey/K.CVPred/K.CV-WF certificate, nonwithdrawal/K $\epsilon$ /K.Drain/CycleClosure, EnvMF/JointFreeze, CompareAdvanceLegal, reset/terminal gates, fixed-instance choices, and NB-Align/K.Low where triggered. A5-S separately supplies total K.RespCov, K.CompleteS, and K.RespClean for K.2c/K.1. A universal RTL claim additionally needs the QSync/QSyncCertInd package-totality route or another valid total-coverage proof. |
| <b>The WFG+/EnvStop/DoneStop starvation taxonomy, K.EnvOrd, or K.Done can replace K.Resp coverage.</b> | No. K.EnvOrd and K.Done are optional diagnostic classifications; they can help explain environment-rooted or completion-rooted response failures but are not part of the soundness-critical Policy-S route, do not discharge A5-L, and never permit omission of a K.Resp monitor. W.Env is likewise not a blanket escape hatch: any environment-rescue or response-monitor waiver must be explicitly scoped and carry the required KObs-closure or response-class coverage-preservation proof.                                                                                                                                                                     |
| <b>Core.LAdm adds a new pipeline-drain requirement beyond Appendix B.</b>                              | No. Core.Phase/Core.LAdm formalize the already-required freeze/drain behavior. Core.16's 'launch' restriction is intentionally broader than message Issue, and K.Drain supplies the finite non-replenishing cleanup evidence; capacity alone does not prove finiteness. Core.SettleKBridge connects the L2 activation to the selected later drain-closed KCut.                                                                                                                                                                                                                                                                                                     |
| <b>The Policy-S reduction is already a certified theorem.</b>                                          | Not as a mechanized and flow-qualified theorem. The scheduling-rules document supplies explicit paper proofs of K.Low.C/K.Low.A5, K.2b.P0/P1/R/P/E/Q and their assembly, including common operation keys, well-founded provenance, typed cycle extension, and nested reset/terminal gates. Lean mechanization, simulator/lowering/QSync qualification, reusable certificate validation, and package-coverage realization remain outstanding.                                                                                                                                                                                                                       |
| <b>Another A5-L route can rescue a serial deadlock.</b>                                                | No. If an admissible Policy-S-shaped Policy-L prefix reaches Dead_K^W, A5-L is false for the unchanged instance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Snooping changes the specification to match RTL.</b>                                                | No. It selects one behavior already admitted by the under-specified source latency/epsilon semantics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>A live AND is enough for snooping.</b>                                                              | No. The wrapper must record per-ordinal RTL events and retain them until source consumption; otherwise events can be dropped or reordered.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Misreading                                                                                      | Correct interpretation                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tool generation automatically discharges the proof obligations.</b>                          | No. The tool or flow must emit documentation, certificates, static checks, monitors, or formal evidence for each named obligation.                                                                                                                                                                                                                                                                                                                                                                |
| <b>A synthesizable and functionally working design is necessarily within the theorem scope.</b> | No. Hidden timing dependence, ambiguous signal anchors, unsupported shared-memory value flow, unmatched reset disposition, fall-through/bypass behavior, or coupled boundary structure can place a correct implementation outside the selected formal model. Appendix F is the architect-facing applicability screen.                                                                                                                                                                             |
| <b>A fall-through FIFO is just an ordinary FIFO with a particular capacity.</b>                 | No. Same-cycle empty-to-consume or full-to-accept behavior changes the channel transition semantics. The bypass or fall-through path must be represented explicitly in Sys_ref and covered by the boundary/elaboration evidence.                                                                                                                                                                                                                                                                  |
| <b>Dynamic reset is covered whenever reset is supported by the design.</b>                      | No. Reset handling is candidate- and phase-specific. The route selects an explicit valid DomResetScope, resolves early Reset then Terminal, and prospectively fixes SFControl/Handoff/Response on nested Continue. R-S/T-S constrain SFControl only; post-handoff reset/terminal are Response dispositions. A scope/domain may not be revised after observing SF.                                                                                                                                 |
| <b>Failed non-blocking polls are invisible, so they cannot affect safety.</b>                   | Their failure is not a committed Sigma event, but the outcome or cadence may select later observable control, values, requests, payloads, or endpoints. Such sites require NB-CF, a valid WC/L.Dir witness, isolation, or an explicitly modeled timing-sensitive protocol.                                                                                                                                                                                                                        |
| <b>Input FIFO depths in a joined pipeline are independent available slack.</b>                  | Not when acceptance is coupled through a join, bundle, shared stage, stateless coupler, or epoch gate. B(c)/occupancy still describes the represented storage legality, but the actual settled handshake is determined by the declared Sys_B^elab network. For K-facing reasoning the same qualified template must also expose the modeled-process ReleaseOwnerSets_E/WaitOwners_E dependency projection; independent input capacities are therefore not automatically independent release slack. |
| <b>K.Resp is an RTL latency guarantee.</b>                                                      | No. It is source-side evidence used to discharge A5-L through serial dominance.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Any epsilon-quiescent state is a KCut.</b>                                                   | No. Core.KCut additionally requires reachability and a settled explicit-boundary request/enablement fixed point. Source L2 SettlePoints satisfy this by Core.SettleIsKCut; Core.QSync-Ref supplies their existence/uniqueness in the standard scope, while Core.QSync-Post derives the RTL KCut-closure interface.                                                                                                                                                                                |

| Misreading                                                               | Correct interpretation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>K.CertCov covers every RTL prefix or every normalization witness.</b> | No. K.CertCov is total only over reachable RTL KCuts. For each KCut it may exhibit one reachable-prefix/NormPost pair and certified package. In the standard QSync theorem-derived route, QualifiedSynchronousCarrierModel_RTL_B/Core.QSync-Post supplies KCut closure from the safety foundation plus QS3-QS5, while full QualifiedSynchronousModel_RTL_B/QS2a-R/RTLKCSettlingWindow places each queried KCut on a selected legal witness and Core.KOriginCarrier-KCut derives its carrier representation. J.QSyncCertInd/J.CertExist-QSync/K.CertCov-QSync then derive package totality. Arbitrary transient prefixes and arbitrary normalization witnesses need not carry packages.                                                                                                                                                                                                                                                |
| <b>QSync all-finite-prefix safety requires the KObs/drain package.</b>   | No. J.QSyncTraceInd runs under QualifiedSafetyCarrierModel_RTL_B and deliberately carries only exact-prefix J.LocalGWR/J.GWR-Comp plus J.HCanon; RTL QS3-QS5, QS2a-R, OfferReach/OfferConf/KObsConf/K.DrainReach/K.RLAdmReach are excluded. It first establishes cycle-aligned packages; Core.RTLPartialCycleClosure plus Core.RTLCycleCarrier/J.CycleIdealCover/J.1-Safe-QSync then extend J.IdealPrefixClosed safety to all reachable finite outer observable prefixes. Core.RTLPartialCycleExt is derived from Closure. If J.QSyncCertInd is already proved for K, J.QSyncCertInd-Trace uses the named qualification projection and stored safety stage to supply the safety induction. The alternate M.7c B1-run route goes further and avoids the RTL QSync carrier/entered-cycle coverage machinery entirely for its safety-only conclusion, while still using source-side QSync on the normal selected J.1 construction route. |
| <b>One observed QSync simulation proves universal K.CertCov.</b>         | No. J.QSyncCertInd requires CI1 for every reachable next-cycle successor and CI2 exhaustive successor coverage. A determinism theorem may collapse that set, but B1 alone does not justify ignoring unobserved internal successors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Delta/tau/idle are just three names for epsilon.</b>                  | No. Delta is same-cycle settling; a SilentCycle is a real CompleteCycle that may advance registered/environment/monitor state; and a B6 idle tick is permitted only after Core.TerminalIdle proves permanent machine+environment no-future-enabling behavior. A present cycle with no selected commit is not automatically terminal stutter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Core.QSync makes the source model deterministic.</b>                  | No. The RTL safety-carrier predicate does not even require QS3-QS5; it supplies cycle/frame/designation structure, not a global settling or execution determinism claim. Under QualifiedSynchronousCarrierModel_RTL_B, Core.QSync-Post gives a unique settled result for a fixed designated origin, inputs and explicit drives/choices, and QS3 ties the operational settling relation to the declared network. QS2a-D is a total designation condition over independently admitted frame/epoch pairs; QS2a-R is a separate witness-relative KCut settling-window condition in the full layer. None is a global execution-determinism theorem or a claim that a bare KCut uniquely determines historical epoch. WC-sensitive outcomes, Core.SelectLatitude under Policy L or Policy S, arbitration, and reactive-environment choices may still admit multiple source executions; SDet is a separate selected Policy-S-run premise.    |

| Misreading                                                                                                           | Correct interpretation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core.KCutClosure_post is still an arbitrary per-instance normalization proof in the standard flow.</b>            | No. The downstream interface is unchanged, but the standard QSync scope derives it through Core.QSync-Post from audited structural facts. An alternative non-QSync settling semantics requires an explicitly defined and separately qualified theorem extension.                                                                                                                                                                                                                                                                              |
| <b>Same-cycle Core.SyncFence requires a physical RTL sub-cycle ordering.</b>                                         | No. The physical E5 requirement is cycle-granular: the predecessor process-facing commit occurs no later than the synchronization commit. The equal-cycle "accounted before interval advance" rule is the proof/certificate attribution convention used to maintain source-interval semantics.                                                                                                                                                                                                                                                |
| <b>K.CV can use the Appendix-G causal graph as its induction order.</b>                                              | No. K.CV is the authoritative design-level causal/value-determinism premise, and Definition K.CVPred/K.CV-WF supplies its separate fact-key provenance certificate with explicit predecessor occurrence closure, deterministic evaluators, comparison-availability, and a well-founded relation/rank. K.CVPred/K.CV-WF do not presuppose K.CV, but their relation also does not inherit the broad Appendix-G causal graph. Same-cycle rendezvous enablement and broad causal/temporal proximity do not become provenance edges automatically. |
| <b>Core.16 'launch' means only endpoint Issue.</b>                                                                   | No. The 15 August clarification makes launch intentionally broader: any source-later preparation/reach or other work capable of replenishing the blocked component is covered by the no-new-later-work discipline.                                                                                                                                                                                                                                                                                                                            |
| <b>"Prefix-closed" means ordinary prefix closure of one linear event log.</b>                                        | No. In the safety theorems it means J.IdealPrefixClosed: downward closure over finite ideals of the retained required-order relation. A cross-interface scoreboard may fail this condition even if an engineer informally calls it a safety check; use direct J.TraceCertCov on the required arbitrary-prefix domain or retain the needed dependency in the observation/order.                                                                                                                                                                |
| <b>Core.IC and Core.SyncFence are trusted axioms that can simply be assumed for RTL.</b>                             | No. They are constitutive legality/invariant clauses of Sys_ref. The corresponding RTL request lifecycle, attribution, process-facing completion, and synchronization-fence behavior must be established by implementation-side conformance evidence and recorded through the M.7a audit entries.                                                                                                                                                                                                                                             |
| <b>Commit is defined as valid-and-ready being high in every proof model.</b>                                         | No. Core.8 defines the abstract boundary commit as the legal E4 channel-commit transition. Simultaneous mapped valid/ready and payload sampling are the RTL_B realization/conformance of that abstract event. Commit(q) is the designated process-facing completion/release occurrence, and returned-result availability remains a later Core.Phase L4/Core.RegSeq matter.                                                                                                                                                                    |
| <b>Finite-prefix J.1 delays an enabled source action by assuming B2 fairness.</b>                                    | No. Core.SelectLatitude makes finite nonselection of otherwise legal L1 issue and L3 boundary actions constitutive source legality under both policies. Core.IssueFair and B2 are separate complete-execution requirements that prevent indefinite deferral; neither is imported into finite-prefix J.1.                                                                                                                                                                                                                                      |
| <b>A Core.SilentCycle used between J Horizon buckets leaves the full source state unchanged.</b>                     | No. SilentCycle is a real CompleteCycle. Registered/internal state, environment state, reset bookkeeping, requests/witness state, and cycle-valued monitors may evolve. J.UnitExt must preserve or re-establish its induction invariant through the typed cycle transition; only the selected-unit ideal/CanonHist is unchanged when no selected J unit commits.                                                                                                                                                                              |
| <b>An empty RTL Horizon becomes an empty J.CanonicalRank bucket, or fixes the same number of source idle cycles.</b> | No. Height/FR/DepRank/DepAcyclic rank actual construction actions in Act_L( $\tau_{post}$ ), so empty Horizons create no rank elements. Any required source-cycle gap is a finite exact-history J.SourceBridgeClosure bridge; Core.SelectLatitude may justify legal nonselection inside a certified bridge cycle but does not establish bridge existence or arrival, and its length is not                                                                                                                                                    |

| Misreading                                                                                                                  | Correct interpretation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                             | the raw RTL Horizon-index difference unless the selected cycle-sensitive evidence requires that alignment.                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Mapped external RAM traffic may always be hidden from <math>\Sigma</math> once a higher boundary is chosen.</b>          | No. Hiding is an admissible use of the existing observation-boundary freedom only when Core.MemFaithful holds against the declared untransformed logical mapping semantics. J.Mem still requires complete representation of every qualifying cross-process memory dependency, and a persistently backpressuring K realization needs a transformation-invariant K-faithful logical boundary (normally Core.MemProfile) or the transaction-changing transformation is outside the covered K instance.             |
| <b>A high ready/valid level can be annotated as whichever BoundaryReq/Issue makes E3 work.</b>                              | No. Core.9 Issue is the operational first-request cycle, not a witness-selected timestamp. Core.10 BoundaryReq is fixed-mapping, process-owned, operation- and boundary-specific. A raw high valid/ready level that represents complementary channel/storage availability or declared coupler logic is not BoundaryReq. Core.IC also requires one live owner for each asserted blocking BoundaryReq and forbids an ownerless back-to-back request interval.                                                     |
| <b>If an operation or boundary action is enabled only intermittently and never selected, Core.IssueFair/B2 is violated.</b> | No. These are weak-fairness conditions with continuous-enablement antecedents. Core.IssueFair applies when a reached operation remains IssueEnabled at every applicable L1 opportunity; B2 applies when an issued boundary action remains continuously ChannelEnabled. Intermittent enablement is outside those weak-fairness guarantees by design, not a violation. If source control never reaches the operation, that is a separate source-control case.                                                     |
| <b>Core.DeclaredTerminal and Core.TerminalIdle are the same terminal condition.</b>                                         | No. Core.DeclaredTerminal is a modeled machine/test termination, abort, or end-of-test predicate; an external host watchdog is not enough unless modeled. Core.TerminalIdle is the stronger B6-eligible global fixed/no-future-enabling condition for machine plus environment. TerminalIdle can also hold for a nonterminating service model proved permanently idle after stimulus exhaustion.                                                                                                                |
| <b>Any persistent nonprogress involving a message channel is the Appendix-K closed internal deadlock.</b>                   | No. K.Dead is a specific reachable drain-closed, release-aware internal wait-cycle predicate. A continuously enabled but unselected boundary action is a B2 fairness failure; lack of required complementary progress is a B3 progress failure; intermittent enablement/starvation can lie outside both weak-fairness antecedents; and environment-only or source-control stalls are also distinct. Appendix K excludes only its defined non-waived closed internal wait-cycle class under the stated premises. |
| <b>DrainClosedNow means there is no enabled non-frontier residual offer.</b>                                                | No. That is only the legacy primitive fast-path intuition. Generalized DrainClosedNowRec is the absence of any finite qualified InFlightStepRec path that changes WaitRoots_D. K.PrimitiveDrainExact gives equivalence with the residual-offer selector only under its restricted primitive profile.                                                                                                                                                                                                            |
| <b>The drain search should consider only actions owned by processes in D.</b>                                               | No. D scopes only the root family being compared. Core.InFlightStepRec deliberately has no D/ownership/relevance filter. An already-admitted action outside D must be explored if a finite path through it can change a D root; the two-stage/join artifacts are direct regressions against a D-scoped action relation.                                                                                                                                                                                         |
| <b>Stage 2 puts synchronization into MsgOf, BoundaryReq, or the literal</b>                                                 | No. Core.7b/Core.7c create a parallel synchronization carrier with SyncOf/SyncTxn/SyncReleaseOwnerSets. Literal message BoundaryReq/Front/WFG/Dead_K remain message-only. SyncFire and                                                                                                                                                                                                                                                                                                                          |

| Misreading                                                                                                      | Correct interpretation                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WFG.                                                                                                            | lowering gate actions participate only in the generalized in-flight/lowering proof infrastructure.                                                                                                                                                                                                                                                                                                              |
| If the serial Sys_K model has Core.CycleClosure, the lowered source Sys_ref path theorem follows automatically. | No. Continuation, profile and search premises are model- and occurrence-indexed. In G2, Sys_K CycleClosure may discharge only the Sys_K IQ3-class continuation field. ModalQual separately binds Sys_ref InFlightQual and paired ProfileCover; real-cycle lowering also requires TimeStutterQual when triggered.                                                                                                |
| Equal KObs proves future in-flight progress.                                                                    | No. KObs reconstructs current state and local Q.2 action/effect applicability only. Core.InFlightCycleExt/J.InFlightPathSound separately establish that reconstructed edges and finite paths have legal operational realizations. B2/B3 cannot bootstrap that existence because their eventuality is interpreted along an execution whose existence is already assumed.                                         |
| If I can prove B1, plain Theorem J.1 needs no source QSync or correspondence certificate.                       | No. Removing B1 from J.1 removes an unnecessary RTL observable-determinism premise; it does not remove the source-witness construction. The normal compositional J.1 route still uses LocalGWR/GWR-Comp plus source-side Core.QSync, WC and all other triggered conformance/construction evidence, or a directly validated J.GWR package must supply equivalent checked settling/bridge evidence.               |
| A terminating RTL simulation is enough to discharge Core.B1RunMax.                                              | No. BRM1 can be checked on the selected run, but BRM2 is universal over Reach_post. Under B1+BRM1 it reduces to ruling out any reachable selected occurrence after the selected final completed cycle (or any selected occurrence in the zero-cycle case). A simulation trace alone does not prove that global negative reachability property unless backed by a completeness/translation/reachability theorem. |
| The B1-run route requires fully deterministic RTL and identical reset/elaboration internals.                    | No. It requires determinism/congruence of the selected outer quotient. Hidden state, epsilon paths and irrelevant micro-order may differ. Dynamic outer-relevant reset is compared through conservative OuterResetSupport + RSC0-RSC3; elaborated outer grouping/order is compared through QPC1-QPC3.                                                                                                           |

## 13. Quick-reference index

|                |                                                                                                                                                                                                                                 |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PURPOSE</b> | <p>Provide a compact map from theorem/definition names to their role in the proof architecture.</p> <p><b>Engineer takeaway:</b> Use this section as a navigation aid when reading the normative scheduling-rules document.</p> |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 13.1 Normative Formal Core and Appendix G

| Name                               | Role                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core.8 Commit / CommitCycle</b> | Abstract boundary commit is the selected model's legal E4 channel-commit transition; RTL mapped valid/ready/payload is the implementation realization. Commit(q) is the designated process-facing completion/release occurrence and CommitCycle(q) its cycle index; returned-result availability remains separate under L4/Core.RegSeq. |



| Name                                                                                                                                              | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core.RegSeq</b>                                                                                                                                | Registered next-cycle dependence rule: cycle-t sequential interface outputs derive from registered state/fixed operands; no same-cycle newly returned interface result feeds a dependent sequential output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Core.Settle / delta-settling</b>                                                                                                               | Complete source L2 same-cycle settling relation and its fixed-point endpoint. The L2 frame freezes cycle index, registered state, selected cycle inputs, and stable sequential drives; Definition Core.Settle itself does not prove termination.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Core.QSync / QSync-Ref / QSync-Post</b>                                                                                                        | Three-level RTL QSync architecture. QualifiedSafetyCarrierModel_RTL_B = QS1 + QS2 + QS2a-D and is the cycle/carrier safety foundation; it does not require RTL QS3-QS5. QualifiedSynchronousCarrierModel_RTL_B extends that foundation with QS3 realization/adequacy and deterministic local evaluation, QS4 finite acyclic settling, and QS5 selected-boundary completeness; Core.QSync-Post then supplies the canonical endpoint of each designated origin and Core.KCutClosure_post without QS2a-R. QualifiedSynchronousModel_RTL_B adds QS2a-R/Core.RTLKCutSettlingWindow for routes that start from arbitrary reachable post KCuts; QSync-KCutCanonical/Core.KOriginCarrier-KCut then recover the represented carrier-relative origin/NormPost package. Core.QualifiedSynchronousCarrierModel-SafetyProjection makes the stronger-to-weaker relationship explicit; its stated second implication carries full qualification through the synchronous-carrier layer to safety.                                                                                                                                                                                                       |
| Core.ExecutionCarrier /<br>ResetEpochKey /<br>RTLFrameEpochReachOn /<br>RTLPartialCycleClosure                                                    | Exact-history carrier/reset/cycle foundation. ExecutionCarrier identifies one legal finite history; ResetEpochKey/Advance canonicalize matched reset provenance; RTLFrameEpochReachOn makes frame/epoch admission occurrence-relative. If no dynamic RESET is permitted after the prescribed base, the same retained epoch coordinates are discharged by a checked no-dynamic-reset fact and fixed-initial-epoch identity advancement rather than a changing reset-key proof. Core.RTLPartialCycleClosure is the finite-completion qualification for an already-entered real cycle and includes the entered-cycle nonterminal fact plus existing-result-or-exact-extension alternative. It is a property of the concrete RTL operational-semantics/translation class, so an independently checked class-level qualification may be reused across instances that are proven to bind to that class/version and satisfy its side conditions. Core.RTLPartialCycleExt is derived from Closure and merely selects that completion witness. RTLCycleCarrier/CycleIdealCover consume Closure on the all-finite-prefix route. None of this machinery is fairness or generic successor totality. |
| Core.ReachPrep / Core.9 /<br>Core.10 / Core.IC /<br>Core.PolicyL / Core.LLegal<br>/ Core.SelectLatitude                                           | Overlap-aware preparation/reach, operational Issue, operation-specific process-owned BoundaryReq/Core.IC, Policy-L step legality and finite non-maximal selection. SelectLatitude cannot create a required source cycle or prove arrival at a later state; exact bridge arrival is J.SourceBridgeClosure/Ext.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Core.Phase / Core.LAdm /<br/>Core.CycleState /<br/>Core.CycleResult /<br/>Core.RTLCycleCarrier /<br/>Core.CycleClosure /<br/>Core.FairPick</b> | L0-L4 source phase structure and liveness-admissible source prefixes/executions. CycleResult/CycleClosure/CycleChoiceExt are occurrence-relative on an exact ExecutionCarrier; FairPick/FairCycleDovetail thread that same history. On RTL, Core.RTLCycleCarrier is a narrow entered-cycle lemma under the safety carrier plus RTLPartialCycleClosure and RC1/RC2; nonterminality follows from Closure + RC1 rather than a duplicate premise. Core.KOriginCarrier-Extend appends a supplied typed RTLCycleStep with no ambient qualification predicate, while AtOrigin/Decomp use the safety carrier. The K-facing Core.L3CloseView is a separate post-construction closure view, not an ordinary L1-L4 scheduling action.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |



| Name                                                                                                                        | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>Core.PolicyS</b>                                                                                                         | Conservative serial-blocking issue policy used by the primary instrumented liveness route. It inherits Core.SelectLatitude: an otherwise legal earlier issue need not be selected at the first finite opportunity. Core.IssueFair prevents indefinite omission on complete admissible executions, while SDet separately fixes the concrete selected Policy-S simulator/scheduler run used by the reduction.                                                                                                                                                                                                                                                                                                                                                   |
| <b>Core.ElabProj / Core.WFG / ReleaseOwnerSets / Core.16 / Core.Drain / Core.SettleKBridge / generalized in-flight Core</b> | Core.ElabProj coordinates explicit elaborated storage/token/accounting, settled handshake, and modeled-process release projection.<br>Core.16/Core.Drain/Core.SettleKBridge govern the reached no-replenishment drain from an L2 blocked activation to later K-facing states. Generalized drain adds Core.InFlightStep (already-admitted complete action units), Core.InFlightDrainWF (finite decreasing admitted work), Core.InFlightCycleExt (restricted same-/cross-cycle opportunity), and Core.18 FlightStateRec/InFlightStepRec reconstruction. The reconstructed step relation has no candidate-D filter; D is introduced later only through K.WaitRoots. Capacity supports finiteness but is not the well-founded decrease or future-existence proof. |
| <b>Core.OrdFrontSingleton</b>                                                                                               | Ordinary non-elaborated source intervals have at most one pending blocking frontier item; genuine multi-frontiers require explicit Sys_B^elab partial-order structure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Core.18 / Core.18c-e KObs / Core.9-Core.10 Issue &amp; BoundaryReq</b>                                                   | Core.18 defines cross-model correspondence as equality of one common typed KObs record <E,w,H,O>; Core.18c defines the source record and Core.18d/e reconstruction functions. J defines KObsPost_J and proves equality through J.KObsConf.<br>Core.9/Core.10 and Appendix C now make Issue/BoundaryReq operational rather than witness-selectable: Issue(q) is the first process-owned request cycle, BoundaryReq is the fixed-mapping operation-specific process-owned request, and Core.IC supplies live-owner/nonwithdrawal/back-to-back attribution. Historical Issue/ReturnedAtCycle timestamps remain outside KObs fields.                                                                                                                              |
| <b>Core.17 <math>\Sigma</math>Match / R1-R5 / E1-E5 / causal dependency</b>                                                 | Core.17 supplies canonical complete selected occurrence/value matching. Appendix G expands it with R1-R5/E1-E5. For sequential signal I/O, each anchor records at most one Read and at most one Write per (process, signal); if several source Writes share the same E2 anchor, the last executed source Write supplies the canonical value and earlier writes are epsilon-internal. R2C fixed-point units match by component identity/local emitted-occurrence ordinal rather than a global cycle bucket; explicitly selected proof-internal RESET units may be carried without adding RESET to $\Sigma$ . The explanatory causal graph may be cyclic and is neither the J construction order nor the K.CVPred/K.CV-WF provenance order.                     |
| <b>Core.KCut / Core.KCutClosure_post / Core.KCutObserve</b>                                                                 | K-facing settled/quiescent comparison domain; downstream RTL observation-domain interface; and persistent-blocker preservation into that domain. In the standard scope QSync-Post derives KCutClosure_post; K.CertCov remains separate package totality.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Core.SyncFence</b>                                                                                                       | Constitutive Sys_ref synchronization-completion legality with a separately checked RTL conformance counterpart: blocking operations in pref_S(s) process-facing commit no later than s; same-cycle completion is permitted and equal-cycle interval accounting is certificate semantics rather than physical sub-cycle timing.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Core.MemSyncOrder</b>                                                                                                    | Array/synchronization commit-order qualification at the schedule/mapping layer; separate from Appendix I/J replay and from J.Mem cross-process shared-memory lowering. Cache/merge/split/reorder must preserve the fence, and one merged storage commit cannot span opposite sides of one synchronization boundary.                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Name                                                                                       | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>Core.MemFaithful / J.Mem / Core.MemProfile</b>                                          | Mapped-memory observation/profile architecture. MemFaithful compares transformed physical realization with the declared untransformed logical mapping semantics at the selected boundary and preserves selected J behavior plus K blocking/dependency facts. J.Mem completely lowers every qualifying cross-process memory dependency or uses the separate memory-state extension. MemProfile packages a reusable transformation-invariant logical boundary, including MemFaithful/MemSyncOrder/J.Mem, K disposition, A6/Q.1 ownership/elaboration, and reset treatment. |
| <b>Core.7b-Core.7e / SyncCarrierWF / SyncFire / EpochGateAdvance</b>                       | Separate synchronization/lowering carrier architecture. SyncOf/SyncTxn/participants/pending-ready-blocked/SyncReleaseOwnerSets type current source synchronization without widening MsgOf or message BoundaryReq. SyncFire is the real already-pending/ready source synchronization transaction. EpochGateAdvance is lowering-only explicit Sys_K gate progress erased by K.Low projection. Core.SyncInFlightOpportunityQual specializes InFlightCycleExt for these constructors.                                                                                        |
| Core.L3CloseView / J.L3CloseView-H / K.NormStable                                          | K-facing source branch from an extendible J.GWR checkpoint. L3CloseView closes the selected L3 action set; J.L3CloseView-H preserves canonical history/root state; K.NormStable NS0-NS6 qualifies the subsequent observation-only KObsNormRef suffix. The closed branch is formed afresh for K-facing use and never extended by the next safety construction step.                                                                                                                                                                                                       |
| <b>Core.KObsSettle / Core.KObsSettleQualified / Core.KObsSettle-Ref / Core.KObsNormRef</b> | K-facing-only post-L3 observation normalization on phase-bearing source models. After the fresh L3CloseView closes the selected L3 action set, KObsSettleQualified checks KSQ1-KSQ4 for the actual post-L3 roots/network/no-feedback discipline; KObsSettle-Ref supplies the finite deterministic observation fixed point used by KObsNormRef and the source KCut. The standard safety-only QSync route does not consume this qualification.                                                                                                                             |
| Core.B1ObsPrefix / Core.B1RunMax / Core.B1RunMax-Cover                                     | Alternate B1-run RTL coverage foundation. B1ObsPrefix is the directional completed-bucket/partial-bucket/no-cycle-migration helper. B1RunMax selects one finite cycle-closed DeclaredTerminal outer-occurrence maximum; under B1+BRM1, BRM2 is equivalent to no selected occurrence after its final completed cycle. B1RunMax-Cover places every reachable selected occurrence into the selected maximum at the same cycle.                                                                                                                                              |
| Core.OuterResetSupport / Core.ResetScheduleCongruence                                      | Triggered B1-run reset congruence. RSC0/ORS1 support conservativeness, RSC1 completed-cycle logical reset-signature agreement, RSC2 partial-current-cycle reset-frontier restriction, and RSC3 reset-owned predecessor-domain closure.                                                                                                                                                                                                                                                                                                                                   |

## 13.2 Appendix I

| Name                      | Role                                                                                                                                              |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>I.A0</b>               | RTL E3 message-issue discipline assumption.                                                                                                       |
| <b>I.ReplayObs</b>        | Logical replay checkpoint determined by committed history, stimulus, and witness; excludes raw preparation and current request state.             |
| <b>I.CutpointOfferObs</b> | Operational current-offer slice carrying Pending_P, Front_P, BoundaryReq, attribution, boundary, direction, epoch, and stable Push payload facts. |

| Name                          | Role                                                                                                                     |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>I.ReplayObs-Congruence</b> | Immediate same-unit replay closure plus paired congruence of the exact selected normalization paths.                     |
| <b>I.DR / I.DR'</b>           | Deterministic equality of logical replay checkpoints at selected normalized and canonical immediate post-unit cutpoints. |
| <b>I.DeferredNBHole</b>       | Carried unresolved shared non-blocking decision with NoDependentUse.                                                     |
| <b>I.3</b>                    | Finite process-local preparation script.                                                                                 |
| <b>I.4</b>                    | Construction of the J.PCert records.                                                                                     |

### 13.3 Appendix J

| Name                                                                                         | Role                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>J.LocalGWR</b>                                                                            | Complete local certificate package; no global replay assumption.                                                                                                                                                                                                                                                                                                                                                               |
| <b>J.DepJ / J.DepSound /<br/>J.HorizonOrderSafe / DepComplete /<br/>DepRank / DepAcyclic</b> | Generate D1-D4 semantic prerequisites and D5 proof-stratification over actual ConstructionAction occurrences. PhaseRole_A5 types genuine A5 actions; ProofUnitConstructionSupport/COrdActPred reflect c-order proof-unit dependencies into executable action order. DepSound applies to D1-D4; HorizonOrderSafe validates D5-only commutation; DepComplete/DepRank/DepAcyclic establish the canonical action order.            |
| <b>Q.1(c-order) / J.OuterCanonHist /<br/>J.ProjGroupOrderAcyclic</b>                         | Every nonempty proof-unit projection has one canonical indivisible outer group whose Flat_Sigma fragment exactly matches pi_Sigma,E and whose strict inter-group required order is backed by prefix-stable normative predecessor rules. J.OuterCanonHist carries groups before flattening; ProjGroupOrderAcyclic and OuterCanon-Ideal prevent contraction/erasure from creating a cross-group cycle or retroactive order edge. |
| <b>J.RegPhaseNF</b>                                                                          | Normalize same-Horizon actions to registered L1/L2/L3 phases. It consumes a finite L2-settle existence fact; Core.QSync-Ref supplies it uniformly on the normal compositional route, while a direct J.GWR route may carry per-Horizon evidence.                                                                                                                                                                                |
| <b>J.NBDecision</b>                                                                          | Decide each deferred non-blocking call from the common settled snapshot.                                                                                                                                                                                                                                                                                                                                                       |
| <b>J.SourceBridgeClosure /<br/>J.SourceBridgeExt / J.UnitExt</b>                             | Exact-history finite source-cycle arrival between nonempty construction buckets when cycle-sensitive evidence requires it. Bridges are occurrence-bound, preserve the exported construction/history invariant and include finite arrival to the exact next cycle-entry state. SelectLatitude is not the existence/arrival premise.                                                                                             |
| <b>J.GWR-Comp</b>                                                                            | Normal local-to-global witness construction from LocalGWR plus PhaseRole_A5, ProofUnitConstructionSupport/SupportInsertionExists, validated action order and exact source bridges. GWR-CompExt uses LocalGWRExtCert's no-retcon exported-checkpoint frontier and literal old-chain reuse.                                                                                                                                      |

| Name                                                                       | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>J.1</b>                                                                 | System-level observable compatibility. Normal signoff is Post -> Ref existence from J.GWR-Comp or a directly validated J.GWR package. The restricted Ref -> Post clause is selected-pair compatibility only: J.RTLConsistentRefPrefix already supplies a selected reachable RTL prefix plus compatible annotations; the theorem does not construct an RTL execution from an arbitrary source run.                                                                                                                                                                                                                                                                                                                                                   |
| <b>J.0</b>                                                                 | Compose an already selected compatible pair. On normal Post -> Ref, J.GWR-Comp supplies that pair; on restricted Ref -> Post, J.RTLConsistentRefPrefix supplies it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>J.TraceCertCov / J.RefProjCover</b>                                     | Safety-side RTL-prefix certificate coverage and optional one-complete-source-run coverage. J.TraceCertCov is route-sensitive; for CycleReach_post the standard QSync route derives it from J.QSyncTraceInd.<br>Core.RTLCycleCarrier/J.CycleIdealCover can then extend J.IdealPrefixClosed safety to non-cycle-aligned finite ideals without per-prefix packages, while properties that fail ideal-prefix closure still need direct broader-domain J.TraceCertCov. J.RefProjCover uses finite canonical ideals; Core.SelectLatitude-legal delay matters to that one-run coverage only when it changes canonical $\Sigma_{DUT}$ history, not for pure unit-empty delay already hidden by the canonical quotient. Neither predicate implies K.CertCov. |
| <b>J.IdealPrefixClosed / J.1-Safe / J.1-Safe-QSync</b>                     | J.IdealPrefixClosed defines safety closure under finite required-order ideals, not just suffix deletion from a linear trace. J.1-Safe transfers such a J.HCanon-invariant property through Post -> Ref under route-sensitive J.TraceCertCov. J.1-Safe-QSyncCycle is the cycle-aligned standard-QSync instantiation; J.CycleIdealCover plus J.1-Safe-QSync extend it to every reachable finite RTL prefix. One-run source use additionally requires J.RefProjCover.                                                                                                                                                                                                                                                                                  |
| <b>J.CycleReach_post / J.QSyncTraceInd / J.CycleIdealCover</b>             | Cycle-aligned RTL safety domain and base/one-cycle/exhaustive-successor package induction under QualifiedSafetyCarrierModel_RTL_B, carrying only LocalGWR/J.GWR-Comp plus J.HCanon; RTL QS3-QS5, QS2a-R and K-only Offer/KObs/drain facts are excluded. J.QSyncCertInd-Trace can supply it via the named qualification projection and stored safety stage. For arbitrary finite-prefix safety, Core.RTLPartialCycleClosure supplies within-entered-cycle completion to Core.RTLCycleCarrier, Core.RTLPartialCycleExt is derived, and J.CycleIdealCover maps every reachable finite outer canonical history to a finite ideal of a reachable cycle-aligned history.                                                                                  |
| <b>J.QSyncCertState / J.QSyncCertInd / J.CertExist-QSync</b>               | Richer K-facing package induction. Each node begins from the extendible safety $\tau_{ref}^G$ , forms a fresh Core.L3CloseView sibling, applies KObsNormRef/K.NormStable NS0-NS6 and the Offer/KObs/Drain package, and never extends an old closed K-facing branch. CertExist-QSync hides the carrier existential semantically; K.CertCov-QSync binds the resulting packages.                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Core.18 / J.HCanon / OfferReach / OfferConf / KObsConf / KObsPost_J</b> | Core fixes the common KObs type/equality seam. J proves common canonical history, source H/O realizability, complete RTL own-side request-to-offer attribution, constructs KObsPost_J, and proves its typed equality with KObs_E,w for the exact selected cutpoint package. The bound E template supplies deterministic settled handshake and ReleaseOwnerSets_E reconstruction; stateless mirror signals are not extra residual offers.                                                                                                                                                                                                                                                                                                            |
| <b>Corollary J.1</b>                                                       | Export equal KObs plus the J.KObs-Proc/Chan/Offers/WFG reconstruction facts, including settled handshake, ReleaseOwnerSets_E/WaitOwners_E, and Core.WFG. It makes no K deadlock, waiver, environment-terminal, or diagnostic conclusion.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Name                                                              | Role                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>J.KObs-SyncCarrier / J.KObs-SyncInFlight / J.KObs-InFlight</b> | Source-side current-state reconstruction from the common KObs. These lemmas recover typed synchronization carrier/readiness and local SyncFire effects, then the selected Sys_ref FlightStateRec/InFlightStepRec action relation. They do not supply future action opportunities.                                                                                                   |
| <b>J.InFlightPathSound</b>                                        | Model-indexed universal operational realization over admitted profile starts. The exact prefix/carrier occurrence returned by one edge is the input to the next. Termination before a remaining declared edge requires separate no-further-edge evidence. K.InFlightProfileCover is a consumer-side premise rather than part of this theorem or KObs.                               |
| <b>J.B1OuterHistCongruence / J.B1PostProjCover-RunMax</b>         | RTL-only alternate coverage machinery. For B1ObsPrefix-related prefixes, ordinary generators plus triggered RSC/QPC yield finite-ideal outer histories. B1RunMax then covers every reachable outer history by an ideal of the selected $p\_post^0$ after trimming any outer-empty suffix; no RTLPartialCycleClosure or fair extension is used.                                      |
| <b>Corollary J.1-Safe-B1Run</b>                                   | Selected-terminal-run safety shortcut. One singleton J.TraceCertCov package matches $p\_post^0$ to $p\_ref^0$ ; if $\Phi$ is extensional on CanonHist_ $\Sigma$ _DUT, J.IdealPrefixClosed, and holds on the selected source history, every reachable RTL history satisfies $\Phi$ . Safety-only; no J.RefProjCover, universal TraceCertCov, QSync carrier induction, or K evidence. |

## 13.4 Appendix K

| Name                                                   | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>K.Dead / K.DeadW / K.ReleaseSCCReduction</b>        | Literal message-rooted reachable generalized-drain-closed internal wait-cycle predicates. K.Dead uses exact ReleaseOwnerSets_E closure plus coarse-WFG connectivity and generalized DrainClosedNowRec. K.ReleaseSCCReduction now relies on K.DrainClosedHeredity: process-indexed WaitRoots restriction gives InFlightEscape monotonicity and therefore drain-closure inheritance to a terminal SCC. Synchronization is not added to the literal WFG.                                                 |
| <b>K.O / K.DeadChar</b>                                | K.O reconstructs the frontier from KObs-derived inputs. K.DeadChar is the source-side/lowered-KCut disabled-channel characterization used by K.Dead and cites Core.12/Core.13; it is not an arbitrary RTL microstate theorem.                                                                                                                                                                                                                                                                         |
| <b>K.KObs-Dead / K.KObs-Ext</b>                        | K-owned factoring and equal-KObs transfer of release-aware deadlock, waiver, environment-terminal, and diagnostic predicates. Equality includes the reconstructed ReleaseOwnerSets_E/WaitOwners_E and handshake facts; Core.WFG remains the authoritative coarse graph definition.                                                                                                                                                                                                                    |
| <b>K.DrainReach / K.DrainReach-Comp / K.RLAdmReach</b> | K.DrainReach supplies nonwithdrawal, K-epsilon, Core.Drain/K.Drain and SettleKBridge for the exact K-facing source branch. K.RLAdmReach is now explicitly four-layered: RLA0 Core.L3CloseView from $\tau\_ref^G$ to $\tau\_ref^{cl}$ , RLA1 RegPhaseReach on the extendible construction checkpoint, RLA2 K.NormStable NS0-NS6 over close-view/KObs normalization, and RLA3 DrainReach on $\tau\_ref^K$ . DrainReach-Comp derives the normal route from DR1-DR4; there is no DR5 history certificate. |
| <b>K.CertifiedCutpoint</b>                             | Complete package for one reachable RTL KCut plus one recorded reachable-prefix/NormPost pair ending at that KCut.                                                                                                                                                                                                                                                                                                                                                                                     |

| Name                                                                  | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>K.CertCov</b>                                                      | Package totality over every reachable RTL KCut. In the standard theorem-derived QSync route, QualifiedSynchronousCarrierModel_RTL_B/Core.QSync-Post supplies KCut closure from the safety foundation plus QS3-QS5; full QualifiedSynchronousModel_RTL_B with QS2a-R/Core.RTLKCutSettlingWindow plus Core.KOriginCarrier-KCut recovers a carrier position for each queried represented KCut; J.QSyncCertInd -> J.CertExist-QSync -> K.CertCov-QSync then derives total certified packages. Alternate independently checked total-extraction routes remain valid.                                                                                                                                                     |
| <b>SDet / K.PolicySProgress</b>                                       | SDet fixes one functional Policy-S artifact and history-affecting choices. PolicySProgress separately binds exact-artifact IssueFair_{Policy S} and applicable B2/B3/reset/terminal progress semantics; K.CompleteS consumes that record. Neither SDet nor K.RespClean implies progress.                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>A5-L</b>                                                           | No admissible Policy-L prefix reaches a source KCut satisfying Dead_K^W. Drain-closedness is part of Dead_K, not an extra A5-L cutpoint restriction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>K.CV / K.OpKey / K.CVPred/K.CV-WF / K.CompareAdvanceLegal</b>      | K.CV is authoritative for design-level serial causal/value determinism. K.CVPred is exact-prefix: Avail_pi, predecessor-before-use, evaluator applicability, ObsKey occurrence/version identity, ResetSupport CV/K.CVResetSupportSound and CVP-M persistence. The serial predecessor relation is directly WellFounded; numeric rank is optional. CompareAdvanceLegal remains the separate comparison launch/Issue legality bridge.                                                                                                                                                                                                                                                                                  |
| <b>Kε / NB-Align / K.SerialRoutePremises</b>                          | K-epsilon is the explicit WFG-inert normalization premise; NB-Align remains separate. K.SerialRoutePremises SR1-SR7 now contains SR3-G generalized-drain consumer coverage and universal SerialCandidateApplicability. Candidate applicability covers environment/release plus reset/terminal closure; it is not an assumption that Policy S itself is a conservative stress test, and it does not assume the later SF blocker or response failure.                                                                                                                                                                                                                                                                 |
| <b>K.JointFreeze-Ord / K.JointFreeze / K.EnvMF / K.EnvMF-Uniform</b>  | Frozen-component no-release/environment applicability. K.JointFreeze-Ord is theorem-derived for the qualifying ordinary or comparability-certified single-frontier/single-singleton-release-support case. Every other admitted StandardEnv candidate - including a single frontier with a multi-owner or multi-alternative ReleaseOwnerSets_E family - requires K.JointFreeze, which may be packaged uniformly for all such candidates. Timed/reactive or otherwise nondefault environments admit only the selected internal frontier plus the K.JointFreeze-Ord release-support shape; K.EnvMF-Uniform is the instance-level sufficient discharge when those conditions hold at every reachable K-facing cutpoint. |
| <b>K.DomResetScope / K.InterruptionGate / K.PolicySCompareDomains</b> | DomResetScope is a predicate on an explicitly selected valid S_dom; DomResetScopeMin/leastness/join supply canonical helpers and only the named transfer facts. Reset is evaluated before Terminal. On nested Continue, one prospective PolicySCompareDomains=<SFControl,Handoff,Response> package is fixed before SF/Q. R-S/T-S constrain only SFControl; post-handoff reset/terminal are Response outcomes.                                                                                                                                                                                                                                                                                                       |

| Name                                                                                   | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| K.RespCov / ClassSet / SFOutputEnvelope / CD3-C-CD4 / RespClean                        | RespCov constructs canonical logical monitors for literal/static-derived dispositions. SFOutputEnvelope is the fixed pre-SF dynamic-instance frontier/provenance/key envelope and predicts neither selection nor triggering. RespSerialBlockerClassSet conservatively covers all possible SF response classes; pruning requires closed RespSerialClassExclusionSound. CD3-C prospectively represents every relevant triggered dynamic monitor's exact first-trigger record; CD4 is derived after SF. RespClean excludes all non-waived RespFail outcomes, including concrete early DirectFail. |
| K.EnvOrd / K.Done                                                                      | Optional diagnostic starvation/completion classifications only. They may simplify WFG+/EnvStop/DoneStop reporting or explain a K.RespFail, but they are not soundness-critical premises, do not discharge A5-L, and cannot justify omitting K.RespCov/K.RespClean or a required response monitor.                                                                                                                                                                                                                                                                                              |
| K.2b / K.2c                                                                            | K.2b/K.2c implement the serial-dominance/response contradiction under the current applicability stack. The frozen serial frontier is handoff-stable/non-advancing only through the certified handoff and thereafter until the first response disposition, not permanently. Response uses reset-first cancelled-pending, terminal maximal-pending, then timeout. K.2c combines this with A5-S; G2 then uses K.Low.A5.                                                                                                                                                                           |
| K.1A / K.1A-Total / K.1                                                                | K.1A is the per-certified-cutpoint theorem and consumes A5-L but not the serial-route premises. K.1A-Total adds Core.KCutClosure_post and K.CertCov. In the standard QSync route those universal-domain facts are derived through QSync-Post/QSync-KCutCanonical and J.QSyncCertInd/J.CertExist-QSync/K.CertCov-QSync. K.1 is the user-facing composition that first obtains A5-L from SDet/A5-S/K.2b/K.2c and then applies the universal layer.                                                                                                                                               |
| K.WaitRoots / K.InFlightEscape / K.DrainClosedHeredity / K.InFlightRelocate            | Generalized drain core plus typed negative-evidence transport. Same-occurrence accepted negatives may be inherited to a nonempty structural subset only by K.InFlightNegativeEvidenceHeredity with a fresh query/status object. K.DeadStructCandidate/Cover fixes the structural candidate family independently of the later contradiction and does not replace occurrence/profile coverage.                                                                                                                                                                                                   |
| K.InFlightQual / Q.2 / K.PrimitiveDrainExact                                           | Q.2 declares current-state action/effect schemas. K.InFlightQual binds soundness, finite drain, occurrence-relative continuation/path soundness and record-level K.InFlightRelationPrecision. Each negative separately binds K.InFlightSearchQuery/Status: ExhaustiveExact requires K.InFlightRelationComplete; OmittedRootIrrelevant is exact-query evidence; UnknownIncomplete is diagnostic. K.InFlightProfileCover is a separate consumer premise, not a CertifiedCutpoint field.                                                                                                          |
| K.Low.ModalQual / LowAdminNormalize / LowWaitRoots / InFlightPathCorr / DeadRootBridge | Qualified G2 lowering transport. TimeStutterQual precedes K.Low.C/K.Low.A5 when real-cycle admin stutter is possible. ModalQual MQ2 is instance-wide: every relevant normalization window must complete before any affecting RESET that would change the projection; LT2 reset mapping alone is insufficient. MQ5 includes both model-indexed InFlightQual and paired-occurrence ProfileCover. ResetStutterMatch/ResetKey-Stutter binds mapped resets; InFlightNegativeEvidenceTransport constructs a fresh target-model negative rather than reusing a query across models.                   |



## 13.5 Appendix L

| Name         | Role                                                                                                                                                                                                                         |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>L.1</b>   | Causal witness selector for epsilon-modeled source choices at generic epsilon-quiescent source choice points. It is intentionally not restricted to Core.KCut, whose additional boundary-evaluation condition is K-specific. |
| <b>L.2</b>   | Conditional coverage/causality/noninterference theorem under which a realizable snoop-selected witness proves WC.                                                                                                            |
| <b>L.3</b>   | NB-CF identity witness; no snooping required for irrelevant failed polls.                                                                                                                                                    |
| <b>L.4</b>   | Cadence-sensitive polling requires isolation/snoop alignment.                                                                                                                                                                |
| <b>L.Dir</b> | Ordinal-preserving source consumption of a free-running RTL snoop stream; benign cycle skew is allowed, while order/kind/payload divergence means witness non-existence and no theorem-backed guarantee.                     |
| <b>L.5</b>   | Online harness realization of an existing abstract witness selector, with W1-W5, L.Dir, coverage, and clean failure reporting as separate obligations.                                                                       |
| <b>W1-W5</b> | Recorded ordinal observation, source-side commit gating, registered sampling, comparison/watchdog with proved-versus-inconclusive timeout treatment, and separate coverage proof.                                            |

## 13.6 Appendix M evidence packaging

| Name                         | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M.7a discharge ledger</b> | Definitive current per-instance trigger/discharge/evidence record. It separates design/implementation-conformance obligations from proof-construction/provenance/coverage artifacts and records whether each triggered item is discharged directly, theorem-derived, or bound to reusable qualification. It includes exact ExecutionCarrier/reset/cycle provenance; the no-dynamic-reset identity disposition where applicable; RTLPartialCycleClosure when consumed (including reusable class-level qualification/binding); SourceBridgeClosure/Ext, PhaseRole_A5, ProofUnitConstructionSupport/c-order schemas; the separately triggered Core.KObsSettleQualified/Core.KObsSettle-Ref record plus L3CloseView/K.NormStable NS0-NS6 on K-facing source packages; W.Env waiver evidence where used; model-specific InFlightQual plus every consuming ProfileCover/query/status; PolicySProgress, K.CV ResetSupport/Avail/CVP-M, SR3-G, SerialCandidateApplicability, selected DomResetScope and prospective response domains/class coverage. An authoritative E4/Core.IC legality discharge may be reused by downstream B-faithfulness, J.CCert, correspondence, and certificate consumers, while those consumers retain their own layer-specific obligations. G2 records TimeStutterQual, instance-wide MQ2 normalization opportunity, paired MQ5 covers, reset-stutter and negative-evidence transport. |

| Name                                                    | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M.7b standard QSync certificate profile</b>          | <p>Four-bundle packaging overlay over the current M.7a ledger. Every triggered M.7a item has an explicit bundle owner for each triggered use even when omitted from the short summary; omission is never an inapplicability disposition. Bundle 1 has a common QualifiedSafetyCarrierModel_RTL_B core (QS1 + QS2 + QS2a-D) with exact history/reset/carrier provenance, A8/A9/A10 where triggered, and the direct-input obligations through their Core.RegSeq/R-E/Core.QSync-QS2 owners; it adds Core.RTLPartialCycleClosure only for the all-finite-prefix route, QualifiedSynchronousCarrierModel_RTL_B/QS3-QS5 only when canonical post settling is consumed, and full QS2a-R only for queried-KCut recovery. For a richer K-facing source-cutpoint profile, bundle 1 additionally carries/references Core.KObsSettleQualified_Sys_ref and may cite Core.KObsSettle-Ref; that qualification is not part of the safety-only profile. Bundle 2 owns J construction/refinement, including exact source bridges, support/c-order and safety coverage; Definition B1 is present only when some separately named consumer triggers it, not merely because J.1/QSync is used; bundle 2 also owns the replay/preparation-normalization use of B4/FPD, plus the K-facing close-view branch when used. Bundle 3 owns K generalized-drain/applicability provenance and G2 transport, and carries or references the same triggered M.7a B4/FPD discharge when B5 charging or the generic Core.CycleClosure source-cycle-existence clause consumes it. Bundle 4 owns the fixed Policy-S response certificate including prospective class/domain/CD3-C evidence and post-SF CD4 checking. Alternate A5-L use is through typed K.DirectA5Proof, not an open-ended "other" route.</p> |
| <b>K.CertHdr / K.CertPkg binding</b>                    | <p>Common theorem-instance/version/trust boundary for generated or manually assembled certificate packages used by certificate-backed Appendix-J safety as well as K-facing/liveness claims. K.CertHdr explicitly carries the authoritative consumed-interface schema manifest and incompatible-schema revalidation status. Safety-only packages contain only common binding fields and subordinate evidence whose M.7a triggers fire; K-only waiver/KObs/A5/response/lowering/liveness fields are absent or inapplicable unless triggered, not fabricated to fill the package. Bind exact history/reset/QSync/bridge/construction/projection/close-view schemas, including the separately triggered Core.KObsSettleQualified source-observation record and Core.KObsSettle-Ref theorem reference for standard K-facing source packages, generalized-drain precision/ProfileCover/query identities, serial CV/progress/applicability/domain evidence and G2 MQ2/MQ5/reset-transport identities as applicable. The 2-September safety-carrier weakening is a named compatible premise projection: prior QualifiedSynchronousCarrierModel_RTL_B evidence may discharge QualifiedSafetyCarrierModel_RTL_B through Core.QualifiedSynchronousCarrierModel-SafetyProjection without MIG1-MIG5 migration solely for that change. Any stored/generated artifact that consumed another incompatibly changed normative interface is invalid for the current claim unless regenerated/rechecked or accompanied by checked K.InterfaceSchemaMigration MIG1-MIG5.</p>                                                                                                                                                                                                                 |
| <b>M.7c alternate B1-run safety certificate profile</b> | <p>Normative safety-only packaging for J.1-Safe-B1Run. It packages the RTL maximum/congruence record, singleton selected-run correspondence, WC variant, and source property/extensionality record. M.7a remains authoritative; M.7c changes only the coverage organization and does not weaken any triggered obligation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## 13.7 Architecture summary

### SUMMARY

Appendix F remains the architect-facing applicability screen, and the Core/G/I/J/K/L decomposition is still the main navigation map. The key safety update is that Appendix J now has a B1-free pairwise correspondence engine plus two distinct universal coverage architectures. The standard QSync route remains the general cycle/carrier path:  $\text{QualifiedSafetyCarrierModel\_RTL\_B} = \text{QS1} + \text{QS2} + \text{QS2a-D}$ ,  $\text{J.QSyncTraceInd}$ , and—only for arbitrary finite prefixes inside an entered cycle— $\text{Core.RTLPartialCycleClosure/J.CycleIdealCover}$ . The alternate M.7c B1-run route is a terminating fixed-stimulus safety shortcut: bucketed B1 at outer  $\Sigma\_DUT$ ,  $\text{Core.B1RunMax}$ , ordinary  $\text{Core.17/R-E}$  conformance, conservative reset congruence  $\text{RSC0-RSC3}$  when triggered, elaboration projection congruence  $\text{QPC1-QPC3}$  when triggered, and one singleton  $\text{J.TraceCertCov}$  package derive  $\text{J.1-Safe-B1Run}$ . It does not require RTL QSync carrier/origin induction,  $\text{RTLPartialCycleClosure}$ ,  $\text{LocalGWRExtCert/GWR-CompExt}$ , universal  $\text{TraceCertCov}$ ,  $\text{J.RefProjCover}$ , or any K-only package solely for that safety conclusion. Source-side  $\text{Core.QSync}$  remains part of the normal selected  $\text{J.GWR-Comp}$  construction in both profiles, unless a direct  $\text{J.GWR}$  route supplies equivalent settling evidence. B1 is determinism of the selected observable quotient rather than full internal execution determinism, and plain  $\text{J.1/A7}$  no longer consume it. Under  $\text{B1+BRM1}$ ,  $\text{BRM2}$  has a cycle-horizon characterization: no reachable selected occurrence may occur later than the selected run's final completed cycle. Appendix K remains a separate liveness/deadlock stack with its existing  $\text{KObs}$ , generalized-drain,  $\text{Policy-S}$ , applicability and response evidence.  $\text{M.7a}$  is the definitive evidence ledger;  $\text{M.7b}$  and  $\text{M.7c}$  are logical packaging overlays. The scheduling-rules document remains normative if any simplified explanation in this guide differs.

Source note: theorem and definition names follow the Catapult HLS User View Scheduling Rules, 3 September 2026 revision